

Information Sharing and Collusion: Theory and Evidence from Poultry Processing

Avner A. Kreps*

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Abstract

Information exchanges among competitors have become more prevalent in recent years. While they may facilitate collusion, they may also affect outcomes in any direction purely by informing firms about their economic environment. Empirically distinguishing between these two forces is crucial for conduct testing and policy evaluation. This paper develops tools to do so and applies them to the Agri Stats information exchange among turkey processors. In the year after the exchange broke down, the retail price of participating processors fell by 5.1% relative to other processors, and descriptive evidence is suggestive of both conduct and information effects. I specify and estimate a structural model that identifies changes to conduct while controlling for information effects. I reject the null hypothesis of no conduct change, and estimate that subscribers acted like they internalized 47% of other participants' profits while sharing information. Information effects of the exchange alone reduce prices by 1.3%, but are outweighed by conduct effects. Bounds on collusive prices under counterfactual restrictions on information sharing indicate that firms could still attain near-monopoly profits even when sharing aggregated information.

JEL: D22, D43, K21, L13, L41, L66

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*Northwestern University, Department of Economics. Email: avner@u.northwestern.edu.

1 Introduction

Information exchanges are a topic of growing interest in antitrust policy. Motivated by advances in data processing and the increasing adoption of common price aggregators and revenue management systems, regulators have taken a more aggressive posture towards information exchanges in the last several years, leading to the recent filing of numerous antitrust cases across myriad industries (Harrington, 2026). In these cases, competing firms share “competitively sensitive” information, such as recent or future indicators of prices and quantities, with each other. Plaintiffs allege the exchange has a *conduct effect* (i.e., facilitates collusion): it makes markets less competitive even without an explicit price- or quantity-fixing agreement. Defendants usually maintain that there is a procompetitive *information effect*: the exchange teaches firms about economic conditions, making markets more efficient.

Evaluating these arguments requires accurately characterizing the information effects of information sharing. Conduct testing—empirically distinguishing between oligopoly solution concepts—has a rich tradition in industrial organization (Bresnahan, 1982). However, workhorse models of competition typically assume complete information, and may therefore misattribute information effects to conduct effects. Given that information effects are ambiguously signed in theory (Kühn and Vives, 1995; Raith, 1996) and may be large relative to conduct effects (Vives, 2002), tests built on complete-information models can be biased in ex ante unforeseeable directions and magnitudes. Further, while regulators have proposed restrictions on information sharing to prevent firms from colluding, the literature has not formally characterized the extent to which different restrictions impede collusion. Thus, there is little economic guidance on either how to detect collusion through information sharing or how best to regulate information exchanges.

In this paper, I specify a structural incomplete-information oligopoly model that allows for conduct testing and policy evaluation in information sharing settings. The model endogenously captures information effects, allowing for identification of conduct effects, and can be identified and estimated using typically available market-level data. I also develop a computational framework to place bounds on outcomes under counterfactual information exchanges, leveraging recent results in the theory of private monitoring games. These bounds show how different restrictions on the kind of information firms share affect the worst-case collusive outcome.

I implement my methods in the context of a unique natural experiment in the poultry processing industry. A third-party data provider named Agri Stats ran a prototypical information exchange among the largest poultry (chicken and turkey) processors in the United States. In May 2019, the information exchange among turkey processors broke down due to

the threat of an antitrust lawsuit. In the year after the exchange ended, Agri Stats clients' prices fell by 5% relative to other producers, with suggestive evidence of both conduct and information effects at play. With this as motivation, I estimate my oligopoly model, test whether the end of the information exchange caused a change in firm conduct, and simulate worst-case collusive outcomes under alternative information structures.

I ultimately reject the null hypothesis that firms maximized their own expected profits with and without the information exchange. I estimate that while sharing information, Agri Stats clients set prices as though they internalized 47% of their rivals' profits. While Agri Stats raised prices in aggregate, the pure information effect reduced average prices by 1.3%; tests relying on complete information therefore underestimate the magnitude of the conduct effect. Finally, to inhibit collusion, preventing full industry participation in an exchange is more important than mandating aggregation: the worst-case average price is nearly at the monopoly benchmark if all firms observe only their own shares and the total share, and is 10.1 percentage points lower if only Agri Stats clients share the fully disaggregated vector of prices and quantities among themselves, as even aggregate sales information is sufficient for firms to effectively detect deviation from rivals.

I begin in Section 2 by reviewing how information exchanges can affect competition. It is relatively well understood how information exchanges facilitate collusion by alleviating monitoring problems, enabling firms to better detect and punish deviation from a dynamic equilibrium (Green and Porter, 1984). However, information exchanges can also shift aggregate outcomes in any direction even when firms maximize expected profits. Intuitively, information sharing allows firms to more effectively price discriminate across unknown states of nature, which, like any form of competitive third-degree price discrimination, has ambiguous effects on expected prices and welfare in both sign and magnitude (Stole, 2007). As such, one cannot infer a change in competitive conduct from a change in average price caused by information sharing. The overall information effect of information sharing depends on the probability distribution over states of nature, how these states enter firms' profit functions, and what firms learn from the additional information they receive from the exchange.

Section 3 details the empirical setting. In the exchange, Agri Stats clients share recent prices and quantities, as well as costs and indicators of future prices and quantities in the form of the number of chicks and poults (baby turkeys). Agri Stats circulates this information among subscribers on a weekly or monthly basis. In 2016, a class of chicken purchasers sued Agri Stats and its chicken subscribers. Concerned about potential litigation risk, turkey processors dropped out of the information exchange in May 2019, causing Agri Stats to suspend its turkey reports. Meanwhile, Agri Stats never stopped producing chicken reports. This quasi-experimental variation in information sharing provides a useful case study for

investigating how and through what channels information exchanges affect market outcomes.

Section 4 begins the empirical analysis by evaluating the aggregate effects of Agri Stats’ exit from turkey. Using a difference-in-differences with turkey processors that did not subscribe to Agri Stats as a control group, I estimate that Agri Stats subscribers lower their retail prices by 5.1% in the year after the information exchange ends, but relative prices slowly rebound to near pre-May 2019 levels in the following two years. This effect is not driven by broader trends in poultry: I estimate a null effect of the turkey information exchange ending on Agri Stats clients’ chicken prices as a placebo. Digging into these aggregate effects reveals hallmarks suggestive of both conduct and information effects: the short-run effects are larger in markets that are structurally more susceptible to collusion, and Agri Stats clients’ prices are more correlated with common demand shocks and with each other while the information exchange is ongoing.

I outline my structural model capturing information effects in Section 5. The model nests workhorse static supply-and-demand models (Berry et al., 1995), but does not assume complete and perfect information. Instead, firms have incomplete information over potentially autocorrelated demand shocks, which are unobserved, and cost shocks, which are private information. They learn about these objects through *signals*: arbitrary functions of past prices and quantities that help them predict demand and their rivals’ prices. When firms share information, they observe more informative signals, giving them more precise information about their residual demand curve. The key tractability assumption is that firms treat their rivals’ prices as following a first-order Markov law of motion, conditional on observable supply and demand variables, meaning that firms need only incorporate period- t signals in order to update their beliefs about their residual demand curve from period t to period $t + 1$. At a competitive solution, which I call a Bertrand-Markov equilibrium, firms set prices to maximize expected profits in each period given beliefs that update in a manner consistent with their rivals’ long-run price distribution.

I then apply this model to the Agri Stats context. I capture information effects with different signal functions before and after the exchange: Agri Stats clients observe each other’s prices and quantities while sharing information, whereas other firms just observe their own. Section 6 demonstrates that the entire system of demand and supply parameters, including beliefs, is identified from market-level data while firms are playing a Bertrand-Markov equilibrium, which I assume is played after the exchange ends. I show that the model can be estimated using a combination of standard methods and Bayesian techniques, and that it provides sensible results.

Section 7 uses these estimates to analyze the effects of the Agri Stats information exchange. I first test whether the data are consistent with expected own-profit maximization

while the information exchange is ongoing. Using a test statistic based on the restriction that there is no discontinuity in the difference between Agri Stats subscribers’ average cost shocks and those of other producers in May 2019, I reject the null hypothesis that information effects alone can explain the effect of Agri Stats’ exit on prices. To quantify the magnitude of the conduct effects, I fit a parameterized conduct model and find that Agri Stats clients set prices as though they internalized 47% of their fellow subscribers’ expected profits while sharing information. The analogous conduct parameter with a complete-information supply side is not statistically distinguishable from zero, underscoring that not accounting for information effects can affect the results of conduct tests. Finally, I simulate equilibrium outcomes if Agri Stats had continued sharing information in the post-period and decompose the overall effect into information and conduct effects. I find that Agri Stats raises average prices by 2.5%, but that this aggregate effect is the composition of a -1.3% information effect and a 3.9% conduct effect. Thus, information provision would make the market operate more efficiently if firms were somehow prevented from using it to collude.

Finally, Section 8 assesses the potential effects of restrictions on the information firms can share. While the estimated conduct parameter allows for computing the conduct effects of the Agri Stats exchange specifically, it may not apply to counterfactual monitoring structures, and finding exact equilibria in repeated games with private monitoring caused by incomplete information exchanges is known to be difficult (Kandori, 2002). Thus, to characterize the potential effects of collusion under data sharing restrictions, I develop a computational framework to bound prices under alternative monitoring structures. The idea is to leverage necessary conditions that a repeated-game equilibrium must satisfy by solving a constrained maximization problem, where the objective is to maximize prices and the constraints are the necessary conditions. This procedure delivers an upper bound on collusive prices for a given information exchange. The main conditions, from Kreps (2026), formalize the intuition that monitoring structures that allow firms to better detect deviation from their rivals enable more collusive outcomes.

Using this method, I assess whether a regulator concerned with minimizing worst-case collusive prices should be more concerned with disaggregation or broad participation: do exchanges of aggregate information among all firms, or precise information among only a subset, allow for more harmful collusion? This question is policy-relevant given that antitrust agencies have become more critical of aggregated exchanges in recent years, while the economics literature has disagreed on the extent to which aggregation inhibits collusion.¹

¹Fudenberg et al. (1994) prove a folk theorem for repeated games with imperfect public monitoring where a key assumption is pairwise identifiability—players can distinguish not only *whether* a player deviates but *who* in particular deviates—which Kandori and Matsushima (1998) extend to private monitoring games with communication. Perhaps influenced by these results, Carlton et al. (1997, p. 434) state that “aggregating

I find that preventing full participation in an information exchange is more important than mandating aggregation of information. If all Agri Stats clients (which comprise roughly two-thirds of the market in my sample) were to continue sharing the full vector of prices and quantities, the upper-bound collusive price attains 85.4% of the distance between competitive and monopoly prices on average. This quantity increases to 95.5% if all firms only observe total quantities and their own prices and shares, while the upper bound is 65.9% under no information sharing.

These results have several policy implications. Information effects can be economically meaningful, and thus it is important to account for them when attempting to detect collusion via information exchange. Further, information provision itself can make markets more efficient if firms were somehow prevented from using it to collude. However, exchanges of even aggregate information can facilitate near-monopoly collusion if the entire market participates. As such, the results support the notion, recently reflected in regulatory policy, that bright-line “safe harbors” for information exchanges satisfying aggregation and timing requirements are a bad idea.

Related literature. The theoretical literature on information exchanges has historically been bifurcated. One literature, dating back to [Stigler \(1964\)](#), studies how information exchanges facilitate collusion by resolving monitoring problems. [Green and Porter \(1984\)](#) provide a formal treatment of how uncertainty impacts collusion, and subsequent contributions study collusion with different monitoring structures ([Athey and Bagwell, 2001; 2008; Harrington and Skrzypacz, 2007; Sugaya and Wolitzky, 2018b](#)) and the impact of self-reported communication ([Kandori and Matsushima, 1998; Harrington and Skrzypacz, 2011; Awaya and Krishna, 2016](#)). More recent papers derive necessary conditions that repeated-game equilibria with private monitoring structures must satisfy ([Sugaya and Wolitzky, 2018a; 2023; Awaya and Krishna, 2019; 2020; Kreps, 2026](#)), providing results that I leverage in this paper. Another literature studying how information sharing affects outcomes in static equilibrium begins with [Novshek and Sonnenschein \(1982\)](#); a series of related papers, reviewed by [Kühn and Vives \(1995\)](#) and [Raith \(1996\)](#), concludes that information effects are ambiguously signed and depend on modeling details such as the type of uncertainty and firm decision variable.² While these two literatures have largely evolved in parallel, I contribute an

the data largely removes the value of information in facilitating collusion.” Conversely, [Awaya and Krishna \(2020\)](#) present conditions under which firms can use aggregate sales information to enforce an equilibrium with near-monopoly profits.

²For example, sharing information about a linear demand intercept is procompetitive in Cournot competition and anticompetitive in Bertrand competition, while these are reversed for sharing information about costs. More recent contributions to this literature include [Asker et al. \(2020\)](#) and [Boyarchenko et al. \(2021\)](#), who study information sharing in auctions; [Bonatti et al. \(2017\)](#) and [Sweeting et al. \(2024\)](#), who incorporate information sharing in dynamic models with signaling incentives; and [Byrne et al. \(2025c\)](#), who study

empirical framework weighing conduct and information effects against one another, enabling aggregate quantification and counterfactual predictions. My model also incorporates more general forms of uncertainty than are typically studied in the information effects literature—while this generality precludes analytical results, it enables characterization of information effects in more realistic settings.

This paper also relates to an empirical literature analyzing case studies of information exchanges. Findings have been mixed: [Albæk et al. \(1997\)](#) and [Byrne et al. \(2025a\)](#) find anticompetitive effects of information sharing among Danish ready-mixed concrete producers and Australian gasoline retailers respectively,³ while [Doyle and Snyder \(1999\)](#) and [Borenstein \(2004\)](#) find procompetitive effects among automakers and airlines respectively.⁴ These papers typically characterize aggregate effects of information sharing without formally testing whether they caused collusion. More recently, [Calder-Wang and Kim \(2025\)](#) and [Calder-Wang et al. \(2025\)](#) test whether adoption of a common pricing algorithm led to changes in conduct among apartment owners. While our structural models are based on similar demand systems, our models of information effects differ due to the setting: it is difficult to model the signal transmitted by the pricing algorithm they study. [Calder-Wang and Kim \(2025\)](#) do not microfound information effects, instead assuming that non-subscribers have complete information but make systematic mistakes, while [Calder-Wang et al. \(2025\)](#) test whether prices are consistent with Bayes-correlated equilibrium under any information structure. In contrast, I study a classical information exchange in which firms directly share actions and outcomes, enabling me to model the information structure and account for information effects more directly.

Finally, this paper relates to the literature on conduct testing and estimation. One important and related early contribution is [Porter \(1983\)](#), which, like this paper, estimates conduct in a setting where firms have incomplete information about demand shocks. This literature has received renewed attention recently, with contributions from [Berry and Haile \(2014\)](#) on identification; [Backus et al. \(2021\)](#), [Duarte et al. \(2024\)](#), and [Dearing et al. \(2026\)](#)

how information sharing can facilitate higher prices through commitment effects. Finally, the literature on third-degree price discrimination ([Holmes, 1989](#); [Corts, 1998](#); [Stole, 2007](#)) draws similar conclusions using broadly similar economics.

³While information sharing among Australian gasoline retailers raised prices overall, [Byrne et al. \(2025b\)](#) find that removing one retailer from the exchange but leaving the rest intact raised prices by even more.

⁴Similar settings studied by prior papers include price transparency regulations, which combine information exchanges among firms with potential reductions in demand-side search costs ([Rossi and Chintagunta, 2016](#); [Luco, 2019](#); [Grennan and Swanson, 2020](#); [Ater and Rigbi, 2023](#); [Montag et al., 2026](#)), and common pricing algorithms, which recommend prices to competing firms using information that is pooled across competitors ([Assad et al., 2024](#); [Calder-Wang and Kim, 2025](#); [Calder-Wang et al., 2025](#)). [Clark and Houde \(2026\)](#) show indirect evidence of collusion via information exchange: firms that were shown to have colluded ex post were more likely to have joined an information sharing platform.

on instrument relevance and model selection; and [Miller and Weinberg \(2017\)](#) providing an important empirical application. These papers primarily restrict attention to complete-information supply-side models; this paper builds on these methods by incorporating incomplete and imperfect information, enabling conduct testing in settings where firm conduct covaries with the information structure.

2 The economics of information effects

It is relatively well understood how information exchanges facilitate collusion by giving firms better information about their rivals' actions, letting them better punish deviation from a collusive equilibrium ([Stigler, 1964](#)). However, information sharing can also affect aggregate outcomes even if firms simply maximize expected profits. When firms share information, they are better able to price discriminate across states of nature, which, like any form of oligopolistic third-degree price discrimination, can move expected prices and welfare in any direction ([Stole, 2007](#)). This section states the result precisely, provides intuition, and discusses the forces that determine the information effects of information sharing.

Suppose that firms $f \in \{1, 2\}$ compete by setting prices. There is a primitive $\theta \in \{A, B\}$ that enters firms' profit functions; θ could shift demand and/or marginal costs, although I am agnostic about its source.⁵ Let $\pi_f^\theta(p)$ be firm f 's profit function given prices $p = (p_1, p_2)$ in state θ ; I assume it is twice differentiable and strictly concave in p_f . Let $b_f^\theta(p_{-f}) = \arg \max_{p_f} \pi_f^\theta(p_f, p_{-f})$ be firm f 's state-specific best-response function, which is single-valued due to strict concavity. I assume that for all f , p_{-f} , and θ ,

$$\frac{\partial^2 \pi_f^\theta(b_f^\theta(p_{-f}), p_{-f})}{\partial p_f \partial p_{-f}} \in \left(0, -\frac{\partial^2 \pi_f^\theta(b_f^\theta(p_{-f}), p_{-f})}{\partial p_f^2}\right),$$

which ensures that best-response functions have positive slopes less than 1, and hence that there exists a unique Nash equilibrium price vector p^θ in each state that satisfies $p_f^\theta = b_f^\theta(p_{-f}^\theta)$ for both firms.

Suppose that without information sharing, firms do not know the realization of θ . Instead, each firm has a prior described by $\mu_f \in (0, 1)$, where firm f believes that $\theta = A$ with

⁵The assumptions of two firms and two states are made for expositional and graphical convenience; the key forces and results are the same with arbitrarily many firms and states.

probability μ_f .⁶ Define firm f 's no-sharing best-response function as

$$b_f^{NS}(p_{-f}) = \arg \max_{p_f} \mu_f \pi_f^A(p_f, p_{-f}) + (1 - \mu_f) \pi_f^B(p_f, p_{-f}).$$

p is a no-sharing equilibrium price vector if and only if $p_f = b_f^{NS}(p_{-f})$ for each $f \in \{1, 2\}$.

What set of prices can arise in a no-sharing equilibrium? For each firm f , define the region U_f as the area between firm f 's best-response curves,

$$U_f = \{(p_1, p_2) : p_f = \alpha b_f^A(p_{-f}) + (1 - \alpha) b_f^B(p_{-f}) \text{ for some } \alpha \in (0, 1)\},$$

and define the area enclosed by all firms' best-response curves as $U = U_1 \cap U_2$. U is the set of potential price vectors sustainable as a no-sharing equilibrium, as the following well-known result from the theory of price discrimination shows.⁷

Proposition 1. p^{NS} is a no-sharing equilibrium price vector for some prior $\mu = (\mu_1, \mu_2)$ if and only if $p^{NS} \in U$.

Proof. See Corts (1998, Propositions 1 and 2). □

Figure 1 displays the intuition for Proposition 1 in graphical form. The firms' state-specific best-response curves are drawn in solid lines, while the no-sharing best-response curves are drawn in dotted lines. The shaded area is U , the area between the firms' best-response curves that defines the set of possible no-sharing equilibrium price vectors. Intuitively, each firm's best-response function when it does not observe the state lies somewhere in between its state-specific best-response functions, with the relative distance between the two curves governed by the firm's prior. The no-sharing equilibrium price vector is the intersection of the two best-response functions; by adjusting the priors, the curves vary continuously between $b_f^A(p_{-f})$ and $b_f^B(p_{-f})$, so one can choose priors such that the curves intersect at any point in U . This result implies that the no-sharing prices could end up below or above the

⁶That firms' priors are potentially heterogeneous is a reduced-form way of capturing the fact that they may observe private information about θ before setting prices. Fully modeling this private-signal process requires firms to form expectations not only about the state but about their rivals' interim beliefs and therefore prices. Adding this private-signal process would not change the overall result, and I incorporate firms forming beliefs about their rivals' prices as well as underlying states of nature in the empirical model.

⁷This result does not rely on heterogeneous priors. Even if firms have common priors (i.e., $\mu_1 = \mu_2$), for any set of best-response functions (which define U) there exist profit functions that generate those best-response functions and whose no-sharing equilibrium price vector is p^{NS} if and only if $p^{NS} \in U$. This is because the state-specific best-response function is defined by the point at which marginal profit in that state is zero, which is invariant to scaling of the marginal profit function, while the no-sharing best-response function is sensitive to the relative scale of the two profit functions. If states differ in terms of the scale of firms' profits—which occurs if, for example, the state is a measure of local market size—then profit functions can be defined such that any point in U is attainable. Corts (1998) states the result in this form as well.

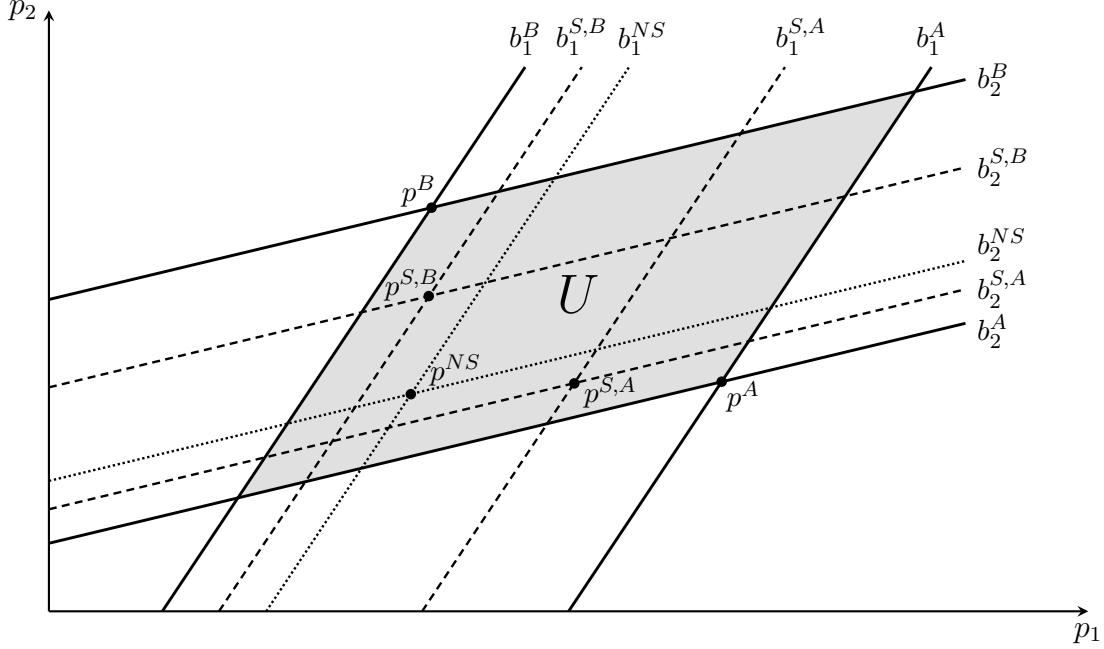


Figure 1: The possible information effects of information sharing

Note: This figure displays possible equilibrium prices under complete information, no information sharing, and information sharing given one specification of profit functions, priors, and signals. The state-specific best-response functions b_f^θ are drawn in solid lines. The shaded area between them defines U , the set of possible no-sharing and with-sharing equilibrium prices. p^A and p^B , the points at which the state-specific best-response functions intersect, are the complete-information equilibrium prices. The no-sharing best-response functions b_f^{NS} are drawn in dotted lines; the no-sharing equilibrium price p^{NS} is at the intersection of these two lines. The with-sharing interim best-response functions, $b_f^{S,y}$, are drawn in dashed lines; the intersections $p^{S,A}$ and $p^{S,B}$ are the with-sharing equilibrium prices.

complete-information prices in *both* states for both firms—as is the case in Figure 1—and hence in expectation.⁸

Now suppose that before firms set prices, they observe a common binary signal $y \in \{A, B\}$, where $\Pr(y = A \mid \theta) = \eta^\theta$ and $0 \leq \eta^B \leq \eta^A \leq 1$. Let $\mu_f(y)$ be firm f 's interim belief that $\theta = A$ after observing signal y , and let $b_f^{S,y}(p_{-f})$ be its interim best-response function. $b_f^{S,y}(\cdot)$ is equal to the no-sharing best-response function if $\eta^A = \eta^B$ and is the corresponding state-specific best-response function if $\eta^A = 1$ and $\eta^B = 0$, varying continuously between the two for intermediate values. The with-sharing prices $(p^{S,y})_{y \in \{A, B\}}$ satisfy $p_f^{S,y} = b_f^{S,y}(p_{-f}^{S,y})$ for each f and y . Because the interim best-response functions lie between the state-specific ones, the equilibrium prices—their intersection—continue to lie in U .

Figure 1 displays the with-sharing best-response functions in dashed lines, with the equilibrium prices marked at their intersections. Information sharing has two effects. First,

⁸The key feature that makes this possible is *best-response asymmetry*: $b_1^A(p_2) > b_1^B(p_2)$ but $b_2^A(p_1) < b_2^B(p_1)$. If firms exhibit best-response symmetry, then information sharing lowers prices in one state and raises them in the other, although it can still raise or lower average prices across states (Holmes, 1989).

firms use the information to make their beliefs more accurate: their interim best-response functions move towards the state-specific ones. Second, it correlates firms' beliefs: firm 1's best-response function moves towards b_1^A precisely when firm 2's best-response function moves towards b_2^A . Together, these forces move the with-sharing equilibrium prices towards the complete-information equilibrium prices, which, in the case depicted in Figure 1, raises prices state-by-state and consequently in expectation. However, the magnitude of the movement in both states depends on the signal probabilities (η^A, η^B) ; if signals are fully informative then information sharing moves prices all the way to the complete-information benchmark, while if signals are completely uninformative then information sharing does not affect equilibrium prices at all.

Ultimately, this example has two implications for the study of information sharing. First, information effects alone can raise or lower prices both state-by-state and in expectation, and can even shift prices in the same direction in all states. Thus, one cannot infer a change in competitive conduct caused by information sharing from a change in average prices, and any econometric model attempting to identify whether information sharing facilitates collusion needs to account for information effects.⁹

The example also elucidates several primitives that drive the sign and magnitude of information effects. First, how profit functions change across states affects the contours of the set of possible prices, U .¹⁰ Second, the underlying probability structure across states and the amount and type of firms' private information affect the heterogeneity of firms' priors when they do not share information. Third, the amount firms learn from information sharing, including whether firms learn differently about different types of shocks, affects how much information sharing moves prices between the no-sharing and full-information benchmarks. Because signals are frequently past prices and quantities, which are themselves equilibrium objects, a model of information effects should capture the extent to which sharing past prices and quantities informs firms about future conditions, which then changes future prices and quantities and hence the underlying signal structure. These primitives—the underlying profit structure across states of nature, the probability distribution of these states, and how much firms learn from information sharing in equilibrium—are all features that a structural model

⁹It is also worth noting that the change in average prices is not a sufficient statistic for the welfare effects of information sharing, as consumer surplus may respond asymmetrically to prices across states. For example, if there is high demand in one state and low demand in another, then a firm that learns about the state will raise prices in the high-demand state and lower them in the low-demand state, which may lower average prices across states but also lower welfare if consumer surplus is more sensitive to prices when demand is high than when it is low, as is the case in many parametric demand systems.

¹⁰For example, states being shocks affecting demand for the entire market implies best-response symmetry (hence the ability of information sharing to raise or lower prices in all states), while states being shocks affecting relative demand across firms implies best-response asymmetry.

of information sharing should attempt to capture as flexibly as possible.

3 Institutional details and data

3.1 Overview of poultry processing

Poultry processing is the preparation of live fowl (primarily chicken and turkey) for human consumption. It includes all steps in the production process starting from a live bird and finishing with a product that is sold to retailers, restaurants, or other food service firms. One firm, known as a processor or integrator, typically handles all steps of this process from slaughter to wholesale, as poultry must be quickly processed after slaughter before being chilled or frozen in order to prevent spoilage.¹¹ The processors are present throughout the supply chain from wholesale to retail; while they contract with, rather than own, poultry farms that grow the birds themselves, the processors decide how many chicks and poults (baby chickens and turkeys) are placed (Zheng and Vukina, 2007). While there are differences between turkey and chicken processing, mostly relating to size and maturity time, the process is similar: feed components are the same, the USDA regulates the processing plants in the same way, and there are many suppliers in common (Fernández-López et al., 2010; USDA).

Wholesale poultry prices are relatively flexible.¹² Retailers and other downstream consumers typically buy processed poultry through purchase orders, which authorize a retailer to purchase a prespecified quantity at a prespecified price, typically expressed as a proportion of a common price index that adjusts for cost shocks. While these purchase orders are occasionally components of longer-term agreements, these agreements are flexibly renegotiated and typically do not constrain either party to purchase or sell any particular quantity of poultry—both processors and retailers generally do not list ongoing purchase orders as contractual obligations in their financial statements, as they are cancelable on short notice. Substantial variation in the terms of these agreements across retailers within a processor is relatively rare (Harjani, 2025).

Poultry processors typically rely on private data providers to learn about market conditions. While the USDA publishes some aggregated price indices and summary statistics, the underlying data are voluntarily reported by processors. These indices are therefore viewed by market participants as less comprehensive than private sources and potentially subject

¹¹The main exception is firms that purchase pre-processed raw poultry and process it further, often turning it into deli meat. Such firms, known as “value-added” processors, produce a very small fraction of the poultry consumed in the US, and I exclude them from my analysis; see Appendix B.1.

¹²This paragraph draws on processor and retailer 10-Ks and financial statements available from the SEC EDGAR database.

(a)			(b)	(c)	(c.1)	(c.2)	(c.3)	(d)	(d.1)	(d.2)	(e)	(e.1)	(e.2)	(f)	(f.1)
											VARIANCE TO NAT'L			ECONOMIC IMPACT DOLLARS	
LINE	PRODUCT CODE	AGRI STATUS/COMPANY	PRODUCT DESCRIPTION	COMPANY POUNDS	COMPANY MIX %	NATIONAL POUNDS	NATIONAL % MIX	COMPANY NET PRICE	NAT'L NET PRICE	NAT'L TOP 25%	NAT'L RANK	NAT'L VAR	TOP 25 VAR	VAR NAT'L	VAR TOP 25
1	508/AD 0000A554	DRUMSTICK, V AD, IW, P A/R,	CHILL, JUM	249,571	6.60	1,293,884	0.26	71.80	75.51	79.54	7-8	-3.71	-7.74	-9,255	-19,319

BREEDER SECTION PARTICIPANTS													
LIN	%	PLA	SP	PLT. NUM.	(a) (a.1) (a.2) (a.3) FEMALE CHICKS-AVG WEEKLY PLACED				(b) (b.1) (b.2) MALE CHICKS-AVG WEEKLY PLACED				
					% LAST YR MO	THIS MONTH	MONTH YR AGO	MALE/ 100 FEM	% LAST YR MO	THIS MONTH	MONTH YR AGO		
1.					1 100	-	29,952	-	9.26	-	2,774	-	
2.					2 97	-	23,813	-	12.74	-	3,034	-	
3.					3 94	-	21,740	-	11.94	-	2,596	-	
4.					4 91	136.44	17,562	12,871	14.06	133.33	2,469	1,852	
5.					5 88	80.51	17,360	21,563	12.67	89.08	2,200	2,470	
6.					6 85	132.70	17,342	13,068	14.33	215.30	2,486	1,155	
7.					7 82	44.16	14,034	36,312	9.12	34.79	1,463	4,204	
8.					8 79	-	14,953	-	10.76	-	1,410	-	
9.					9 76	GRNV	165.41	14,468	8,747	11.68	172.66	1,690	979
10.					10 73	201.26	14,338	7,124	12.25	193.10	1,757	910	
11.					11 70	85.85	12,750	14,851	16.16	104.17	2,080	1,978	
12.					12 67	166.82	11,727	7,038	12.97	165.72	1,521	1,818	
13.					13 64	98.19	11,098	11,302	13.64	98.66	1,514	1,535	
14.					14 61	-	10,506	-	21.57	-	2,266	-	
15.					15 58	68.33	10,250	15,000	12.20	61.73	1,250	2,025	
16.					16 55	-	9,690	-	12.78	-	1,239	-	
17.					17 52	91.30	9,668	10,589	15.15	90.75	1,465	1,614	
18.					18 48	-	9,286	-	12.98	-	1,206	-	
19.					19 45	128.21	9,180	7,160	11.11	120.28	1,020	848	
20.					20 42	91.82	7,894	8,597	14.47	91.30	1,142	1,251	
21.					21 39	91.82	7,894	8,597	14.18	91.30	1,120	1,226	
22.					22 36	81.89	5,813	18,066	10.99	108.45	1,090	968	
23.					23 33	-	5,800	-	14.22	-	825	-	
24.					24 30	ARCADA	100.00	5,675	5,675	11.23	100.00	638	638
25.					25 27	82.00	5,625	6,860	12.00	84.38	675	800	
26.					26 24	65.43	4,995	7,634	13.14	74.62	656	879	
27.					27 21	126.36	4,743	3,754	12.22	126.47	580	458	
28.					28 18	-	4,717	-	12.09	-	570	-	
29.					29 15	71.58	4,716	6,589	13.65	72.66	644	886	
30.					30 12	-	4,488	-	14.09	-	632	-	
31.					31 9	88.28	3,278	3,713	14.73	97.64	483	494	
32.					32 6	-	0	5,508	0.00	-	0	1,367	
33.					33 3	-	0	8,597	0.00	-	0	1,251	
34.	33			Total	145.80		347,353	238,248	12.86	145.37	44,631	30,703	

to strategic nonreporting (Leonard, 2017; Boussios, 2023).

3.2 Agri Stats

Agri Stats is a data analytics and benchmarking firm that collects and exchanges price, quantity, cost, and inventory information among the 15–20 largest poultry processors in the US. These processors collectively accounted for 80% and 95% of US turkey and chicken production in the 2010s (Holdheim and Tameez, 2022).¹³ Agri Stats has been sued several times recently—most notably, by the Department of Justice in late 2023—on the grounds that its information exchange facilitated collusion.¹⁴

Agri Stats exchanges information through printed reports. While these reports are gen-

¹³It also operated among pork and egg processors, although its clients accounted for a smaller share of those markets.

¹⁴Much of the information below is taken from documents filed in these lawsuits. All footnote citations to legal documents are from *US et al. v. Agri Stats, Inc.*, case no. 0:23-cv-03009-JRT-JFD, D. Minn.

erally unavailable to the public, some excerpts have been made available in legal filings, examples of which are shown in Figures 2 and 3. Figure 2 displays price and quantity information for one product. Columns (c) through (c.3) give the subscribing firm information on its production relative to its rivals, while columns (d) through (d.2) compare the firm’s price to the industry-wide average and 75th percentile. Column (e) shows that the firm’s price for this product is seventh-highest out of eight firms, and the remaining columns show Agri Stats’ calculation of the loss from not pricing at the average or 75th percentile. Figure 3 contains plant-level information on the number of chicks placed at farms serving different processing plants. Other parts of the reports, examples of which are not available publicly, include plant-product-level prices, quantities, product mixes, and input usage (Leonard, 2017; Tunheim, 2026). The reports are disseminated on a weekly or monthly basis, depending on the type of data, and contain data from the most recent time period.

According to Agri Stats and its clients, processors primarily use the information exchange to more accurately forecast market conditions. For example, in a 2009 earnings call, a poultry executive cites Agri Stats data on prices and hens placed by rivals as the basis for saying that “I see a lot of information from Agri Stats that tells me that nobody’s going to ramp up” production, and that therefore prices would remain elevated (Leonard, 2017). On the same earnings call, an Agri Stats executive says that “the goal is obviously to provide the fact based forecasting and analysis that we talked about, that covers the supply, demand, price conditions and so forth,” and the reports themselves state that one purpose is “to help forecast Broiler [chickens] & pounds produced for future months.”¹⁵ Agri Stats and its clients also claim that the reports help processors find production efficiencies: for example, Agri Stats’ website states that “protein producers depend on Agri Stats’ reports to help them identify opportunities to reduce production costs to keep prices low” (Agri Stats, 2023).¹⁶

Certain details of the reports that have been made public have implications for the analysis to come. First, while aspects of the reports were either aggregated (as in Figure 2) or anonymized, the plant-level data could be deanonymized, and processors were allegedly successful in doing so; as such, I will treat the reports as revealing information tied to specific firm and product identities. Second, the reports are based on automatic data transfers from internal accounting systems that Agri Stats regularly audited, not self-reports; therefore, participants exchange true information without any truth-telling constraints. Third, the reports were only available to subscribing processors that provided their own data, while Agri Stats refused to sell to poultry purchasers or workers; thus, the exchange only affected

¹⁵Second Amended Complaint, p. 21.

¹⁶Some of these purported efficiencies may not be benign: separate lawsuits allege that processors used the Agri Stats reports to reduce costs by fixing wages for processing plant employees, as well as payments to poultry growers (Scarcella, 2026).

the supply side of the market, without potential countervailing effects of price transparency on the demand side.

Agri Stats stopped publishing its turkey reports in May 2019, testifying that the stoppage was caused by a lack of data due to subscribers dropping out. This occurred after the first lawsuit against chicken processors began in late 2016, but before the first turkey-related lawsuit in late 2019. An Agri Stats executive testified that processors who dropped out “cited concerns regarding the broiler [chicken] litigation,” which was echoed in a 2020 internal Agri Stats presentation, suggesting that turkey processors had hoped to avoid legal trouble (which chicken processors already were in) by exiting Agri Stats before facing litigation.¹⁷ Meanwhile, Agri Stats continued publishing its chicken reports without any changes until it agreed to behavioral remedies as part of a settlement with the Department of Justice in May 2026.¹⁸ The clean variation in information sharing in turkey provides an ideal setting for studying the effects of information sharing, while chicken provides a useful placebo given the overlap in the set of suppliers, inputs, and regulatory structures in the two industries.¹⁹

3.3 Data

My primary dataset is the NielsenIQ retail scanner data, provided by the [Kilts Center](#) for Marketing at Chicago Booth. This dataset contains weekly prices and quantities from a nationally representative set of retailers. I use data from two product modules (the most granular product category in the data): fresh meat and sliced deli meat. The predominant turkey product in the fresh meat module I use is ground turkey, which, unlike whole turkey, is not a seasonal product—the module with whole turkey only has data beginning in 2020, foreclosing its inclusion in the analysis. The sample period is June 2017 to May 2022, which spans two years before to three years after the end of the turkey information exchange.²⁰

I augment the scanner data with several auxiliary data sources. First, I incorporate the NielsenIQ consumer panel data, which contain purchase-level information from a panel of households, into my demand estimation procedure. Second, I use the [USDA Meat, Poul-](#)

¹⁷Plaintiffs’ Opposition to Defendant’s Motion to Dismiss.

¹⁸The settlement prevents Agri Stats from sharing any nonpublic price information, requires that it aggregate any exchanged quantity or cost data, prevents sharing any data more recent than 45 days old, and requires it to sell the reports to any potential purchaser, including poultry buyers or workers.

¹⁹Agri Stats also shut down most aspects of its pork reports around this time. I focus on turkey for my analysis as there is cleaner variation in the information available to turkey processors than to pork processors, Agri Stats had a wider reach in turkey, and chicken provides a more natural comparison for turkey than for pork.

²⁰I start my analysis in June 2017 to avoid contamination from a prior antitrust claim of poultry market manipulation in late 2016–early 2017 ([Welshans, 2017](#)), and end my analysis in May 2022 due to the outbreak of the Highly Pathogenic Avian Influenza (i.e., avian flu), which caused the death of millions of birds and substantially impacted poultry prices in mid-2022 ([Grossen, 2023](#)).

	Fresh turkey		Deli turkey	
	Full sample	Est. sample	Full sample	Est. sample
Number of regions	202	110	205	118
Total monthly revenue	\$42.0m	\$40.4m	\$22.6m	\$21.5m
Number of products	19.8	8.7	43.3	15.5
	[12, 26]	[7, 10]	[30, 57]	[13, 18]
Number of firms	4.4	5.0	9.5	10.7
	[3, 6]	[4, 6]	[7, 12]	[9, 12]
Number of Agri Stats clients	2.7	3.2	4.7	5.0
	[2, 4]	[2, 4]	[4, 6]	[4, 6]
Agri Stats client share	0.67	0.61	0.77	0.71
	[0.38, 0.96]	[0.34, 0.89]	[0.63, 0.92]	[0.59, 0.83]
HHI	6,113	5,410	3,557	3,088
	[4,582, 7,440]	[4,312, 6,559]	[2,709, 4,023]	[2,522, 3,457]

Table 1: Summary statistics

Note: Est. sample refers to the sample used for the structural model. Number of products, firms, Agri Stats clients, Agri Stats client share, and HHI each feature the average followed by the interquartile range, both across region-months. Products are defined as a UPC for the full sample and, for the estimation sample, as a unique value of brand- $\mathbb{1}$ (breast meat)- $\mathbb{1}$ (whole smoked turkey) for fresh turkey and brand- $\mathbb{1}$ (low fat) for deli turkey. Shares are computed using sales volume in pounds.

try, and Egg Product Inspection Directory, which contains ownership, addresses, licenses, and “doing business as” information of all USDA-regulated poultry processing plants. The current directory is publicly available online; I use monthly snapshots from throughout my sample period obtained from a Freedom of Information Act request. I geocode all processing plant locations and compute driving distances between processing plants and stores using the Google Maps API. Third, I obtain price indices of inputs, including poultry feed and diesel, as well as the CPI-U (which I use to deflate all prices to May 2019 levels), from the [Federal Reserve Bank of St. Louis](#) FRED database. Fourth, I hand-collect ownership information of brands in the scanner data from the internet. Fifth, I identify which firms subscribed to the Agri Stats reports using information from legal filings.

Table 1 displays summary statistics of the sample I use for the reduced-form analysis (Section 4) and that used for structural model estimation and counterfactuals (Sections 6–8).²¹ Both fresh and deli turkey are concentrated, but fresh turkey more so: the average

²¹To move from the full sample to the estimation sample, I remove tiny-share products, aggregate products from the UPC level to the brand-characteristics level, and keep regions that have at least one panelist purchase in every month. Throughout the paper, I use Nielsen Designated Market Areas as the main geographical delineation, and refer to them as regions. A Nielsen DMA is a collection of counties that receive the same television stations and advertising. They are commonly used to define geographic markets in studies of consumer packaged goods (Nevo, 2001; Miller and Weinberg, 2017; Bhattacharya et al., 2026). Finally, I aggregate and report all prices and quantities on a per-pound basis throughout. See Appendix B.1 for additional details.

market (region-month) has an HHI of roughly 3,500 and around 10 competing firms for deli turkey, as compared to an HHI of 6,000 and four to five competing firms for fresh turkey. Agri Stats clients dominate both markets, representing 60–70% of the fresh meat market and more than 70% of the deli market by volume.²² Notably, however, there is substantial variation across markets in the Agri Stats share, particularly in fresh turkey; markets in the top quartile are nearly completely dominated by Agri Stats subscribers, with a share of more than 96%, while Agri Stats clients collectively have a share of less than 40% in the bottom quartile. Finally, while the filters used to construct the estimation sample result in nearly half of the regions being excluded, the remaining regions comprise more than 95% of the total turkey sold, and the concentration and penetration of Agri Stats subscribers are broadly similar in both samples.

4 Motivating evidence

This section analyzes how Agri Stats’ exit affected turkey prices and quantities and evaluates whether patterns in the data are consistent with conduct and information effects. If firms used the exchange to collude, then the conduct effect should lower average prices of participants after the exchange ends, and may do so more in markets whose structure is more susceptible to collusion. Meanwhile, the effect on average prices of firms learning about economic fundamentals through the exchange is directionally ambiguous, although firms’ prices should become less attuned to demand shocks and to each other’s prices after the exchange ends. The conduct effect should be immediate; when the exchange ends, firms internalize that in the future their rivals will not be able to detect deviation, which by backward induction makes them want to deviate as soon as possible. Conversely, the information effect may take more time to realize, as information from just before the end of the exchange may still be predictive of the economic environment at first but become stale over time.

4.1 Aggregate effects

Figure 4 shows the timeseries of average turkey prices (panel (a)) and total sales quantity (panel (b)) for Agri Stats clients in the orange solid line, and other producers in the blue dashed line. The vertical dashed line is May 2019, the month in which Agri Stats stopped

²²This is slightly lower than the 80% figure cited in the complaint and several other sources. One may not expect this figure to line up exactly, as I analyze only a subset of turkey sold at retail, and grocery store markets only represent a subset of the turkey sold in the US; a substantial fraction of turkey is purchased by the food service industry, including restaurants, catering services, and cafeterias at institutions like schools, hospitals, and prisons.

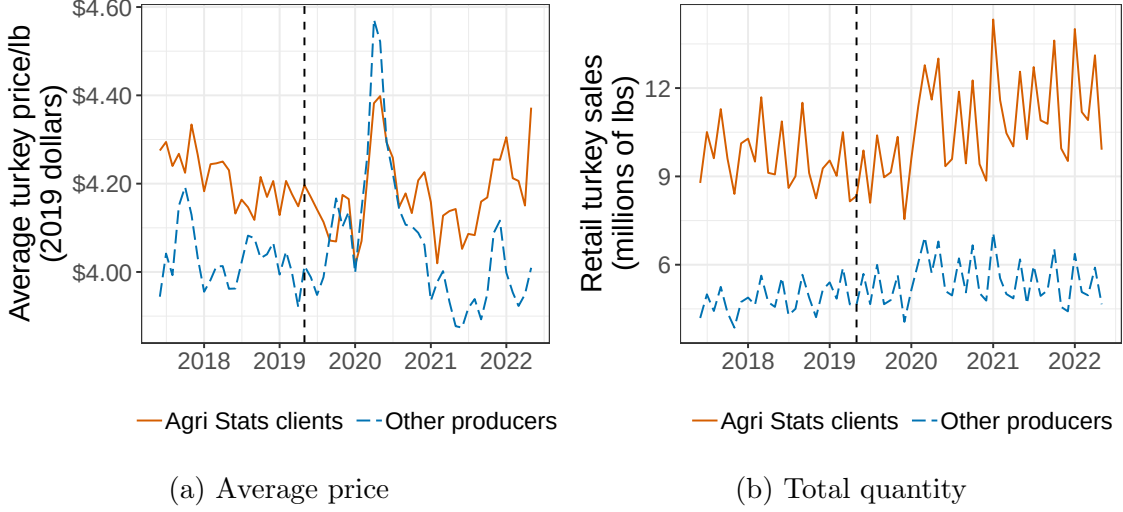


Figure 4: Average turkey price and sales quantity over time

Note: Average prices are weighted by total within-UPC-store quantity throughout the time period. The vertical dashed line represents May 2019.

producing its turkey reports. While the information exchange was ongoing, participants' turkey was roughly 20 cents (5%) more expensive than other producers' turkey. After the exchange ended, participants' prices fell to around the level of their competitors within several months. Prices of both groups evolved relatively similarly for the next 18 months, including during a price spike at the beginning of the COVID-19 pandemic. Finally, former Agri Stats clients' relative prices began rising again towards the end of the period. Meanwhile, the quantity sold by Agri Stats clients slightly rose after the end of the exchange, both in absolute terms and relative to others, suggesting that the initial relative price decline was not solely due to relative demand trends.

To quantify the change in relative prices and verify that it holds within-product and region, I estimate an event study. For UPC j in store s , region $r(s)$, and month t , the estimating equation is

$$\log(p_{jst}) = \alpha_{jr(s)} + \gamma_t + \beta_t \text{AS}_j + \varepsilon_{jst}, \quad (1)$$

where p_{jst} is UPC j 's price in store s and month t , $\alpha_{jr(s)}$ is a UPC-region fixed effect, γ_t is a month fixed effect, and AS_j is an indicator for whether UPC j is produced by an Agri Stats subscriber. β_t , the coefficient of interest, represents the average price difference between Agri Stats clients and other producers in month t ; I normalize its value in May 2019 to zero. While non-Agri Stats clients' prices are used to control for common demand and cost shocks, this exercise is fundamentally descriptive; one expects non-Agri Stats clients' prices to respond to price changes of Agri Stats clients, inhibiting any causal interpretation.²³

²³If prices are strategic complements, then this force in particular would cause this regression to underesti-

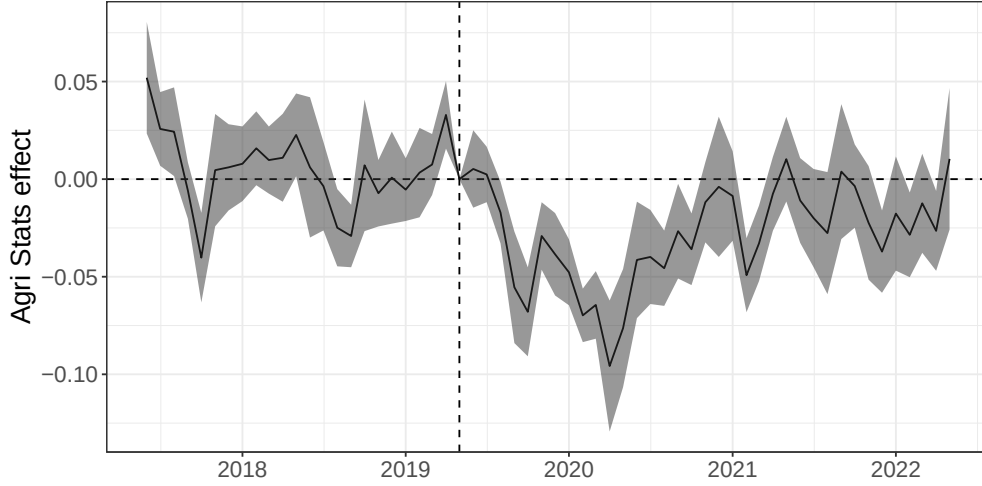


Figure 5: Event study results

Note: Regression is weighted by total within-UPC-region quantity throughout the time period. The shaded gray region is the 95% confidence interval, with standard errors clustered by UPC-region. The vertical dashed line represents May 2019.

Figure 5 shows the results, which demonstrate similar patterns to the raw price timeseries. The black line tracks the evolution of β_t over time, and the shaded gray region is the 95% confidence interval based on UPC-region clustered standard errors. Agri Stats clients' prices followed roughly parallel trends to those of their competitors for the two years leading up to May 2019. Then, within several months of May 2019, subscribers' relative prices fell by about 5%, remained depressed for about 18 months, and then slowly climbed back up to slightly below the pre-May 2019 level. This result independently replicates a legal finding that Agri Stats' turkey clients' prices decreased upon the end of the information exchange.²⁴

I quantify the effect further by estimating two-by-two difference-in-differences specifications, which replace the time-varying β_t in (1) with one β multiplied by a post-May 2019 indicator. Results are in Table 2. Columns (1) and (2) show that after the exchange ended, participants' relative prices fell by 5.1% and 3.4% in short-run (one-year) and long-run (three-year) time horizons respectively. In column (3), I add an interaction term between the treatment variable $AS_j \times Post_t$ and a linear time trend to separately estimate an immediate price drop followed by growth over time, and find that Agri Stats clients' prices immediately fell by 5.2% and then rose by 1.2% per year in the three years thereafter.

I next check whether similar trends occur in chicken, which acts as a useful placebo because there is substantial overlap between the markets—90% of turkey producers and 85% of Agri Stats turkey producers by volume also produce chicken—and because the chicken

mate the price effects of Agri Stats.

²⁴Plaintiffs' Opposition to Agri Stats' Motion to Exclude and Strike Expert Testimony. The magnitude of the decline is redacted from the public version of this filing.

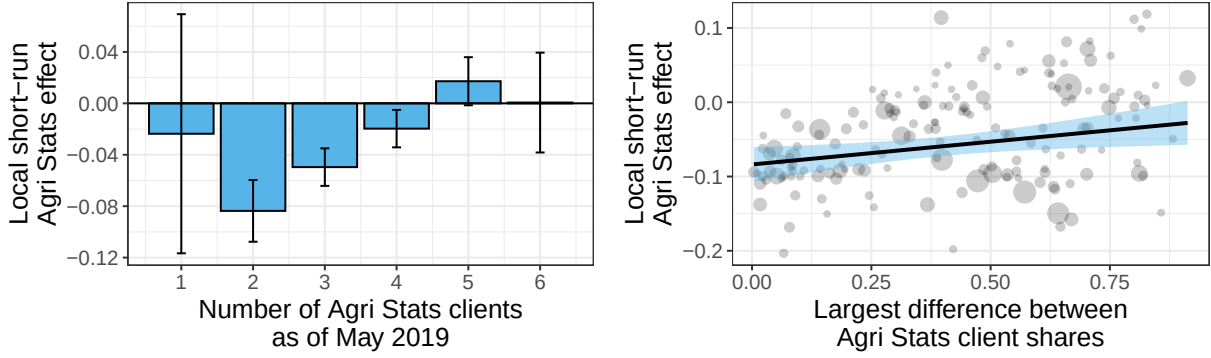
	(1)	(2)	(3)	(4)	(5)
$AS_j \times Post_t$	-0.051*** (0.006)	-0.034*** (0.006)	-0.052*** (0.007)	4.96×10^{-6} (0.021)	4.96×10^{-6} (0.021)
$AS_j \times Post_t \times t$			0.012*** (0.004)		
$AS_j \times Post_t \times Turkey_j$					-0.034 (0.021)
UPC-region FEs	✓	✓	✓	✓	✓
Month FEs	✓	✓	✓	✓	
Month-meat type FEs					✓
Observations	8,926,344	15,388,358	15,388,358	4,307,021	19,695,379
Meat	Turkey	Turkey	Turkey	Chicken	Both
Post-period	1 year	3 years	3 years	3 years	3 years

Table 2: Difference-in-differences results

Note: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. Standard errors clustered by UPC-region are in parentheses. Regressions are weighted by total within-region quantity throughout the time period. AS_j is an indicator for whether UPC j is produced by an Agri Stats client. $Post_t$ is a post-May 2019 indicator. t is a linear time trend variable, denominated in years. $Turkey_j$ is an indicator for whether UPC j is a turkey product.

information exchange continued throughout the sample period. Figure A.1 shows that the chicken prices of Agri Stats clients relative to others remained flat around the end of the turkey exchange. Column (4) of Table 2 runs the same specification as column (2) on chicken and finds a precisely estimated null, while column (5) runs the full triple-differences specification and finds an estimated relative effect between turkey and chicken that is the same as the turkey-only difference-in-differences in column (2), although the third level of differences adds imprecision to the estimate. I also estimate the triple-differences counterpart to the event study depicted in (1), where the time-varying coefficients of interest represent the turkey-chicken difference in the Agri Stats-other producer difference over time (Figure A.2), and find similar overall patterns to Figure 5: Agri Stats clients' relative prices dropped in turkey relative to chicken after May 2019 but slowly increased thereafter.

These results tell a consistent story: when the exchange ended, relative prices of participants immediately fell, but eventually increased back to roughly the with-sharing level. While the rebound could be due to a procompetitive information effect that takes a relatively long time to realize, as previously hypothesized, it could also be due to secular demand trends (as the contemporaneous relative increase in Agri Stats clients' quantity late in the period in Figure 4(b) might suggest), forgone efficiencies causing Agri Stats clients to become relatively less efficient without the reports, or unrelated supply-side factors. The goal of the



(a) Effects by number of Agri Stats clients (b) Effects by symmetry of Agri Stats clients

Figure 6: Heterogeneity in local Agri Stats effects by collusion plus factors

Note: Panel (b) only includes markets with two or three Agri Stats clients as of May 2019. Market-by-market regressions are weighted by total within-UPC-region quantity, and standard errors are clustered by UPC. In panel (a), estimates are averaged, weighted by total within-market quantity. The crossbar represents the 95% confidence interval. In panel (b), each bubble is a market, and its size is proportional to total quantity in that market. The black line and shaded region are the predictions and 95% confidence intervals from a weighted regression of local short-run Agri Stats effect on asymmetry. Both panels feature second-step standard errors computed using the Murphy-Topel correction.

rest of this section is to demonstrate that other patterns in the data are consistent with the presence of both conduct and information effects.

4.2 Suggestive evidence of conduct effects

Tacit collusion is easier to sustain when there are fewer competitors, and when those competitors are more symmetric (Ivaldi et al., 2003; Garrod and Olczak, 2018). As such, if the information exchange had a conduct effect, one might expect it to be larger in markets where these “collusion plus factors” are more prevalent. To investigate, I first compute market-level short-run Agri Stats effects by regressing $\log(p_{jst})$ on a UPC fixed effect, a month fixed effect, and a post-May 2019 indicator interacted with an Agri Stats indicator separately by market (unique combination of region and module), using only data from June 2017 to May 2020. Second, I aggregate the treatment coefficient across markets based on the number of Agri Stats clients in the market as of May 2019. Third, I define an asymmetry measure for each market as the largest within-market difference in two Agri Stats clients’ shares, and assess its correlation with the short-run Agri Stats effect. If there are conduct effects that are predicted by collusion plus factors, then one would expect that markets with fewer (but more than one) Agri Stats clients, and markets whose Agri Stats clients are more symmetric, had more negative short-run Agri Stats effects.

Figure 6 shows the results, which indicate that both hypotheses hold. Panel (a) finds that the short-run Agri Stats effect was -8.4% and -5.0% on average in markets with two and

three subscribers respectively, but -1.1% in all other markets. Panel (b) scatters market-level symmetry (which increases as points move to the left) against the short-run Agri Stats effect and finds a clear correlation: moving from the 25th to 75th percentile of symmetry is associated with a 2.6pp more negative short-run Agri Stats effect ($SE = 1.1pp$). Both results show that collusion plus factors correlate with the aggregate effects of information sharing in a manner consistent with conduct effects.

4.3 Suggestive evidence of information effects

I next evaluate two predictions of the model of information effects presented in Section 2. First, Agri Stats clients' beliefs should become more accurate when sharing information, causing their prices to respond more to demand; in particular, if firms learn about a common-value demand shock, then their prices should correlate more with that shock. Second, Agri Stats clients' beliefs should become more correlated when sharing information, causing comovement in their prices: prices are strategic complements, so learning a firm's price incentivizes rivals to move their prices in the same direction.

I begin by estimating constant-elasticity total demand curves for each module:

$$\log(Q_{rmt}) = \gamma_{rm} + \delta_{\text{month}(t)} + \alpha_m \log(P_{rmt}) + \xi_{rmt}, \quad (2)$$

where Q_{rmt} is total quantity in region-module-month rmt , γ_{rm} and $\delta_{\text{month}(t)}$ are region-module and month-of-year fixed effects, and P_{rmt} is a fixed-weight price index instrumented with poultry feed prices, an input cost.²⁵ ξ_{rmt} represents a demand residual common to all producers in the market, some portion of which firms may learn by sharing information. I recover elasticities of -2.12 in fresh turkey and -7.03 in deli turkey, with first-stage F -statistics of 276 and 1,109 respectively.

With the estimated residuals $\hat{\xi}_{rmt}$ in hand, I first test whether prices in markets with a greater Agri Stats presence tracked common demand shocks more while the information exchange was ongoing. Letting AS_{rm} be the share of production by Agri Stats clients in region-module rm as of May 2019, I estimate

$$\begin{aligned} \log(P_{rmt}) = & \alpha_{rm} + \gamma_t + \beta \hat{\xi}_{rmt} + \zeta AS_{rm} \text{Post}_t + \\ & \theta \hat{\xi}_{rmt} \text{Post}_t + \eta \hat{\xi}_{rmt} AS_{rm} + \mu \hat{\xi}_{rmt} AS_{rm} \text{Post}_t + \varepsilon_{rmt}. \end{aligned} \quad (3)$$

I standardize $\hat{\xi}_{rmt}$ by its within-module standard deviation so that β can be interpreted as

²⁵Prices at the UPC-store-month level are aggregated to the region-module-month-level index by taking a weighted average, using total UPC-store-level quantity sold over the entire time period as weights.

	(1)	(2)
$\hat{\xi}_{rmt}$ (demand residual)	0.042*** (0.014)	0.039*** (0.015)
$AS_{rm} \times Post_t$	-0.024* (0.014)	-0.022* (0.013)
$\hat{\xi}_{rmt} \times Post_t$	0.023* (0.012)	0.021* (0.012)
$\hat{\xi}_{rmt} \times AS_{rm}$	0.031*** (0.009)	0.040*** (0.011)
$\hat{\xi}_{rmt} \times AS_{rm} \times Post_t$	-0.044*** (0.015)	-0.041*** (0.015)
Region-module FEs	✓	✓
Month FEs	✓	✓
Observations	14,486	24,152
Post-period	1 year	3 years

Table 3: Responsiveness of Agri Stats clients' prices to demand

Note: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. Observations are weighted by total region-module level quantity over the sample. Standard errors clustered by region-module, corrected for first-step uncertainty in $\hat{\xi}_{rmt}$ using the delta method, in parentheses.

the increase in log prices associated with a 1 SD increase in the demand residual. If the exchange had information effects, one would expect $\eta > 0$ (before May 2019, prices in Agri Stats-heavy markets correlated more with the demand residual) and $\mu < 0$ (Agri Stats-heavy markets became less responsive to the demand residual after May 2019).

The results in Table 3 indicate that these hypotheses hold. The first three rows show that prices are correlated with the demand residual, prices decreased by more after May 2019 in Agri Stats-heavy markets (although imprecisely so at this level), and non-Agri Stats clients' prices became slightly more correlated with demand shocks after the end of the exchange. $\hat{\eta}$ (fourth row) is positive and significant: using the full post-period time horizon, a 10pp increase in the Agri Stats share is associated with a 0.004 log-point increase in the demand residual-price association before May 2019, which is 10.3% of the baseline association of 0.039. Finally, $\hat{\mu}$ (fifth row) is roughly $-\hat{\eta}$, suggesting that Agri Stats clients' and rivals' prices were not differentially correlated with demand after May 2019. The pattern is similar in one-year and three-year time horizons, suggesting that this effect occurred quickly after the information exchange ended.

Next, I test whether Agri Stats clients' prices became less correlated with each other after the end of the information exchange. To do so, I estimate the following regression separately before and after May 2019, at the firm-store-month level:

$$\log(p_{fst}) = \alpha_{fs} + \gamma_t + \sum_{G \in \{AS, ASC\}} \beta_G \mathbb{1}\{f \in G\} \log(\bar{p}_{AS \setminus f, st}) + \varepsilon_{fst} \quad (4)$$

	(1)	(2)	(3)	(4)
β_{AS} (Agri Stats clients)	0.667*** (0.097)	0.613*** (0.100)	0.328*** (0.062)	0.329*** (0.063)
β_{ASC} (other producers)	0.548*** (0.053)	0.481*** (0.054)	0.531*** (0.045)	0.531*** (0.045)
$\hat{\xi}_{rmt}$ (demand residual)		0.081*** (0.015)		-0.002 (0.006)
Firm-store FEs	✓	✓	✓	✓
Month FEs	✓	✓	✓	✓
Period	Pre	Pre	Post	Post

Table 4: Correlation of prices with same-store average Agri Stats client price

Note: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. Only includes store-module-months with at least two Agri Stats firms. Standard errors clustered by region-module, corrected for first-step uncertainty in $\hat{\xi}_{rmt}$ using the delta method, in parentheses. Average leave-out prices are weighted by total quantity throughout the time period.

where α_{fs} and γ_t are firm-store and month fixed effects, AS and ASC are the sets of Agri Stats subscribers and other producers, and $\bar{p}_{AS \setminus f, st}$ is the average price of all Agri Stats products in the same store and module, excluding firm f . The coefficients of interest, β_{AS} and β_{ASC} , represent the change in log prices for Agri Stats and non-Agri Stats firms associated with a 1 log-point change in the average price of other Agri Stats firms at the same store. I also reestimate this specification while controlling for the common-value region-month demand residual $\hat{\xi}_{rmt}$ from (2), which isolates correlation due to information about this demand residual from observation of rivals' prices themselves.

The results, shown in Table 4, suggest that Agri Stats clients' prices became significantly less correlated after the exchange ended. Before May 2019, the elasticity of a firm's price with respect to the average Agri Stats client price in the same store was 12pp higher for Agri Stats clients than for other producers, but roughly 20pp *lower* after May 2019. Some of the pre-period effect is due to correlated information about the demand state—conditioning on $\hat{\xi}_{rmt}$ reduces the pre-period Agri Stats subscriber elasticity by roughly 5pp—but the gap between pre-period subscriber and non-subscriber elasticities grows slightly wider, while the post-May 2019 correlations are unaffected.

Overall, firms seem to have used the Agri Stats reports to learn about their economic environment in a manner consistent with theoretical models of information effects. When combined with the evidence of conduct effects, there appear to be multiple forces affecting overall outcomes, which reduced-form methods are not equipped to separate. Further, given that information effects seemed to occur soon after May 2019, I do not know why Agri Stats

clients' prices rebounded slowly after May 2019, and whether this pattern can be explained by relative demand or supply trends. Finally, as discussed in Section 2, average prices are not sufficient statistics for the welfare effects of information sharing. These factors motivate the development of a structural model to estimate how firms' information sets changed with the end of the exchange, test and quantify changes in conduct, assess welfare effects, and simulate how outcomes might change under counterfactual exchanges.

5 Model

This section presents a structural model of oligopolistic competition that captures the information effects of information sharing. In the model, firms have incomplete information about demand shocks and rivals' cost shocks, and potentially imperfect information about rivals' prices and quantities. After each period, each firm observes a signal: a general function of the vector of prices and quantities. When firms share information, their signal contains more information. Firms use their signals to form beliefs about demand and their rivals' prices in each period, updating their beliefs from period to period based on what they observe. In a competitive Bertrand-Markov equilibrium, firms set prices that maximize their expected profits in each period given beliefs that are consistent with the signals they receive and the long-run autocorrelation of rivals' prices.

Throughout, I use j to index products, f firms, r regions, and t months. I write a variable without certain subscripts to mean a vector stacking that variable over all omitted subscripts, while I add a minus sign to denote all entries except the one(s) corresponding to that subscript. For example, if x_{jrt} is a product-region-month-specific variable, then x_{rt} is the vector stacking values of those variables across products and within that region-month, while $x_{(-f)rt}$ is that vector excluding entries of products owned by firm f .

5.1 Supply

I begin with the supply side, which is novel; the demand side, which is described next, is a standard mixed logit (Berry et al., 1995). Firms $1, \dots, F$ compete in regions $1, \dots, R$ for an infinite horizon of time periods $1, 2, \dots$. I refer to a unique combination of a region and time period as a *market*; for the remainder of this subsection, I condition on a particular region and omit r subscripts for brevity. Each market is characterized by an exogenous tuple $(\mathcal{J}_t, \theta_t, x_t, w_t, \xi_t, \omega_t)$, where $\mathcal{J}_t \in 2^{\mathcal{J}}$ is the set of available products (and $\mathcal{J}$ is the set of all products ever sold), $\theta_t \in \Theta$ is a vector of consumer characteristics that includes the market size I_t , $x_t \in \mathcal{X}$ is a vector of observable product characteristics for each product, $w_t \in \mathcal{W}$ is a

vector of observable cost shifters for each product, $\xi_t \in \Xi$ is a scalar demand shock for each product, and $\omega_t \in \prod_{j \in \mathcal{J}} \Omega_j = \Omega$ is a scalar mean-zero cost shock for each product. Product j in market t has market share $s_j(p_t; \mathcal{J}_t, \theta_t, x_t, \xi_t)$, where $p_{jt} \in \mathcal{P}_j$ is product j 's price in market t , and constant marginal cost $c_{jt} = h(w_{jt}) + \omega_{jt}$. For brevity, I write $s_{jt}(p_t; \xi_t)$ as the demand for product j in market t , fixing $(\mathcal{J}_t, \theta_t, x_t)$. Firm $f \in \mathcal{F}$ owns products $\mathcal{J}_f$, where $\bigcup_{f \in \mathcal{F}} \mathcal{J}_f = \mathcal{J}$; firm f 's product assortment in market t is $\mathcal{J}_{ft} = \mathcal{J}_f \cap \mathcal{J}_t$.²⁶ I assume that $\{(\mathcal{J}_t, \theta_t, x_t, w_t, \xi_t)\}_{t \geq 1}$ is generated by a stationary ergodic process.

Given a vector of prices p_t , demand shocks ξ_t , and cost shocks for firm f 's products $\omega_{ft} = \{\omega_{jt}\}_{j \in \mathcal{J}_{ft}}$, firm f 's profit in market t is

$$\pi_{ft}(p_t; \xi_t, \omega_{ft}) = I_t \sum_{j \in \mathcal{J}_{ft}} s_{jt}(p_t; \xi_t)(p_{jt} - h(w_{jt}) - \omega_{jt}). \quad (5)$$

Information and timing. The following steps, which I discuss in turn below, take place in each period.

1. Primitives $(\mathcal{J}_t, \theta_t, x_t, w_t, \xi_t, \omega_t)$ are drawn. All primitives except (ξ_t, ω_t) are commonly known to all firms; ξ_{jt} is unobserved by all firms, while ω_{jt} is observed by firm f if $j \in \mathcal{J}_{ft}$.
2. Firm f observes $\omega_{f't}$ for a subset of firms $f' \in \mathcal{F}_f^{\text{obs}} \subseteq \mathcal{F} \setminus \{f\}$.
3. All firms simultaneously set prices p_t ; shares s_t are realized.
4. Firms observe a signal of prices and shares, where firm f 's signal is $y_{ft} = y_f(p_t, s_t)$.

Step 1 states the fundamental uncertainty firms face: the vector of demand shocks ξ_t and rival cost shocks $\omega_{(-f)t}$. These, particularly ξ , traditionally correspond to characteristics that are unobserved by the economist but known to market participants, such as style or innate product quality (Berry, 1994). It may therefore seem odd to assume that firms do not observe them. However, as I explain below, firms learn about these objects through the information they observe about the market, and thus still have more information about them than the economist. Therefore, these variables' typical interpretation is still valid. Furthermore, the existence of Agri Stats implies there is economically significant information about the market that firms do not naturally observe.

Step 2 allows firms to share information about their costs with a subset of their rivals before prices are set, enabling them to conjecture their rivals' prices more accurately. In the Agri Stats case, a key component of the costs shared by Agri Stats is inventories, which affect the shadow cost of selling additional units, and the case record indicates that firms used the

²⁶The indices and dimensions of product-level vectors such as ξ_t can change due to variation in product assortment. To avoid notational issues, I index these vectors by the full universe of products $\mathcal{J}$, with dummy values for $j \notin \mathcal{J}_t$ that are ignored by all operators.

exchanged cost and inventory information to better project their rivals' prices. In the event that firms that share information have common knowledge of each other's costs and priors over unknown objects—as will be the case—sharing costs allows firms to conjecture each other's prices perfectly.

Step 3 is a standard price-setting assumption, but there is nothing special about prices; the model extends to any setting with one flexible decision variable and where firms share information about both their decisions and their outcomes. Step 4 encodes sharing general information about past prices and shares. For example, a firm only observing its own prices and share has $y_f(p_t, s_t) = (p_{ft}, s_{ft})$, while observing the full vector of prices and shares corresponds to the identity function $y_f(p_t, s_t) = (p_t, s_t)$. Aggregates, such as average prices or total quantities, are also supported.²⁷ I consider deterministic signal functions here, but the model could accommodate stochastic signals as well.

Beliefs. The payoff-relevant uncertainty firm f faces in market t is summarized by its beliefs, a probability distribution ρ_{ft} over $(\xi_t, p_{(-f)t})$. Notice that firms do not directly form beliefs over $\omega_{(-f)t}$, as they do not enter into firm f 's profits conditional on $p_{(-f)t}$. In each period, firms have *prior beliefs* (i.e., their beliefs as they are setting prices) and *posterior beliefs* (i.e., their beliefs after they set prices and observe the signal of prices and shares). Sharing information about past prices and shares affects posterior beliefs by giving firms a more precise signal of what occurred in the market, which affects their prior beliefs for the following period to the extent that ξ_t and $p_{(-f)t}$ are autocorrelated.

To be precise about this mechanism, suppose firm f has prior belief ρ_{ft} in market t , sets prices p_{ft} , and receives signal $y_{ft} = y_f(p_t, s_t)$. If the firm shares information about costs with a subset of its rivals and knows their beliefs, then this prior will be degenerate over these rivals' prices. It uses Bayes' rule to form its posterior as the regular conditional distribution $\rho_{ft}(\cdot | y_{ft})$, with a more precise signal tending to result in a more precise posterior. Firms use their posteriors to form their priors for the next period. For a more precise posterior to imply a more precise prior, demand and prices must be autocorrelated; otherwise, period t outcomes would be uninformative of period $t + 1$. I therefore place slightly more structure on ξ_t than is typical, by assuming that ξ_t follows a stationary first-order Markov process with a law of motion $F^\xi(\cdot | \xi_t)$. This nests as special cases common value and independent private value demand shocks, the prevailing assumptions in the information effects literature (Raith, 1996).

While ξ is a structural primitive, p is endogenous; its evolution over time is ex ante unclear and can be influenced by the structure of how firms' beliefs themselves evolve. For example,

²⁷Because quantities are just shares multiplied by market size and market size is commonly known, I use “quantities” and “shares” interchangeably here and below.

if firm 1 conjectures that firm 2's price depends on what happens more than one period ago, firm 1's price may then optimally depend on its beliefs about what happened more than one period ago, which then self-confirms the conjectured history-dependence. Without any restrictions, firms could in theory need to condition on the entire signal history as a state variable, creating a curse of dimensionality on firms' state spaces.

I therefore assume that each firm f conjectures that $p_{(-f)t}$ follows a stationary conditional first-order Markov process, with kernel $F_f^p(\cdot \mid p_t, \xi_t, x_{t+1}, w_{t+1})$. This assumption essentially says that if firms knew all the prices and demand state last period, along with the demand and cost observables this period, then there would be no informational benefit of knowing prices more than one period ago in predicting the prices rivals set today.²⁸ Despite rival firms not knowing ξ_t and p_t , the transition kernel depends on them because they affect s_t , which enters $y_{(-f)t}$, which then affects $\rho_{(-f)t}(\cdot \mid y_{(-f)t})$, and hence $\rho_{(-f)(t+1)}(\cdot)$ and $p_{(-f)(t+1)}$. The distribution of period- $t+1$ prices also depends on x_{t+1} and w_{t+1} because they are commonly known and shift demand and costs respectively.

Given these assumptions, firm f views the vector $(\xi_t, p_{(-f)t})$ as following a first-order Markov process, with law of motion

$$F_f(\cdot \mid \xi_t, p_t, x_{t+1}, w_{t+1}) = F^\xi(\cdot \mid \xi_t) F_f^p(\cdot \mid \xi_t, p_t, x_{t+1}, w_{t+1}). \quad (6)$$

Then, given a posterior belief $\rho_{ft}(\cdot \mid y_{ft})$, the prior belief for the following period is

$$\rho_{f(t+1)}(\cdot) = \int F_f(\cdot \mid \xi_t, p_t, x_{t+1}, w_{t+1}) d\rho_{ft}(\xi_t, p_{(-f)t} \mid y_{ft}). \quad (7)$$

The conjectured law of motion F_f^p and signal function y_f therefore pin down how firm f 's beliefs update from period to period.

Pricing rules and long-run consistency. To close the model, I must specify (a) firms' objectives and (b) an equilibrium condition pinning down firms' conjectured laws of motion. Let $\Lambda = 2^{\mathcal{J}} \times \Theta \times \mathcal{X} \times \mathcal{W}$ be the space of all commonly known exogenous variables, $\Omega_f^{\text{obs}} = \prod_{f' \in \{f\} \cup \mathcal{F}_f^{\text{obs}}} \prod_{j \in \mathcal{J}_{f'}} \Omega_j$ be the space of all cost shocks observed by firm f , $\mathcal{P}_f = \prod_{j \in \mathcal{J}_f} \mathcal{P}_j$ be the space of firm f 's prices, and $\mathcal{B}_f = \Delta \left(\Xi \times \prod_{f' \neq f} \mathcal{P}_{f'} \right)$ be the space of firm f 's beliefs. A *pricing rule* for firm f is a function $\sigma_f : \Lambda \times \Omega_f^{\text{obs}} \times \mathcal{B}_f \rightarrow \mathcal{P}_f$ that maps a set of observable exogenous variables, f -observed cost shocks, and beliefs to firm f 's prices.

²⁸This bears similarity to the central assumption in Markov perfect equilibrium (Ericson and Pakes, 1995; Maskin and Tirole, 2001), wherein agents conjecture that their rivals' actions combine to form a Markov process. There, the dependence over time in firms' actions is due to dynamics in firms' problems; here, firms face a static problem, but there is time-dependence coming from incomplete information and autocorrelation in the state over time.

The pricing rule summarizes the outcome of firm f 's period-by-period optimization problem. One special example is expected own-profit maximization. This rule satisfies

$$\sigma_f(\cdot) \in \arg \max_{\tilde{p}_f} \int \pi_{ft}(\tilde{p}_f, p_{(-f)t}; \xi_t, \omega_{ft}) d\rho_{ft}(\xi_t, p_{(-f)t}). \quad (8)$$

When firms share cost information with some rivals, their pricing rules can be measurable with respect to the cost shocks of those rivals, therefore allowing firms to incorporate rivals' expected profits into their objectives directly. While these static pricing rules do not accommodate directly modeling dynamic incentives, this measurability allows scope for collusive behavior. I return to this issue in Section 8.²⁹

To move towards an equilibrium concept, I must ensure that firms' conjectured price laws match reality. Fix a profile of pricing rules, signal functions, period-1 beliefs, and conjectured price kernels $\{\sigma_f, y_f, \rho_{f1}, F_f^p\}_{f \in \mathcal{F}}$. These, along with Bayes updates of beliefs in each period, induce an augmented process over $(\mathcal{J}_t, \theta_t, x_t, w_t, \xi_t, p_t)$. I call an initial belief vector $\{\rho_{f1}\}_{f \in \mathcal{F}}$ *regular* with respect to $\{\sigma_f, y_f, F_f^p\}_{f \in \mathcal{F}}$ if this process converges to a unique invariant distribution Π . Intuitively, this means that the impact of the initial distribution disappears over time.

Regularity implies that the notion of a long-run distribution of realized prices and primitives is well-defined. Let $\Pi^+(\cdot \mid p_t, \xi_t, x_{t+1}, w_{t+1})$ be the conditional distribution of p_{t+1} induced by the stationary joint distribution of consecutive observations under Π . Firm f 's conjectured price kernel is *long-run consistent* if, for every Borel set $A \subseteq \prod_{f' \neq f} \mathcal{P}_{f'}$ and Π -almost every conditioning state $(p_t, \xi_t, x_{t+1}, w_{t+1})$,

$$F_f^p(A \mid p_t, \xi_t, x_{t+1}, w_{t+1}) = \Pi^+(\mathcal{P}_f \times A \mid p_t, \xi_t, x_{t+1}, w_{t+1}). \quad (9)$$

In other words, the conjectured pricing law of motion must equal the actual long-run law of motion of prices that is induced by the pricing rules and those conjectured laws of motion.

Equilibrium definition. Given a set of initial beliefs, signal functions, and pricing rules $\{\rho_{f1}, y_f, \sigma_f\}_{f \in \mathcal{F}}$ and a data-generating process for primitives, a *Markov equilibrium in pricing rules* consists of a vector of prices p_t for each market, prior and posterior beliefs $\rho_{ft}(\cdot)$ and $\rho_{ft}(\cdot \mid y_{ft})$ for each firm in each market, and a conditional first-order Markov kernel $F_f^p(\cdot \mid p_t, \xi_t, x_{t+1}, w_{t+1})$ for each firm such that the initial beliefs are regular with respect to the kernel and the following hold:

²⁹Focusing on static pricing rules precludes signaling behavior in which firms set statically suboptimal prices in order to influence their rivals' beliefs in a manner beneficial to the firm's future profits. Capturing this dynamic incentive would add significant complexity, so I focus on static expected profit maximization as the main competitive benchmark.

1. *Consistency of within-group price conjectures.* Before prices are set, firm f 's prior ρ_{ft} places probability one on the event that $p_{f't} = \sigma_{f'}(\mathcal{J}_t, \theta_t, x_t, w_t, \{\omega_{f''t}\}_{f'' \in \mathcal{F}_f^{\text{obs}}, \rho_{f't}})$ for all $f' \in \mathcal{F}_f^{\text{obs}}$.
2. *σ -optimality of prices.* Given prior beliefs ρ_{ft} , $p_{ft} = \sigma_f(\cdot)$ for all ft .
3. *Updating of beliefs.* For all ft , given signals $y_{ft} = y_f(p_t, s_t(p_t; \xi_t))$, $\rho_{ft}(\cdot | y_{ft})$ is derived from $\rho_{ft}(\cdot)$ using Bayes' rule, and the marginal of $\rho_{f(t+1)}(\cdot)$ over ξ_{t+1} and the prices of firms outside $\mathcal{F}_f^{\text{obs}}$ is derived from $\rho_{ft}(\cdot | y_{ft})$ using (7), with F_f derived from F^ξ and F_f^p as in (6).
4. F_f^p satisfies long-run consistency (9) for all f .

A special case is *Bertrand-Markov equilibrium*: a Markov equilibrium in pricing rules with the own-profit maximization pricing rule, (8).

5.2 Demand

As previously mentioned, I parameterize demand using a mixed logit model with linear utility in the spirit of [Berry et al. \(1995\)](#) for my empirical application. This is the standard demand framework in consumer packaged goods markets ([Nevo, 2001](#); [Miller and Weinberg, 2017](#); [Backus et al., 2021](#); [Bhattacharya et al., 2026](#)). Reintroduce the region subscript r . For a mass of consumers $i \in [0, I_{rt}]$ with unit demand over options in $\mathcal{J}_{rt}$ plus an outside option, which is purchasing a different meat, the utility enjoyed by consumer i from purchasing product j is

$$u_{ijrt} = \alpha_i p_{jrt} + x'_{jrt} \beta_i + \gamma_{\text{brand}(j)} + \gamma_r + \gamma_{\text{month}(t)} + \xi_{jrt} + \varepsilon_{ijrt}, \quad (10)$$

where $\gamma_{\text{brand}(j)}$, γ_r , and $\gamma_{\text{month}(t)}$ are brand, region, and month-of-year fixed effects, and ε_{ijrt} is drawn independently and identically from a type-I extreme value distribution. θ_{rt} —the set of commonly observed consumer characteristics—includes these fixed effects and the distribution of (α_i, β_i) in market rt . The product characteristics are indicators for white meat and, in fresh turkey only, whole-smoked turkey. The outside option's utility is normalized to $u_{i0rt} = \varepsilon_{i0rt}$. (10) yields individual choice probabilities s_{ijrt} following the typical logit functional form, and integrating over (α_i, β_i) yields the demand functions s_{jrt} .

5.3 Discussion and application to Agri Stats

Before continuing, it is useful to describe how I will apply the model to test whether Agri Stats facilitated collusion, and how it relates to the conceptual framework presented in Section 2. I assume a stationary distribution of ξ throughout the sample period, but I

allow the periods before and after May 2019—the month the information exchange ends—to correspond to two separate Markov equilibria in pricing rules. As is standard in the conduct testing literature, I assume that firms interacted competitively (here, according to a Bertrand-Markov equilibrium) while not sharing information (Miller and Weinberg, 2017; Calder-Wang and Kim, 2025). The assumption of a Markov equilibrium in pricing rules before May 2019 allows me to recover and propagate firms’ beliefs from the realized price and share history without specifying the objective that generated these prices. Firms therefore enter June 2019 with beliefs over ξ that are carried over from the information sharing period, capturing the potential timing mismatch between conduct and information effects discussed in Section 4.

I also leverage the institutional details reviewed in Section 3 to specify signal functions with and without information sharing. I assume that Agri Stats shared the entire vector of past prices and quantities, as well as costs and inventories, with its clients: that is, for all $f \in AS$, $\mathcal{F}_f^{\text{obs}} = AS \setminus \{f\}$ and $y_f(p_{rt}, s_{rt}) = \{(p_{f'rt}, s_{f'rt}) : f' \in AS\}$. This implies that subscribers have common knowledge of one another’s marginal costs and, because they begin with common beliefs and observe a common signal, a common belief over the residual uncertainty—the demand shock ξ_{rt} and non-subscribers’ prices. The inputs to each subscriber’s pricing rule $\sigma(\cdot)$ are therefore commonly known among subscribers, so Agri Stats clients perfectly conjecture the prices of fellow subscribers while sharing information. Meanwhile, due to the relative lack of alternative data sources in poultry markets, I assume that all firms that are not sharing information do not observe costs of any other competitors, and only observe their own prices and quantities: $\mathcal{F}_f^{\text{obs}} = \emptyset$ and $y_f(p_{rt}, s_{rt}) = (p_{frt}, s_{frt})$. This applies both to non-subscribers while the exchange is ongoing and to all firms after the exchange ends.

This model is an instance of the framework presented in Section 2, with two key generalizations. That framework has three main primitives: the set of states and profit functions $\{\pi_f^\theta(p)\}_\theta$, the beliefs μ_f , and the signal informativeness η^θ . Here, the underlying state space is $\Xi \times \Omega$, the set of all possible vectors of demand and cost shocks, and the profit functions are derived from the mixed logit demand system and marginal cost functions. The first generalization made by the structural model is that the beliefs and signal informativeness, which are primitives in Section 2, are derived endogenously from the signals firms observe and the underlying data-generating process of the state. The second generalization is that rivals’ prices also act as a state variable: there is residual noise in rivals’ prices due to previously observed private information, which can be interpreted as the source of heterogeneous priors in Section 2. In both models, firms have static decision rules, which means that rivals’ cost shocks are strategically irrelevant conditional on rivals’ prices; hence, firms’ beliefs need

only be formed over these prices and the vector of demand shocks, which continue to be payoff-relevant even when conditioning on rivals' prices. This is akin to firms forming beliefs over rivals' best responses directly, as opposed to the underlying state generating these best responses. Finally, in this model the realized prices come from firms maximizing expected profits over all possible states, whereas in a typical complete-information Bertrand-Nash model firms set prices to maximize profits in the realized state. In the language of Section 2, prices are in the interior of U in a Bertrand-Markov equilibrium but on the boundary of U in the complete-information model. This distinction will be important for interpreting the conduct tests in Section 7.

By placing minimal restrictions on the state space and deriving the priors from the underlying data-generating process of shocks, the model allows the data to generate a relatively wide range of potential information effects without imposing their existence or direction. For example, if demand shocks are estimated to be either not autocorrelated or completely predictable, then learning past outcomes will not teach firms anything about upcoming unknowns; if demand shocks of one firm are associated with higher future demand shocks of another firm, then those firms learning each other's quantities and prices may be particularly valuable; and knowledge of different rivals' quantities may be differentially useful at predicting a firm's own demand shock depending on how closely these firms compete.³⁰ The model makes limited assumptions about these factors, instead allowing the data to determine the informativeness of firms' signals about market fundamentals. It also incorporates both best-response symmetry and best-response asymmetry, in the language of the price discrimination literature; in particular, common-valued demand shocks imply best-response symmetry while independent demand shocks imply best-response asymmetry.

The model does, on the other hand, impose what types of signals firms observe both when sharing information and when not. Although these assumptions come from details of the setting and industry, it may seem stark to assume that firms do not directly observe rivals' prices or shares when not sharing information, or that firms fully share unobserved marginal costs, giving rivals sufficient information to perfectly conjecture their prices. In addition, the initial conditions for firms' beliefs need to be calibrated during estimation.³¹ To alleviate

³⁰The mixed logit demand system may place some parametric restrictions on the way in which demand curvature and cross-price elasticities change across states, which can impact information effects. Mixed logit is known for having relatively flexible cross-price elasticities among parametric demand systems, and I verified by simulation that information effects can have any sign with even simple logit demand models. Regardless, incorporating nonparametric demand estimation procedures (e.g., [Compiani, 2022](#)) may be a fruitful avenue for future study.

³¹As discussed in Section 6, I will calibrate the initial beliefs at the stationary distribution implied by the data-generating process of shocks and the equilibrium law of motion of prices. The initial conditions are unlikely to matter much empirically, as the exchange is ongoing at the beginning of my sample, and hence Agri Stats subscribers learn about the state relatively quickly regardless of their initial beliefs. The

concerns that these assumptions drive the results of the forthcoming conduct tests, I perform several robustness checks, described in Appendix C. First, I assume that firms not sharing information observe either average prices and total quantity in each market from government indices, or the full vector of past prices and quantities from retail data. Second, I assume that the shared cost information is not fully informative of rivals’ prices, instead constituting a noisy but unbiased signal of rivals’ prices. Third, rather than assuming that firms shift between equilibrium price kernels immediately in May 2019, I allow for a buffer period in which firms continuously transition between the pre-May 2019 and post-May 2019 kernels over a period of up to a year. Fourth, I vary the location and precision of firms’ period-zero beliefs. The results are robust to these assumptions.

One additional assumption warrants a caveat. My data contain retail prices and quantities, but the exchange was among upstream processors. Manufacturers setting retail prices is a standard assumption in studies of consumer packaged goods (e.g., Nevo, 2001; Miller and Weinberg, 2017; Backus et al., 2021), and there is evidence this assumption fits scanner data well (Duarte et al., 2024). Moreover, bargaining power heterogeneity played a limited role in the Agri Stats litigation—Agri Stats did not circulate retailer-specific information, and all turkey purchasers were certified as one class for damages assessment—suggesting that retailer-processor bargaining is second-order for understanding the effects of Agri Stats.³² However, I accept as a limitation that my model does not capture this vertical structure.

6 Identification, estimation, and results

6.1 Identification

I begin with identification of demand. When firms have complete information, the typical identification problem is that firms set prices knowing the realizations of the unobserved (to the econometrician) demand shocks ξ , causing price endogeneity. While I assume firms do not directly observe realizations of ξ , this endogeneity problem still arises: firms set prices having beliefs over ξ that are correct in expectation, causing correlation between prices and ξ . Thus, identification still requires instruments z that shift prices while being exogenous to demand shocks (Berry and Haile, 2014).

To resolve this endogeneity, I use three types of instruments. First, I use input and transportation costs: feed prices and diesel prices multiplied by the distance to the nearest

robustness check in Appendix C.3, which perturbs the initial conditions, demonstrates that this is indeed the case.

³²In addition, Appendix C.5 provides some reduced-form evidence that vertical structure played a small role, if any, in explaining the Agri Stats effect.

poultry processing plant. I interact the latter with a store-brand indicator, reflecting that retailers may be differentially able to account for fluctuations in diesel prices by adding or reducing inventory at the store. Second, I exploit the quasi-experimental price variation from the end of the information exchange by using a pre-May 2019 indicator interacted with an Agri Stats indicator. This instrument is valid if the relative distribution of demand shocks is mean-independent of the end of the information exchange, while its relevance is demonstrated by the reduced-form results. Third, I use the differentiation instruments of [Gandhi and Houde \(2020\)](#), which shift prices by changing the degree of local competition in characteristics space analogously to traditional “BLP instruments.” For two characteristics x and x' (where x and x' could be the same characteristic), the instrument is

$$z_{jrt}^{GH,x,x'} = \sum_{k \in \mathcal{J}_{rt}} (x_{jrt} - x_{krt})(x'_{jrt} - x'_{krt}).$$

I also estimate an exogenous “predicted price” characteristic conditional on all characteristics and instruments $\hat{p}_{jrt} = \mathbb{E}[p_{jrt} \mid x_{jrt}, z_{jrt}]$ using a random forest, and construct differentiation instruments based on $\hat{p}_{jrt}$.

Once demand is identified, the full distribution of (p, ξ, x, w) is identified. Hence, the joint distribution of $(p_{r(t+1)}, p_{rt}, \xi_{r(t+1)}, \xi_{rt}, x_{r(t+1)}, w_{r(t+1)})$ is identified for each region r , meaning the conditional distributions of $p_{r(t+1)} \mid p_{rt}, \xi_{rt}, x_{r(t+1)}, w_{r(t+1)}$ and $\xi_{r(t+1)} \mid \xi_{rt}$ are also identified. The equilibrium long-run consistency condition therefore identifies the equilibrium pricing laws of motion for each firm both before and after the exchange ends, along with the law of motion of ξ_r .

Supply-side identification, given the Bertrand-Markov equilibrium assumption after May 2019, rests on identifying the components of firms’ first-order conditions:

$$c_{jrt} = h(w_{jrt}) + \omega_{jrt} = p_{jrt} + \left(\mathbb{E}_{f_{rt}} [D_{p_f} s_{f_{rt}}(p_{rt}; \xi_{rt})'] \right)^{-1} \mathbb{E}_{f_{rt}} [s_{f_{rt}}(p_{rt}; \xi_{rt})] \quad \forall j \in \mathcal{J}_{f_{rt}}, \quad (11)$$

where $\mathbb{E}_{f_{rt}}[\cdot]$ is the expectation operator with respect to firm f ’s beliefs over $(\xi_{rt}, p_{(-f)rt})$ in market rt , and $D_{p_f} s_{f_{rt}}(\cdot)$ is the Jacobian of firm f ’s demand with respect to its own products’ prices. In the typical complete-information model, the entire right-hand side of (11) is known after demand is identified, which identifies the vector of marginal costs, c_{jrt} . Here, however, I must know firms’ beliefs in order to evaluate the expectations on the right-hand side, presenting a fundamental identification problem: one observed price may be consistent with a firm having either high costs and pessimistic beliefs about its demand, or low costs and optimistic beliefs about its demand.

Nevertheless, I proceed to argue that given a set of initial conditions for firms’ beliefs,

subsequent beliefs are indeed identified by the structure of the model, and that therefore costs are identified from the first-order conditions as usual. I argue by induction: if beliefs in period t are identified, then beliefs in period $t + 1$ are also identified. The first step is to identify posterior beliefs given prior beliefs in period t . With demand identified, $s_{rt}(p_{rt}; \xi_{rt})$ is known; hence, given a prior for firm f in market rt , all functions of s_{rt} and its inputs correspond to a known probability distribution. This includes the (known) signal function $y_f(p_{rt}, s_{rt})$. Second, since realized prices and shares come from data, the signal realization $y_{f,rt}$ itself is known. Hence, firm f 's posterior distribution over $(p_{(-f)rt}, \xi_{rt})$ conditional on having received signal $y_{f,rt}$ is directly implied by Bayes' rule, identifying posteriors. Firms then use the equilibrium laws of motion over $(\xi_{rt}, p_{(-f)rt})$ which, as discussed above, are identified because demand is identified, to move from the period- t posterior to the period- $t + 1$ prior. As such, firms' prior beliefs in period $t + 1$ are identified from the updating rule, equation (7). In addition, notice that the assumption of a Markov equilibrium in pricing rules allows me to identify beliefs in the exact same way before May 2019; this process did not rely on imposing the Bertrand-Markov pricing rule (or any pricing rule in particular).

This leaves an initial conditions problem: period-zero beliefs must be specified in order to recover subsequent beliefs. To be internally consistent with a Markov equilibrium in pricing rules, they must be regular—that is, they must eventually converge in expectation to the stationary distribution implied by the previously identified laws of motion. I take the natural approach of calibrating firms' beliefs at the beginning of my sample (June 2017) to that stationary distribution itself, which is clearly regular. This calibration is justified by the fact that the information exchange had been ongoing for more than ten years by this point, with a relatively stable membership of firms and minimal entry or exit in the market. Iteratively recovering beliefs in period $t + 1$ from beliefs in period t then identifies beliefs throughout the entire time period. Finally, with beliefs identified, marginal costs, $h(\cdot)$, and ω are identified by firms' pricing rules; in a Bertrand-Markov equilibrium (as is the case after May 2019), the pricing rules are given by firms' first-order conditions for expected profit maximization, (11).

6.2 Estimation

Estimation proceeds in four steps, closely following the identification argument but with additional parametric structure to account for the relatively high dimension of firms' belief spaces. The estimation procedure is implemented separately for deli and fresh turkey. Additional details are available in Appendix B.2.

I begin with demand estimation, which uses the micro-BLP estimator of [Berry et al. \(2004\)](#).

I parameterize the random coefficients $(\alpha_i, \beta_i) = (\alpha, \beta) + \Sigma\nu_i + \Pi d_i$, where Σ is diagonal, $\nu_i \sim \mathcal{N}(0, I)$ is conformable with Σ and independent of d_i , and d_i includes indicators for three household income groups (less than \$50k, between \$50k and \$100k, over \$100k) and whether children live in the household. Letting $\theta = (\alpha, \beta, \Sigma, \Pi)$ be the vector of structural demand parameters to estimate, I form two types of moments: instrument-exogeneity moments of the form $\mathbb{E}[z'\xi(\theta)]$, where $\xi(\theta)$ are the values that make predicted and actual shares match, and micro-moments of the form $\mathbb{E}\left[\sum_j s_{ijrt}(\theta)x_{jrt} \mid d_i = d\right]$ minus their empirical counterparts from the consumer panel data for each characteristic x and demographic d .³³ Finally, given the relatively large number of Gandhi-Houde instruments available, I reduce their dimension to principal components explaining 95% of their variance, a strategy also used in [Backus et al. \(2021\)](#). I use `pyblp` for estimation ([Conlon and Gortmaker, 2020; 2025](#)).

I next estimate the law of motion of ξ and firms' equilibrium updating rules for p . I parameterize these as linear VAR(1) processes with product-region-specific intercepts, common slopes across regions, and Gaussian innovations; that is,

$$\begin{aligned}\xi_{rt} &= \bar{\xi}_r + B^\xi \xi_{r(t-1)} + e_{rt}^\xi, & e_{rt}^\xi &\stackrel{\text{iid}}{\sim} \mathcal{N}(\mathbf{0}, \Sigma^\xi), \\ p_{rt} &= \bar{p}_r + g(w_{rt}, z_{rt}^{GH}) + B^p(\xi_{r(t-1)}, p_{r(t-1)}) + e_{rt}^p, & e_{rt}^p &\stackrel{\text{iid}}{\sim} \mathcal{N}(\mathbf{0}, \Sigma^p).\end{aligned}\tag{12}$$

Off-diagonal elements of B^ξ , B^p , Σ^ξ , and Σ^p reflect potential timeseries dependence and correlation of firms' demand and prices with one another. I include the Gandhi-Houde instruments in $g(\cdot)$, which nonparametrically shifts prices, as a reduced-form way of including the entire vector of product characteristics shifting each firm's price, and I assume a linear timeseries in ξ_{t-1} and p_{t-1} because it will result in conjugate priors, as I explain below.

To estimate (12), I use the system GMM estimator of [Blundell and Bond \(1998\)](#). This estimator is well-suited for so-called “small- T , large- N ” settings, as is the case here because my sample has fewer time periods than regions, and is robust to unbalanced panels caused by the presence of regional products ([Roodman, 2009](#)). I use system GMM directly to estimate the law of motion of ξ using data throughout my sample period; to estimate the price timeseries, I first partial out $g(w_{rt}, z_{rt}^{GH})$ using a random forest before applying the system GMM following [Robinson \(1988\)](#), and I estimate separate timeseries before and after May 2019 to account for the change in equilibria when the information exchange ends.

Third, I estimate beliefs using the procedure outlined in Algorithm 1. I parameterize beliefs as jointly normal over (ξ, p) , starting in $t = 1$ at the stationary distribution implied by the estimate of (12). For each market rt and firm f , I define the signal functions and set of firms sharing costs with firm f based on whether the information exchange is ongoing

³³To identify demographic interactions with the constant, I use micro-moments of the form $\Pr(d_i = d \mid j > 0)$.

Algorithm 1 Belief estimation

For each region $r = 1, \dots, R$:

1. *Initialize priors* for all firms as $(\xi_{r1}, p_{r1}) \sim \mathcal{N}(\mu_{fr1}, \Sigma_{fr1})$, where μ_{fr1} and Σ_{fr1} are given by the stationary distribution implied by the timeseries estimates.
 2. For each time period $t = 1, \dots, T$ where t^* corresponds to May 2019, and each firm $f \in \mathcal{F}_{rt}$:
 - (a) *Adjust signals for information sharing.* If $t \leq t^*$ and f is an Agri Stats client, then define the signal function $y_f(p_{rt}, s_{rt}) = \{(p_{jrt}, s_{jrt}) : j \in \mathcal{J}_{f'rt}, f' \in AS\}$ and the set of firms sharing costs $\mathcal{F}_f^{\text{obs}} = (\mathcal{F}_{rt} \cap AS) \setminus \{f\}$; otherwise, define $y_f(p_{rt}, s_{rt}) = \{(p_{jrt}, s_{jrt}) : j \in \mathcal{J}_{frt}\}$ and $\mathcal{F}_f^{\text{obs}} = \emptyset$.
 - (b) *Adjust priors given shared information about future prices.* Change elements of μ_{frt} corresponding to prices of firms in $\mathcal{F}_f^{\text{obs}} \cup \{f\}$ to the realized prices, and set all corresponding rows and columns of Σ_{frt} to zero.
 - (c) *Approximate posteriors* $\mathcal{N}(\mu_{frt}^{\text{post}}, \Sigma_{frt}^{\text{post}})$ given the realized signal using Laplace's method.
 - (d) *Compute next period's prior* $\mathcal{N}(\mu_{fr(t+1)}, \Sigma_{fr(t+1)})$ using the estimated timeseries process for (ξ, p) (12).
-

and the firm's membership in Agri Stats as described in Section 5.3. Then, given firm f 's prior and realized signal, I approximate firm f 's posterior using Laplace's method, a classical Bayesian method for approximating complicated posteriors as Gaussians (Tierney and Kadane, 1986).³⁴ According to this method, the posterior is approximated as a joint normal with mean equal to the maximum a posteriori estimate of (ξ_{rt}, p_{rt}) conditional on the signal and covariance equal to the inverse Fisher information of the signal. Finally, given the linear-Gaussian laws of motion, firm f 's prior in market $r(t+1)$ is also normally distributed with a closed-form expression. Looping this process over all firms, periods, and regions results in belief estimates for all firms in all markets.

Finally, I back out costs from the first-order conditions (11), evaluating expectations of shares and demand Jacobians with respect to each firm's beliefs using quasi-Monte Carlo integration. I then estimate $h(w_{jrt})$ using a random forest, after residualizing product, region, and month fixed effects. w_{jrt} contains all cost shifters included as demand instruments—feed prices and diesel prices times distance to the nearest processing plant, with the latter interacted with store-brand indicators—although feed prices are absorbed by the month fixed effect. With the cost function estimated, I back out the residuals $\hat{w}_{jrt}$. I compute standard errors using a parametric bootstrap of demand parameters combined with a nonparametric

³⁴This is an approximation because the demand function is nonlinear. It would be exact otherwise.

Parameter	Fresh turkey				Deli turkey		
	Constant	Price	Breast	WS	Constant	Price	Low fat
β		-2.904 (0.231)	3.574 (0.313)	-5.669 (0.875)		-0.260 (0.018)	0.061 (0.303)
diag(Σ)	3.153 (0.503)	0.617 (0.049)	0.975 (0.351)	0.695 (1.355)	0.466 (0.275)	0.000 (0.092)	0.202 (2.327)
II: Medium income	-2.436 (0.388)	1.031 (0.104)	-3.669 (0.219)	4.778 (1.240)	2.731 (0.178)	-0.278 (0.029)	-1.125 (0.120)
II: High income	-4.827 (0.340)	1.513 (0.099)	-3.578 (0.206)	5.052 (1.232)	1.077 (0.155)	-0.046 (0.025)	-1.041 (0.142)
II: Children	1.637 (0.250)	-0.029 (0.050)	-1.326 (0.112)	0.828 (0.239)	2.280 (0.118)	-0.259 (0.017)	-0.439 (0.049)
Number of observations		57,616				109,820	
Median own-price elasticity		-5.11				-2.78	
Median aggregate elasticity		-0.63				-0.55	
First stage R^2		0.70				0.83	
First stage F -statistic		3,197.7				408.5	

Table 5: Demand estimates

Note: Heteroskedasticity-robust standard errors in parentheses. “WS” refers to whole/smoked.

bootstrap sampling over regions with replacement to account for asymptotic variance in the panel estimates, following [Kapetanios \(2008\)](#).

6.3 Results

Table 5 shows the estimated demand parameters. Mean price coefficients are negative and significant; the median own-price elasticity is -5.11 and -2.78 for fresh and deli turkey respectively, but there is substantial heterogeneity in products’ price sensitivity.³⁵ The median aggregate elasticity, or the percentage change in inside share from a 1% increase in all prices, is around -0.6 for both categories, similar to the elasticity of turkey in the demand model used by the USDA ([Maples et al., 2022](#)). Demographic information is important in explaining taste heterogeneity: fresh turkey is most appealing to low-income consumers (echoing a USDA finding that turkey is an inferior good), and high-income consumers are less price-sensitive than medium-income consumers in both markets. Finally, the instruments predict prices well, with first-stage R^2 s of 0.7 or more and F -statistics above conventional thresholds for instrument relevance.

The timeseries results are summarized in Table 6. Both demand and prices are auto-

³⁵The distribution of own-price elasticities is plotted in Figure A.3. I get somewhat more price-elastic results using feasible approximations to the [Chamberlain \(1987\)](#) optimal instruments; see Appendix C.4.

Slope matrix								
	Diagonal			Off-diag. mean		Shock correlation		
	Mean	SD	IQR	Same firm	Diff. firm	Mean	SD	IQR
<i>A. ξ (demand shock)</i>								
Fresh	0.56	0.24	[0.44, 0.75]	0.033	-0.004	0.12	0.14	[0.03, 0.20]
Deli	0.57	0.26	[0.46, 0.76]	0.011	0.000	0.03	0.15	[-0.06, 0.11]
<i>B. Pre-period prices</i>								
Fresh	0.37	0.24	[0.18, 0.57]	0.020	-0.002	0.01	0.16	[-0.06, 0.10]
Deli	0.45	0.26	[0.32, 0.62]	0.011	0.003	0.02	0.16	[-0.05, 0.07]
<i>C. Post-period prices</i>								
Fresh	0.42	0.22	[0.26, 0.58]	-0.003	-0.003	0.03	0.16	[-0.04, 0.11]
Deli	0.49	0.27	[0.25, 0.68]	0.028	0.003	0.02	0.13	[-0.04, 0.07]

Table 6: Summary of timeseries results

Note: “Diagonal” and “Off-diag. mean” summarize the diagonal and off-diagonal elements of B^ξ and the price coefficients of B^p , respectively. “Shock correlation” summarizes the off-diagonal correlation coefficients in Σ^ξ and Σ^p , where $\text{Cor}_{jk}^\xi = \Sigma_{jk}^\xi / \sqrt{\Sigma_{jj}^\xi \Sigma_{kk}^\xi}$, only including the lower triangle as $\text{Cor}_{jk}^\xi = \text{Cor}_{kj}^\xi$, and similarly for Σ^p . “Off-diag. mean” and “Shock correlation” columns only include pairs of products that compete against each other in at least six markets.

correlated, as one might expect, with the autocorrelations not changing much between pre- and post-periods or between fresh and deli turkey. The average autocorrelations are between 0.37 and 0.57, meaning that a shock to demand or prices mostly decays within several months. Therefore, the late-sample increase in the relative prices of former Agri Stats clients highlighted in Section 4 cannot be attributed to information effects that take over a year to realize. Off-diagonals of the slope matrix are on average zero but tend to be slightly positive among products owned by the same firm, which could reflect common within-firm shocks or advertising. Shocks tend to be nearly independent across firms on average, with correlation coefficients near zero, except for demand shocks in fresh turkey, which have a small common-value component. This implies that firms largely have differentially ordered best-response functions across states of nature, opening up the possibility for a wider range of information effects. Finally, recall that the price timeseries first residualizes a nonparametric function of observables; I find that this function predicts prices well, explaining 92.4% of the variation, but there is still some residual variation left to be predicted by the information firms observe.³⁶

To summarize the estimated beliefs, I assess how accurately firms perceive their own-price

³⁶Figures A.4 and A.5 show the full distribution of diagonal elements and off-diagonal elements of B^ξ and B^p respectively, with off-diagonals broken out by whether the two firms have the same owner. Figure A.6 is a scatter plot of the predictions of prices made by $g(\cdot)$ against actual prices.

Time period	Average RMSE _{jrt}	
	Agri Stats clients	Other producers
June 2017–May 2018	0.270	0.373
June 2018–May 2019	0.270	0.320
June 2019–May 2020	0.445	0.307
June 2020–May 2021	0.467	0.297
June 2021–May 2022	0.485	0.310

Table 7: Accuracy of beliefs about own-price elasticities

Note: RMSE_{jrt} is computed using a quasi-Monte Carlo simulation with 1,000 draws per firm-market, then averaged across firm-markets within a year.

elasticity, and how that changes over time for Agri Stats clients and other producers. For each firm in each market, I take D draws from that firm’s beliefs over (ξ, p_{-f}) and, for each draw, compute the implied own-price elasticity of that firm’s products given p_f . This gives an approximation of firm f ’s beliefs over its products’ own-price elasticities, $\{\tilde{\epsilon}_{jrt d}\}_{d=1}^D$. I then compute the scaled root mean squared error of the beliefs as

$$\text{RMSE}_{jrt} = \sqrt{\frac{D^{-1} \sum_{d=1}^D (\tilde{\epsilon}_{jrt d} - \hat{\epsilon}_{jrt})^2}{\text{Var}(\hat{\epsilon}_{jr})}}, \quad (13)$$

where $\hat{\epsilon}_{jrt}$ is the estimated own-price elasticity for product j in market rt at the realized (ξ, p_{-f}) , and $\text{Var}(\hat{\epsilon}_{jr})$ is its variance over time. I normalize the RMSE by the within-product-region standard deviation because it provides a natural benchmark: if firms simply guess that their own-price elasticity is equal to the time average in each market, then the average RMSE_{jrt} for each product-region is 1 by construction.³⁷

Table 7 shows the average RMSE_{jrt} by year for Agri Stats clients and other producers. The averages are far below 1: both groups do better than simply guessing the average. While the information exchange is ongoing, Agri Stats clients more accurately perceive their own-price elasticity than other producers. The end of the exchange causes Agri Stats clients’ accuracy to worsen by approximately 0.15–0.20 standard deviations of the within-product-region elasticity. Echoing Table 3, the bulk of this effect occurs within the first year after the information exchange ends. To better understand what firms actually learn from the exchange, I compute the change in RMSE of Agri Stats clients’ beliefs over their own ξ , their rivals’ ξ , and their rivals’ prices separately, without aggregating them into elasticities. I find that firms do not lose much predictive power over their own demand—the RMSE for

³⁷In the taxonomy of Section 2, the change in RMSE_{jrt} for Agri Stats clients when sharing information is analogous to η^θ , the extent to which the signal firms gain when sharing information tightens their beliefs.

Time period	Average marginal cost		
	Agri Stats clients	Other producers	% difference
June 2019–May 2020	\$2.92	\$3.15	7.8%
June 2020–May 2021	\$2.99	\$3.05	2.1%
June 2021–May 2022	\$3.05	\$3.02	-0.9%

Table 8: Average post-May 2019 marginal costs for Agri Stats clients and other producers

Note: Average costs are unweighted within a year-product-region and weighted by total within-year-product-region quantity across product-regions within a year.

own- ξ is 5.1% lower with the information exchange than without—and that the bulk of the effect happens in rival ξ and rival prices, with post-May 2019 RMSE increases of 14.9% and 23.6% respectively. This result suggests that Agri Stats primarily informed firms about rival demand shocks and prices, previewing a finding in Section 8.

Finally, I find that supply-side changes not captured by information effects are at least in part responsible for the relative growth in Agri Stats clients’ prices towards the end of the sample period. As shown in Table 8, Agri Stats clients had 7.8% lower recovered marginal costs on average than other producers in the year after the information exchange ended, but were no more efficient by the end of the sample period. One possible interpretation is that, as claimed by Agri Stats, information sharing resulted in investment efficiencies, the loss of which caused marginal cost convergence in the long run. Of course, as mentioned in Section 3.2, this convergence could be due at least in part to input and labor markets becoming less monopsonistic after the exchange ended, and hence should not necessarily be credited as a positive effect of information sharing. Another interpretation is that firms slowly began attempting to collude in the years following the end of the exchange, although collusion should in theory have been less effective without the monitoring provided by Agri Stats. Regardless of the ultimate explanation, potential long-run changes in marginal costs after the end of the exchange will be important to account for when testing for product market changes in conduct caused by Agri Stats’ exit.

7 The economic effects of Agri Stats

This section uses the estimated supply-and-demand system to analyze the effects of the Agri Stats exchange. I first assess whether the data are consistent with firms playing a Bertrand-Markov equilibrium with and without information sharing and estimate the magnitude of conduct effects using a conduct parameter model. I then simulate equilibrium outcomes under information sharing in the post-period and decompose aggregate effects into information

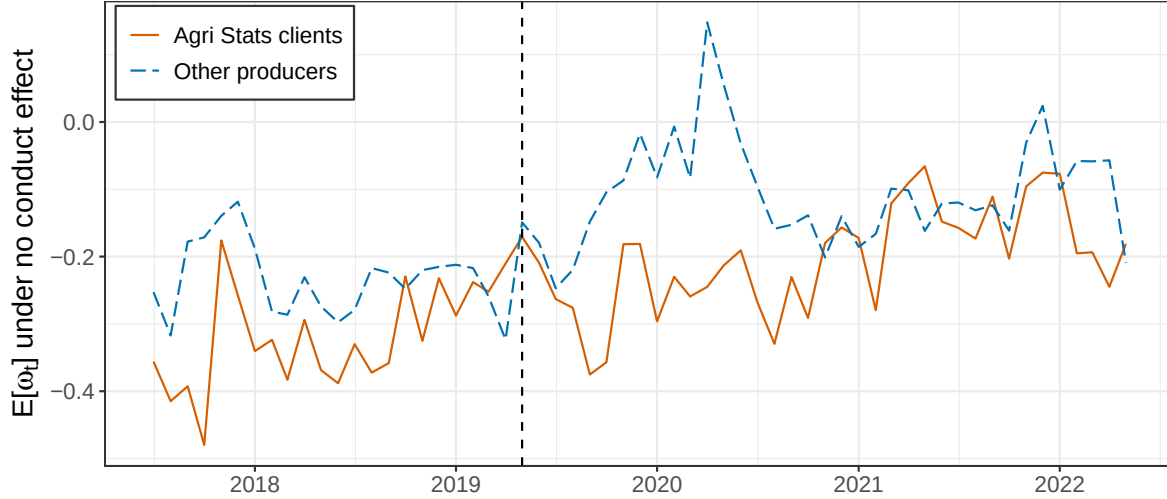


Figure 7: Average cost shocks over time under Bertrand-Markov conduct

Note: Cost shocks are weighted by total product-region-level quantity.

and conduct effects.

7.1 Conduct testing

Are prices consistent with firms playing a Bertrand-Markov equilibrium with and without information sharing? As in complete-information conduct models, each vector of prices maps to a unique vector of marginal cost shocks, ω . Thus, following the conduct testing literature, I can exploit a restriction on how ω should behave under the true conduct model to formulate a statistic to test whether and to what extent conduct changed when the information exchange ended (Berry and Haile, 2014).

To motivate such a restriction, consider the evolution of average cost shocks over time assuming that firms played a competitive Bertrand-Markov equilibrium throughout the sample period (Figure 7). The cost shocks for Agri Stats clients are on a relatively consistent trend up until May 2019, but they immediately drop after May 2019, while the drop in other producers' costs is far smaller. However, like overall marginal costs, the Agri Stats client cost shocks converge back to the level of other firms' shocks towards the end of the sample period, suggesting that these trends are not explained by changes to observables in the marginal cost function. Further, this immediate effect post-May 2019 is unlikely to have been caused by changes to unobservable marginal costs, as long as any forgone efficiencies would not have immediately affected costs once the information exchange ended.

I formalize this idea in the following moment condition, which states that there is no discontinuity in the average difference between Agri Stats clients' cost shocks and other

Conduct model	Discontinuity in ω			Test statistic	p -value
	AS clients	Others	Difference		
Bertrand-Markov	0.213 (0.085)	0.000 (0.043)	0.213 (0.108)	1.98	0.047
Bertrand-Nash	0.062 (0.038)	0.008 (0.027)	0.054 (0.049)	1.11	0.267

Table 9: Baseline conduct test

Note: Each row tests the null hypothesis that the specified conduct model was followed both before and after the information exchange ended, under the assumption that equation (14) holds. Bootstrap standard errors, computed with 500 bootstrap draws, are in parentheses. Discontinuities are estimated using a local linear regression before and after the cutoff, with an Epanechnikov kernel and a bandwidth computed to minimize placebo discontinuities over June 2017–February 2019 and December 2019–May 2022, which results in a bandwidth of 11 months.

producers’ cost shocks when the information exchange ended:

$$m := \mathbb{E} \left[\left(\lim_{t \uparrow t^*} \omega_{AS,rt} - \omega_{AS^C,rt} \right) - \left(\lim_{t \downarrow t^*} \omega_{AS,rt} - \omega_{AS^C,rt} \right) \right] = 0, \quad (14)$$

where t^* corresponds to May 2019, and AS and AS^C correspond to Agri Stats clients and other producers respectively. m is a “difference-in-discontinuities,” and can therefore be estimated using standard regression discontinuity techniques (Grembi et al., 2016). To do so, I regress $\hat{\omega}_{jrt}$ on the full set of pairwise interactions between a time trend, a post-May 2019 indicator, and an Agri Stats indicator, weighting observations by $k\left(\frac{t-t^*}{h}\right)$ for a kernel function k and positive scalar bandwidth h . $\hat{m}$ is the coefficient on the interaction between the Agri Stats indicator and the post-May 2019 indicator.³⁸ Given a consistent estimate of m ’s standard error $\hat{\sigma}$, which I compute using the bootstrap, the test statistic $t = \hat{m}/\hat{\sigma}$ is asymptotically standard normal under the correct conduct model. I select the bandwidth by minimizing the out-of-sample estimated difference-in-discontinuities among periods outside the window from three months before to six months after the end of the exchange, which returns a bandwidth of 11 months.

The first row of Table 9 displays the estimates. Under the null hypothesis of Bertrand-Markov conduct throughout the sample period, the estimated discontinuity in Agri Stats clients’ relative cost shocks at May 2019 is 0.213 (signifying higher costs with than without the information exchange), compared to a discontinuity of zero for other producers. This

³⁸This moment condition falls in the broader category of restrictions of the form $\mathbb{E}[f(z)'\omega] = 0$, where $f(\cdot)$ is a function of instruments z that shift the way markups depend on conduct that is excluded from ω (Backus et al., 2021). Here z is either 1 or -1 depending on Agri Stats and pre-post status, which leverages the timing of the information exchange as a direct conduct shifter (similar to Miller and Weinberg, 2017), while $f(\cdot)$ is chosen to weight observations closer to May 2019 higher so that any long-run relative changes in ω (for example, due to forgone investment efficiencies) do not invalidate the moment restriction.

results in an estimated moment value of $\hat{m} = 0.213$, which, normalized by the standard error, rejects the null hypothesis that conduct was Bertrand-Markov throughout.

A question one might have had throughout the paper is whether information effects are empirically relevant enough to be worth modeling. To illustrate how much information effects matter, I reestimate marginal costs and recompute the conduct test under the standard complete-information Bertrand-Nash model, where firms maximize their own static profits in every period with complete information over demand shocks and rivals' prices. The estimated discontinuities and associated p -value are listed in the second row of Table 9. While assuming complete information results in more precise estimates, the estimated moment value is much smaller than the Bertrand-Markov value, and Bertrand-Nash conduct results in a statistically insignificant estimated difference-in-discontinuities.³⁹

While the preceding single-hypothesis test of Bertrand-Markov conduct is valid under correctly specified demand and cost functions, the conduct testing literature has emphasized making comparisons between different conduct models to add robustness to model misspecification (Backus et al., 2021; Duarte et al., 2024). In this spirit, I consider a parameterized family of Markov equilibria in pricing rules in which Agri Stats subscribers maximized a weighted sum of their own expected profits and their fellow subscribers' expected profits while sharing information. That is, before May 2019, Agri Stats clients' prices solve

$$p_{f,rt}(\tau) = \arg \max_{p_{f,rt}} \mathbb{E}_{f,rt} \left[\pi_{f,rt}(p_{rt}; \xi_{rt}) + \tau \sum_{f' \in AS \setminus \{f\}} \pi_{f',rt}(p_{rt}; \xi_{rt}) \right], \quad (15)$$

where the conduct parameter $\tau \in [0, 1]$ describes the fraction of other subscribers' profits internalized by Agri Stats clients. $\tau = 0$ corresponds to Bertrand-Markov conduct, while $\tau = 1$ corresponds to joint expected profit maximization among subscribers.⁴⁰ Each candidate τ results in a different vector of cost shocks and therefore a different estimate of $\hat{m}(\tau)$, which can be used to compare different conduct models. I then estimate a conduct parameter by searching for the τ^* minimizing the distance between $\hat{m}(\tau^*)$ and zero.⁴¹

³⁹Another potential test is whether there is a discontinuity in Agri Stats clients' cost shocks, without differencing other producers' cost shocks. These estimated discontinuities are shown in the first column of Table 9. This test results in an even lower p -value for the test of Bertrand-Markov conduct, while Bertrand-Nash conduct is again not rejected at 5% significance.

⁴⁰Corts (1999) shows that this family of conduct models does not necessarily correspond to the reduced form of a complete-information dynamic game, and there is no reason to suspect that it does so in the present incomplete-information setting. This exercise should therefore be viewed as a quantification of conduct effects, rather than a fully specified model of how information sharing affects collusion. I discuss the challenges of modeling conduct effects of information exchanges further in Section 8.

⁴¹In this case, there exists an interior τ^* such that the estimated discontinuity is exactly zero; I find that $\hat{m}(\tau)$ is monotone in τ , so τ^* is unique, and I search for it using bisection. An alternative approach to comparing conduct models is Rivers and Vuong (2002)-style pairwise testing based on differences in

Model	Estimated conduct parameter	95% confidence interval
Incomplete information	0.475	[0.178, 0.724]
Complete information	0.179	[0.000, 0.488]

Table 10: Conduct parameter estimates

Note: 95% confidence intervals are computed with 500 bootstrap draws. $\tau^* \in [0, 1]$ is estimated to minimize the square of the left-hand side of equation (14), assuming Agri Stats subscribers maximized the objective in (15). Discontinuities are estimated using a local linear regression before and after the cutoff, with an Epanechnikov kernel and a bandwidth computed to minimize placebo discontinuities over June 2017–February 2019 and December 2019–May 2022, which results in a bandwidth of 11 months.

As shown in Table 10, I estimate $\tau^* = 0.475$, demonstrating that Agri Stats clients acted as though they internalized nearly half their fellow subscribers’ profits while sharing information. The 95% confidence interval is bounded away from zero. For comparison, I compute τ^* under the analogous complete-information model and find a conduct parameter of 0.179, with a 95% confidence interval including zero. These results suggest two broad takeaways. First, information effects cannot explain the price changes stemming from the end of the exchange. Even when considering alternative conduct models, the best-fitting model is one in which firms exhibited some degree of coordination in prices while sharing information. Second, ignoring these information effects would have led to underestimating the extent to which firms were coordinating. This both emphasizes the importance of modeling these forces and points to the procompetitive nature of information effects in this context.

7.2 The information and conduct effects of Agri Stats

To assess the impact and mechanisms of the Agri Stats exchange, I use the conduct parameter estimate to simulate what would have happened had Agri Stats continued operating after May 2019. I first simulate a Bertrand-Markov equilibrium in which each firm maximizes its expected profits with the Agri Stats signal structure. Comparing these simulated outcomes to the realized outcomes gives the information effects of Agri Stats without any conduct effects. I then simulate a Markov equilibrium in pricing rules with the estimated conduct parameter, incorporating the conduct effects of Agri Stats.

The primary challenge in computing counterfactual Markov equilibria in pricing rules is simulating the counterfactual long-run law of motion of prices. In equilibrium, firms set prices to optimize their objective given beliefs, update these beliefs given the conjectured

implied moment values between two candidate values of τ . Such tests have different limiting distributions depending on whether the values of τ being considered result in moment values close to zero (Hall and Pelletier, 2011), and here there is a unique interior value of τ such that $\hat{m}(\tau) = 0$, making the limiting distribution of the Rivers-Vuong test statistic unclear. Hence, I adopt conduct parameter estimation as my main approach.

Scenario	Average prices			Profits (\$m/year)			CS (\$m/year)	TS (\$m/year)
	AS	Others	Total	AS	Others	Total		
No sharing (actual)	4.24	4.10	4.19	210.2	81.6	291.8	396.1	687.9
Just accuracy	4.18	4.10	4.15	209.0	80.4	289.3	404.9	694.2
Accuracy + correlation	4.17	4.09	4.14	208.1	80.2	288.4	406.9	695.3
Information effect	4.17	4.09	4.14	208.0	80.3	288.3	407.3	695.6
Info + conduct effect	4.39	4.11	4.30	217.3	83.3	300.6	378.5	679.1

Table 11: Decomposing the impact of Agri Stats

Note: Average prices are weighted by no-sharing (actual) total quantity. In all scenarios except no sharing, Agri Stats clients share prices, quantities, costs, and inventories, and other producers just observe their own prices and quantities. In the “Information effect” scenario, firms maximize their own expected profits. In the “Just accuracy” scenario, firms update their beliefs using rivals’ no-sharing prices and shares. In both the “Just accuracy” and “Accuracy + correlation” scenarios, firms update beliefs using the estimated long-run law of motion of prices from the no-sharing equilibrium rather than the information effect equilibrium. In the “Info + conduct effect” scenario, Agri Stats clients maximize (15) using the estimated conduct parameter, while other producers continue to maximize their own expected profits.

long-run price kernel, and conjecture a long-run price kernel that is consistent with the prices rivals actually set. For the realized information environment, the long-run distribution can be directly estimated from the data. Conversely, finding a counterfactual equilibrium requires finding a long-run price kernel Π^+ such that $\Pi^+ = \Phi(\Pi^+)$, where $\Phi(\Pi^+)$ gives the law of motion of the optimal prices given the beliefs generated by Π^+ . For more details on how I do this in practice, see Appendix B.3.⁴²

Table 11 shows Agri Stats’ impact on average prices, profits, and welfare through both the conduct and information channels. Comparing the first and last rows, I find that Agri Stats raises average prices by 2.5% (11 cents/lb) and lowers consumer surplus by 4.5% (\$17.6 million annually). This aggregate effect is the composition of a procompetitive information effect and an anticompetitive conduct effect: the information effect lowers prices by 1.3% and increases consumer surplus by 2.8%, and the conduct effect raises prices by 3.9% and lowers consumer surplus by 7.1%. The impact on Agri Stats clients’ prices alone lines up closely with the reduced-form results (Table 2): the conduct effect of 5.4% is close to the short-run reduced-form estimate of 5.1%, and the overall effect of 3.5% to the long-run estimate of 3.4%. Agri Stats increases profits, primarily for Agri Stats clients (3.4%) but also for other producers (2.1%), which benefit from the softened competition and higher prices from Agri Stats clients. However, information effects alone lower Agri Stats clients’ profits. While this

⁴²Unlike the standard supply-side model, this problem is not separable by market: equilibrium outcomes in one time period affect the long-run price distribution, thereby affecting equilibrium outcomes in all other periods. In addition, I keep marginal costs fixed at their estimated values in all simulations. If, as claimed by Agri Stats, information sharing directly causes marginal cost efficiencies, then estimated counterfactual prices with information sharing may be too high.

may suggest a revealed preference argument that Agri Stats facilitates collusion, in the sense that firms would not rationally pay for Agri Stats if there were no conduct effect, it may also be individually rational for firms to subscribe to Agri Stats even though its overall effect is to reduce profits. Agri Stats causes deadweight loss, reducing total surplus by 1.3%.

To dig into the mechanisms behind information effects, I perform the following decomposition motivated by Section 2. Let $p(y^{NS}(p^{NS}, s^{NS}), \Pi^{+, NS})$ be the vector of expected profit-maximizing prices at beliefs derived using the no-sharing signal functions $y^{NS}(\cdot)$ —evaluated at the no-sharing equilibrium outcomes (p^{NS}, s^{NS}) —and the no-sharing equilibrium price kernel $\Pi^{+, NS}$. The first effect of information sharing is that firms’ signals tell them more about the state, making their beliefs more accurate and moving their best-response functions. This is captured by changing $y^{NS}(\cdot)$ to $y^S(\cdot)$, the with-sharing signal function. The second effect is that firms’ beliefs become more correlated, thereby causing movement along best-response functions and hence changes to equilibrium prices and shares. This is captured by changing the inputs to the signal function to (p^S, s^S) , the equilibrium values with information sharing. Finally, the remaining effect is the change in the equilibrium distribution of signals, captured by changing $\Pi^{+, NS}$ to $\Pi^{+, S}$. $p(y^S(p^{NS}, s^{NS}), \Pi^{+, NS})$ incorporates just the accuracy effect of information sharing, while $p(y^S(p^S, s^S), \Pi^{+, NS})$ incorporates the accuracy and correlation effects without the signal distribution effect.

The results of the decomposition are shown in the middle rows of Table 11. The accuracy effect accounts for 84.7% of the overall information effect on Agri Stats clients’ prices, with the correlation effect and the signal distribution effect contributing 15.3% and close to zero respectively. The accuracy effect on Agri Stats clients’ profits, while a smaller proportion at 55.2%, is still negative; the business-stealing externality caused by rivals responding to the signal outweighs firms being better able to maximize their profits due to the additional information provided by the signal. The accuracy effect on non-subscribers’ prices is zero by definition; the remainder of the effect (although small) is roughly evenly split between the correlation effect—which is nonzero because information sharing changes Agri Stats clients’ prices, which changes other firms’ prices in equilibrium—and the signal distribution effect.

These results also provide insight into how much and what firms learn from the exchange, and how this learning depends on primitives. Table 12 shows how Agri Stats affects the accuracy of its clients’ beliefs, as measured by the reduction in average RMSE over own-price elasticities, own ξ , rival ξ , and rival prices. Agri Stats improves its clients’ beliefs over their own-price elasticities by 42%, with similar magnitudes for both fresh and deli turkey. However, the two modules differ in *what* firms learn about. In both modules, firms have a relatively good sense of their own ξ without Agri Stats. Agri Stats provides relatively more information about rival demand for deli turkey than for fresh turkey. Given the esti-

Module	Reduction in average RMSE for Agri Stats clients			
	Own-price elasticity	Own ξ	Rival ξ	Rival price
Fresh turkey	43.1%	11.9%	17.9%	40.5%
Deli turkey	37.4%	4.7%	37.4%	19.9%
Overall	41.9%	8.4%	28.0%	23.3%

Table 12: How much and what firms learn from Agri Stats

Note: RMSEs of own-price elasticities are normalized by the within-product SD of own-price elasticities before aggregating and taking percentage differences. For each of own ξ , rival ξ , and rival prices, RMSEs of beliefs over each applicable product-market combination are averaged within each firm-market combination, weighted by quantities, and then averaged across firm-markets.

mated primitives, this makes sense: demand shocks are more common-valued in fresh turkey (Table 6), suggesting that fresh turkey producers can use knowledge of their own shock to better forecast their rivals’ shocks without Agri Stats. Conversely, Agri Stats reduces RMSE over rivals’ prices more for fresh turkey producers than for deli turkey producers. Table A.1 decomposes information and conduct effects by module and finds that while information effects alone lower prices by 2.1% in fresh turkey, they slightly raise prices in deli turkey, highlighting that these differences in primitives are ultimately economically important for information effects.

8 Bounding collusion via information exchanges

The final goal of the paper is to assess outcomes under counterfactual restrictions on what kind of information firms can share with one another. This task presents a conceptual challenge. While the estimated conduct parameter can credibly be used to simulate the effects of the Agri Stats exchange, it cannot be used to simulate the effects of other exchanges; different information structures likely correspond to different conduct parameters, and the lack of microfoundation of the conduct parameter model (Corts, 1999) makes selecting counterfactual parameters an ad hoc process at best. However, solving a fully microfounded dynamic collusive game with counterfactual information structures is difficult. Exchanges that do not fully reveal every firm’s information lead to repeated games with private monitoring, a class of games for which finding optimal strategies is known to be intractable (Kandori, 2002). The problem stems from firms being able to condition their actions on their private history; to best-respond to rivals, firms must therefore form beliefs over the entire history of signals, rendering the recursive methods used in the analysis of public monitoring games inapplicable (Abreu et al., 1990; Fudenberg et al., 1994). Further, while Sugaya (2022) proves a folk theorem for private monitoring games, implying that infinitely patient firms can collude

if signals have a sufficiently large support, this result is less useful for characterizing the potential for collusion in real-world situations with finitely patient firms.

While the theory literature has not characterized exact optimal strategies in repeated games with imperfect monitoring, it has derived necessary conditions that equilibrium distributions of actions must satisfy. Thus, my approach is to bound the distribution of prices under different information structures by solving a constrained optimization problem that maximizes average prices subject to these necessary conditions holding. This section first summarizes the necessary conditions and computational framework, then implements the framework to study the extent to which aggregating the data shared in an information exchange affects the upper bound on the worst-case collusive price.

8.1 Constructing the bounds

Let $\alpha \in \Delta(\mathcal{P})$ be a stationary distribution over prices for all firms with support bounded by $\mathcal{P}$, and let $\Psi(\alpha) = \mathbb{E}_\alpha \left[\sum_j \psi_j p_j \right]$ be the weighted average price of a particular price distribution given weights ψ_j that sum to unity.⁴³ Suppose any equilibrium must satisfy the inequality $C(\alpha; y) \leq \mathbf{0}$, where $C(\cdot)$ is a vector of necessary conditions and $y = \{y_f\}_{f=1}^F$ is a counterfactual set of signal functions of interest. A bound on average prices is found by solving

$$\max_{\alpha \in \Delta(\mathcal{P})} \Psi(\alpha) \quad \text{s.t.} \quad C(\alpha; y) \leq \mathbf{0}. \quad (16)$$

If the necessary conditions $C(\cdot)$ are also jointly sufficient, then this bound is sharp.

The immediate question is what necessary conditions are included in $C(\cdot)$. One standard condition is that it cannot be profitable for a firm to deviate if rivals can punish it with full Nash reversion (Sugaya and Wolitzky, 2018a). Let p^C be the competitive price vector that simultaneously maximizes each firm's own expected profits, $\mathbb{E}_f[\pi_f(p)]$ be firm f 's expected profits given its beliefs over ξ and a price vector p , and δ be a common discount factor. Then for all prices $p \in \text{supp}(\alpha)$, all firms f , and all potential deviations $\tilde{p}_f$, a necessary condition for α to be an equilibrium price distribution is

$$\overbrace{(1 - \delta) (\mathbb{E}_f[\pi_f(\tilde{p}_f, p_{-f})] - \mathbb{E}_f[\pi_f(p)])}^{=C_f^{\text{Nash}}(p, \tilde{p}_f, \alpha)} + \delta \left(\mathbb{E}_f[\pi_f(p^C)] - \int \mathbb{E}_f[\pi_f(p')] d\alpha(p') \right) \leq 0. \quad (17)$$

The first term is the profit gain the firm gets from deviating to $\tilde{p}_f$ in the current period, when it is supposed to set p_f . The second term is the decline in expected future payoffs from

⁴³Here I focus on bounds on share-weighted average prices. One could also bound expected profits, consumer surplus, or total surplus by adjusting $\Psi(\cdot)$ appropriately.

rivals punishing that deviation with Nash reversion in all future periods. Of course, if firms cannot perfectly observe their rivals' actions, then they may not be able to perfectly punish any deviation with Nash reversion; that makes this a necessary but generally insufficient condition.⁴⁴

The complete-information Nash condition does not depend on the monitoring structure, and hence will not characterize how changing the information firms share affects possible collusive outcomes. To do so, I add two additional sets of conditions from Kreps (2026, Theorems 1 and 2) that bound the short-run gain from a particular deviation by a function of the discount factor, the potential long-run punishment from the deviation, and the change to the distribution of signals caused by the deviation.⁴⁵ I briefly summarize each condition in turn below.

Suppose that after each period, firms collectively observe signals $y = (y_1, \dots, y_F) \in \mathcal{Y} = \prod_f \mathcal{Y}_f$ with conditional distribution $h_y(\cdot | p)$ given a price vector $p \in \mathcal{P} = \prod_f \mathcal{P}_f$. The leading example is that the signals are functions of shares, which are stochastic conditional on prices due to demand uncertainty. In addition, recall two standard measures of statistical distance between two distributions with densities g and h : the χ^2 -divergence is defined as $\chi^2(h||g) = \int (h(x) - g(x))^2 / g(x) dx$,⁴⁶ and the total variation distance is defined as $TV(h||g) = \frac{1}{2} \int |h(x) - g(x)| dx$. Each of these distance measures will be used by one of the conditions to quantify the extent to which a price change affects the distribution of signals, and hence the extent to which firms can detect deviation by their rivals.

Suppose that after each period, all firms communicate a message about their signal; then, a mediator tells the firms what price to set based on the messages it receives. Fix a firm f , and define a *manipulation* as a pair of functions $m_f : \mathcal{P}_f \rightarrow \Delta(\mathcal{P}_f)$ and $q_f : \mathcal{P}_f^2 \times \mathcal{Y}_f \rightarrow \Delta(\mathcal{Y}_f)$. The particular deviation considered by this bound is that in one period, when firm f is supposed to set price p_f , it instead deviates to price $\tilde{p}_f$, drawn from $m_f(\cdot | p_f)$. Then, if it receives signal y_f , it tries to “cover up” its deviation by reporting signal $\tilde{y}_f$, drawn from $q_f(\cdot | p_f, \tilde{p}_f, y_f)$. Firm f is obedient and truthful in all other periods.

Let $h_{\tilde{y}}(\tilde{y} | p, \tilde{p}_f, q_f) = \int h_y(y | \tilde{p}_f, p_{-f}) q_f(\tilde{y}_f | p_f, \tilde{p}_f, y_f) dy_f$ be the distribution of reported signals $\tilde{y} = (\tilde{y}_f, y_{-f})$ if firm f manipulates according to (m_f, q_f) and firm f 's rivals are obedient and truthful. Then, given a putative equilibrium price distribution α , define the χ^2 -detectability of a manipulation (m_f, q_f) as the expected χ^2 -divergence of the distributions

⁴⁴The condition is sufficient when all firms' signals include the full price vector. In this case, it reduces to the typical complete-information incentive compatibility constraint.

⁴⁵The two conditions are based on similar results originally derived by Sugaya and Wolitzky (2023) and Awaya and Krishna (2019). Kreps (2026) strengthens both conditions and extends them to cases in which a portion of the signals are observed on a delay.

⁴⁶If there exists an x where $h(x) > 0$ and $g(x) = 0$, then $\chi^2(h||g) = +\infty$.

of y without the manipulation and $\tilde{y}$ with the manipulation, and the *gain* from m_f as the expected difference in profits:

$$\begin{aligned}\chi_f^2(m_f, q_f, \alpha) &= \mathbb{E}_{p \sim \alpha} \left[\mathbb{E}_{\tilde{p}_f \sim m_f(\cdot | p_f)} \left[\chi^2(h_{\tilde{y}}(\cdot | p, \tilde{p}_f, q_f) \| h_y(\cdot | p)) \right] \right], \\ g_f(m_f, \alpha) &= \mathbb{E}_{p \sim \alpha} \left[\mathbb{E}_{\tilde{p}_f \sim m_f(\cdot | p_f)} [\mathbb{E}_f[\pi_f(\tilde{p}_f, p_{-f})] - \mathbb{E}_f[\pi_f(p)]] \right].\end{aligned}$$

A necessary condition for α to be an equilibrium price distribution (implying that this deviation is unprofitable) is that, for all firms f and manipulations (m_f, q_f) ,

$$g_f(m_f, \alpha) \leq \sqrt{\frac{\delta}{1 - \delta} \chi_f^2(m_f, q_f, \alpha) \text{Var}(\pi_f(\alpha))}, \quad (18)$$

where $\text{Var}(\pi_f(\alpha)) = \text{Var}_{p \sim \alpha}(\mathbb{E}_f[\pi_f(p)])$ is the variance in firm f 's expected profits if the price distribution is α .⁴⁷ The intuition is the following. Collusion is impossible if a firm can deviate profitably (high g_f) in a way that cannot be detected (low χ_f^2). In particular, the ratio of the two must be bounded by the product of how much firms weigh future punishment relative to present gain and the size of that punishment, with the latter bounded by the on-path variance of profits.⁴⁸ If an exchange shares verified information about quantities, then firms can no longer lie about what quantity they actually sold, meaning any deviation becomes more detectable; hence, larger one-shot gains from deviation can be sustained in equilibrium.

The final condition is based on a different deviation: firm f sets a single price $\tilde{p}_f$ in all future periods instead of following its equilibrium strategy. As before, let $g_f(\tilde{p}_f, \alpha)$ be the expected gain from that deviation, and define the *TV-detectability* of the deviation as $TV_f(\tilde{p}_f, \alpha) = \mathbb{E}_{p \sim \alpha} [TV(h_{y_{-f}}(\cdot | \tilde{p}_f, p_{-f}) \| h_{y_{-f}}(\cdot | p))]$, the expected total variation distance between the conditional distribution of rivals' signals under p and under $\tilde{p}_f, p_{-f}$. Finally, let $\|\pi_f\|_\infty$ be the sup-norm of firm f 's expected profits. Then, α is an equilibrium price distribution only if, for all f and $\tilde{p}_f$,

$$g_f(\tilde{p}_f, \alpha) \leq \frac{2\delta}{1 - \delta} \|\pi_f\|_\infty TV_f(\tilde{p}_f, \alpha). \quad (19)$$

The intuition for this bound is similar to that of the last bound, although the two formulas differ because the bounds consider different deviations. Because in this bound firm f permanently deviates to $\tilde{p}_f$, instead of deviating according to m_f for one period and complying in

⁴⁷Technically, in both this condition and the following one, α is an occupation measure over prices, where an occupation measure is the discount-weighted average of each period's distribution of prices. If the equilibrium distribution of prices is stationary, then the occupation measure simply equals that distribution.

⁴⁸Theorem 2 of [Sugaya and Wolitzky \(2023\)](#) demonstrates that this condition is tight up to a constant factor in public monitoring games, in the sense that cooperation is possible if $\chi_f^2(\cdot)/(1 - \delta)$ is sufficiently large for all manipulations.

future periods, this condition bounds punishment with the worst-case difference in payoffs, $\|\pi_f\|_\infty$, instead of the on-path payoff variance.

Constraints (18) and (19) are nonnested. Abstracting away from the difference between χ^2 -divergence and total variation distance, several factors point towards (18) being the tighter bound. First, (18) scales with $\sqrt{\delta/(1-\delta)}$, instead of $\delta/(1-\delta)$, which makes it relatively tighter as δ increases. Second, (18) allows each firm to tailor its deviation to different prices based on the conditional distribution of rivals' prices given its own, while (19) only allows for one deviation price. Third, (18) bounds punishment with the on-path standard deviation of payoffs, $\sqrt{\text{Var}(\pi_f(\alpha))}$, which is sharper than bounding it with $\|\pi_f\|_\infty$.⁴⁹ However, (19) uses the statistical detectability of a deviation in rivals' signals, which is smaller than the statistical detectability of a deviation in rivals' signals plus firm f 's manipulated signals (Kreps, 2026, Proposition 1). I therefore incorporate both conditions into the bound computation.⁵⁰

To implement (16), I approximate α as a discrete distribution with N support points $\{p_n, a_n\}_{n=1}^N$. Thus, if there are J products, then there are $N(J+1)$ choice variables: locations of N price vectors of length J each, and N weights summing to 1. I bound the support of this distribution by the competitive price vector p^C and joint expected profit-maximizing price vector p^M , which together define $\mathcal{P}$. To implement the necessary conditions, which must hold for all deviations, I write them as inner maximization problems enumerated over all possible deviations. The program is therefore

$$\begin{aligned}
\alpha^*(y) = \arg \max_{\alpha=\{p_n, a_n\}} & \sum_{n,j} a_n \psi_j p_{n,j} \\
\text{s.t.} & \sum_n a_n = 1, \quad a_n \geq 0, \quad p_n \in [p^C, p^M] & \forall n, \\
& \max_{\tilde{p}_f} C_f^{\text{Nash}}(p_n, \tilde{p}_f, \alpha) \leq 0 & \forall n, \forall f, \\
& \max_{m_f, q_f} \frac{g_f(m_f, \alpha)}{\sqrt{\chi_f^2(m_f, q_f, \alpha)}} \leq \sqrt{\frac{\delta}{1-\delta} \text{Var}(\pi_f(\alpha))} & \forall f, \\
& \max_{\tilde{p}_f} g_f(\tilde{p}_f, \alpha) - \frac{2\delta}{1-\delta} \|\pi_f\|_\infty TV_f(\tilde{p}_f, \alpha) \leq 0 & \forall f.
\end{aligned}$$

This problem is feasible; a point mass at the competitive price satisfies all constraints.⁵¹

⁴⁹This holds because the standard deviation of $\mathbb{E}_f[\pi_f(p)]$ is at most $(\sup_p \mathbb{E}_f[\pi_f(p)] - \inf_p \mathbb{E}_f[\pi_f(p)])/2$, which is at most $\|\pi_f\|_\infty$.

⁵⁰In practice, I find that the χ^2 -detectability bound tends to be tighter than the TV -detectability bound at the discount factor I use in my analysis. Across all counterfactual monitoring structures with nontrivial detectability constraints, the χ^2 -detectability constraint is binding to a relative tolerance of 0.001 for 60.4% of firm-market pairs, as compared to 7.1% for the TV -detectability constraint.

⁵¹At the competitive price, there is no gain from deviation, trivially satisfying the Nash IC constraints. By

While this is a nonlinear and potentially nonconvex nested maximization problem, several factors make it tractable. First, the Nash IC maximization problem simply amounts to finding a best response for firm f to a rival price vector, and hence is solvable using standard iteration techniques (Morrow and Skerlos, 2011). Second, all functions of p in the problem are moments of expected profits, except for $\chi_f^2(\cdot)$ and $TV_f(\cdot)$; to evaluate these, I closely approximate the distribution of shares conditional on prices with additive logistic normal distributions. The basis for this approximation is that the inside share vector from a (plain) logit demand system with a normally distributed ξ is exactly additive logistic normally distributed—here it is an approximation due to the random coefficients—while the approximation enables a closed-form expression for $\chi_f^2(\cdot)$ and a standard closed-form bound on $TV_f(\cdot)$.⁵² All model objects therefore have analytic gradients, while gradients of the constraints with respect to elements of α can be derived using the envelope theorem. Third, the optimal distributions appear well-approximated with relatively few support points; I present results with $N = 5$ below, and across all counterfactual exchanges I consider, 93% of upper bounds have at least one node with a probability less than 0.1%.⁵³

This method provides an upper bound on average prices under a given information structure for one market, taking as inputs beliefs over ξ for each firm in that market. To compute an aggregate bound, I compute the bound separately in each region-module, using the most common product assortment in that region, average marginal costs, and beliefs at the stationary distribution implied by the timeseries estimates.⁵⁴ I calibrate $\delta = 1.0801^{-1/12}$, reflecting the monthly analogue of the food processing industry’s 2021 weighted average cost of capital

definition, the gain from any deviation, and hence the left-hand sides of (18) and (19), are bounded above by zero at the competitive price, while the right-hand sides of both are bounded below by zero, so both these conditions are also satisfied. In evaluating the left-hand side of the χ^2 -detectability constraint, I adopt the conventions that $0/0 = 0$ and $k/0 = \pm\infty$ with the same sign as k for $|k| > 0$, so that the constraint is satisfied if $g_f \leq 0$ and violated if $g_f > 0$ and detectability is zero. Finally, note that if firm f shares its price directly with another firm, then the last two constraints are vacuously satisfied, in which case only the Nash IC constraints bind on firm f .

⁵²When computing the χ^2 -detectability constraint, there is an additional measurability restriction that a message manipulation q_f for firm f must only affect signals for firm f and not other firms. This requires slightly modifying the transformation so that exactly one of firm f ’s signals and one of firm f ’s rivals’ signals are not conditionally normal in a plain logit model. I bound the total variation distance, which does not have an analytic formula in the Gaussian case, with $\sqrt{2}$ times the Hellinger distance, which does; this strategy is also used by Awaya and Krishna (2020). For more details, see Appendix B.4.

⁵³Table A.2 contains points from the CDF of the minimum a_n across upper bounds for each counterfactual monitoring structure I consider.

⁵⁴Because I collapse the time dimension, I refer to a region-module as a market for the remainder of the paper. I note three caveats about this strategy. First, the bound is computed separately in each region, meaning there is no scope for multimarket contact effects (Bernheim and Whinston, 1990). Second, the bound does not dynamically account for how expected future movement in ξ changes deviation constraints (Rotemberg and Saloner, 1986). Third, setting common stationary priors over ξ does not take into account the indirect effect that information sharing may have on collusive outcomes by homogenizing firms’ beliefs themselves, affecting perceived deviation payoffs.

of 8.01% (Damodaran, 2021). Additional details, including formulas and gradients of all objects, are available in Appendix B.4.

8.2 Aggregation versus participation in information sharing

With these upper bounds, one can simulate the potential effects of counterfactual restrictions on data sharing. I focus on two aspects of exchanges: *aggregation* (allowing information exchanges to share only aggregates, instead of specific or disaggregated information) and *participation* (the fraction of the market that contributes to and receives the information exchange). Whether aggregation prevents collusion via information exchange has been a matter of broader policy debate: the FTC and DOJ allowed a “safe harbor” for sufficiently aggregated information exchanges until rescinding this policy in 2023, and have since submitted statements of interest in multiple cases stating that aggregated information exchanges can still harm competition.⁵⁵ Meanwhile, the DOJ’s 2025 settlement with RealPage, a rent-setting software provider, prevents it from using any nonpublic competitor information when determining a firm’s rental price “regardless of form, aggregation, anonymization, or age” (US v. RealPage, Inc., 2025). However, while the Agri Stats settlement prevents it from exchanging any price information, the settlement allows sharing aggregates of other information, including quantities (US et al. v. Agri Stats, Inc., 2026). The settlement also mandates that Agri Stats allow anyone to purchase the reports, suggesting that participation may increase.

To investigate whether regulators should prioritize aggregation or participation, I compute the upper bound under several different signal structures. I start with firms not observing rivals’ prices and shares, as is the case under no information sharing. I then compute bounds under the pre-settlement status quo: product-level quantities and/or prices are shared only among Agri Stats clients. Finally, I compute the bounds if all firms participate, but have access only to the total quantity—which complies with the Agri Stats settlement—and progressively add more disaggregated information. I scale average prices in each market between zero and one, where the competitive price is zero and the monopoly price is one; all upper-bound prices are expressed as percentages of this distance.

Figure 8 shows the distribution of the scaled worst-case average price across markets for each of these signal structures. The first observation is that collusive prices can already be relatively high without information sharing: the no-sharing upper bound is 65.9% on

⁵⁵For example, the DOJ’s statement of interest in *In Re Frozen Potato Products Antitrust Litigation* says that “to the extent that Defendants suggest that exchanges of aggregated, anonymized, or backward-looking information are incapable of causing anticompetitive effects, Defendants fail to grapple with the ways that such information can harm the competitive process” (Department of Justice, 2026, p. 6).

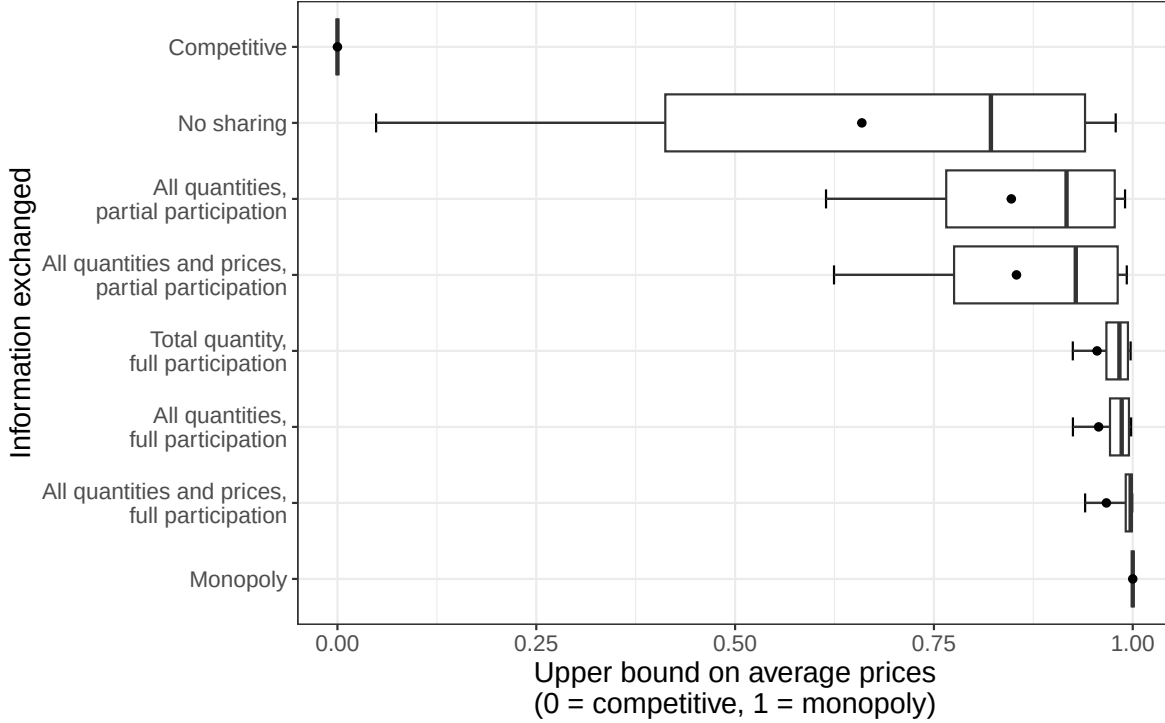


Figure 8: Distribution of upper bounds on average prices across markets

Note: I compute the upper bound on average prices within each market (region-module), then plot the distribution across markets. The whiskers are the 10th and 90th percentiles, the edges of the box are the 25th and 75th percentiles, the middle line is the median, and the point is the average.

average. This suggests that even without observing rivals' prices directly, firms can already detect deviation somewhat well using their own shares. There is a long left tail, however; the 10th percentile bound is 4.9%, suggesting that information sharing is necessary to sustain full collusion in many markets. Information sharing, as implemented by Agri Stats, makes a difference: Agri Stats clients sharing the full vector of prices and quantities among themselves results in an upper bound of 85.4% on average. Notably, once Agri Stats clients share the full vector of quantities, sharing prices does not matter much; price deviations are nearly perfectly detectable with precise information about quantities.

The key result, however, is that all firms sharing aggregate information results in the possibility of near-perfect collusion. Even if firms only observe total quantity, the average upper bound increases to 95.5%, and the 10th percentile increases to 92.5%.⁵⁶ While sharing

⁵⁶The settlement also requires that any data Agri Stats shares are no less than 45 days old. [Kreps \(2026\)](#) shows that when part of the signal is observed on a delay of L periods, the correct bound replaces detectability with effective detectability, defined as $(1 - \delta^L)(\text{detectability of contemporaneously observed signal}) + \delta^L(\text{detectability of entire signal})$, in both (18) and (19). Given that a period is a month, this means that the effective detectability reduces by at most $1 - \delta^2 \approx 1.3\%$, but almost certainly less, as firms always observe their own quantities with no delay. I ultimately find that incorporating this requirement reduces the upper bound under publicly shared total quantities by 0.13% on average.

disaggregated quantities and prices does increase the bound to 96.7% under full participation, the increase is not as large as under partial participation.⁵⁷

What explains why sharing aggregate information with all participants has such a large impact? This computational result is similar to the analytical result of [Awaya and Krishna \(2020\)](#), who find that aggregate information is more useful for detecting deviation when shocks are uncorrelated among firms. The intuition, in their words, is that “whereas individual sales can be rather variable, aggregate sales are usually much less variable. Thus, if a firm cuts its price from an agreed upon level, this will cause, with high probability, an increase in aggregate sales observed by all” (pp. 422–423). Conversely, if demand is correlated across firms, then sharing aggregate information does not matter as much—firms can already use their own sales information to get a sense for the common demand shock. Indeed, Figure [A.8\(b\)](#) plots the distribution of the bounds separately for fresh turkey, for which demand shocks are more correlated across products, and deli turkey, for which demand shocks are more independent. Without information sharing, fresh turkey is more susceptible to collusion than deli turkey, with average bounds of 87.5% and 45.9% respectively. This pattern continues under partial participation. However, under total quantity sharing with full participation, the fresh turkey bound is eclipsed by the deli turkey bound, at 94.2% and 96.8% respectively. While there are other differences between the modules—fresh turkey is more concentrated and more price-elastic than deli turkey, affecting deviation payoffs and detectability of deviation through own shares—this finding reinforces the notion that the structure and correlation of shocks are key contributors to how information exchanges affect competition.

I close with several points about the interpretation of these results. First, these results are upper bounds, not exact equilibria. While one hopes that the extent to which these bounds are sharp is roughly consistent across counterfactual exchanges, there is no formal guarantee. In addition, information exchanges may be important for equilibrium selection, not just for broadening the set of equilibria, and may thus make attaining the upper bound more likely. Second, the level of each bound is a function of the discount factor, and the effective discount factor for sustaining collusion via information exchanges that might be subject to antitrust scrutiny could be lower than that implied by firms’ capital costs ([Harrington and Chang, 2009](#)). Thus, comparisons across counterfactuals may be more reliable than levels within a counterfactual. Third, the comparison between partial- and full-participation information exchanges takes the set of participating firms as given. Endogenizing participation in infor-

⁵⁷Figure [A.8\(a\)](#) disaggregates the distribution based on firms that participate in any exchange (i.e., Agri Stats clients) versus those that only participate in full-participation exchanges (other producers). It shows that full participation allows the bound to increase for all firms, although (perhaps obviously) substantially more for firms that do not participate in the partial exchange.

mation exchanges may be a worthwhile area for future study. Finally, one could interpret these results as suggesting that the DOJ erred in allowing Agri Stats to share aggregate quantity information and make its reports more broadly available for purchase. However, the intent behind the latter clause was to allow poultry purchasers and workers to acquire the reports, which may have additional positive effects outside the model. In addition, the only protein for which Agri Stats currently produces reports is chicken, where its clients comprise nearly all of the market, limiting the scope of the unintended consequence of broadening participation in the exchange.

9 Conclusion

There is rising concern among antitrust enforcers, policymakers, and observers about whether firms use common vendors or software to collude with one another, sparking a multitude of cases alleging anticompetitive information sharing. This paper presents a new framework for empirically evaluating the effects of such exchanges, combining a workhorse industrial organization supply-and-demand model with an information structure that captures how information sharing changes firms' information sets, and therefore actions. The model can be used to test whether firms are using information exchanges to collude, as well as to simulate the information effects of counterfactual exchanges. I also leverage recent theory results to bound counterfactual outcomes in order to assess the worst-case collusive scenario under different restrictions on information sharing.

I apply this machinery to the Agri Stats natural experiment, in which the largest turkey processors' prices fell by 5% in the year after they suddenly stopped sharing price, quantity, and inventory information with one another. This price change cannot be rationalized by firms continuing to maximize expected profits with more precise signals while sharing information. Instead, I estimate that subscribing firms set prices as though they internalized 47% of fellow subscribers' profits while sharing information. Both conduct tests with complete-information models and counterfactual simulations show that information effects countervailed conduct effects; Agri Stats would have lowered prices by 1.3% had firms used the information to compete. Finally, the bounds suggest full participation in information exchanges substantially raises worst-case collusive prices, even if firms just share aggregate quantity information, while sharing disaggregated information within a smaller group has a smaller impact.

These results have several policy implications. First, while the results are estimated in the context of the poultry processing industry, the model can be used to test conduct and simulate information exchanges in other industries and contexts, as it captures the

most common forms of information sharing. Second, conduct tests and counterfactuals reinforce the notion that complete-information conduct tests may be biased in the presence of changing information sets. Finally, the counterfactual results suggest that recent policy movement away from “safe harbors” for aggregated information exchanges is a good idea. Instead, information exchanges should be judged on a case-by-case basis with the economic fundamentals of the market in mind.

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Online Appendix for “Information Sharing and Collusion: Theory and Evidence from Poultry Processing”

Avner A. Kreps

September 9, 2026

A Additional figures and tables

This section contains additional figures and tables that are referenced in the main text.

1. Figure A.1 contains the evolution of prices and quantities of chicken products over time in the two modules I study.
2. Figure A.2 contains the results of the triple-differences event study that benchmarks the turkey event study against the placebo chicken event study. This figure shows the estimates and 95% confidence interval of β_t from the regression

$$\log(p_{jstm}) = \alpha_{jr} + \gamma_{mt} + \mu_t \text{AS}_{jm} + \beta_t \text{AS}_{jm} \text{Turkey}_m + \varepsilon_{jstm}, \quad (20)$$

where the m subscript refers to the meat type (turkey or chicken), and Turkey_m is an indicator for turkey products. β_t is normalized to zero in May 2019.

3. Figure A.3 shows a kernel density plot with the unweighted distribution of estimated own-price elasticities at realized prices, separately for fresh and deli turkey.
4. Figure A.4 shows the distribution of diagonal elements of the estimated timeseries slope matrices.
5. Figure A.5 shows the distribution of off-diagonal elements of the estimated timeseries slope matrices, separately by whether the entry corresponds to products made by the same firm. The vertical dashed lines represent the means of these distributions.
6. Figure A.6 is a scatter plot containing predictions of $g(w_{jrt}, z_{jrt}^{GH})$ on the horizontal axis plotted against actual prices on the vertical axis, with a nonparametric line of best fit. A linear regression of actual on predicted prices yields an R^2 of 92.4%.
7. Figure A.7 binscatters the expectation of firms’ beliefs about their products’ own-price

elasticity against the actual own-price elasticity given realized prices. As in the main text, I simulate the expected value of firms' beliefs about their own-price elasticity by taking draws from their beliefs over (ξ, p_f) , computing the elasticity, and taking the mean across draws.

8. Figure A.8 shows the distribution of average price upper bounds across markets separately for participants and non-participants in partial information exchanges, and by fresh and deli turkey.
9. Table A.1 presents the decomposition of Agri Stats' overall effects into information and conduct effects, separately for fresh and deli turkey.
10. Table A.2 shows the distribution of the smallest-probability price node (out of five) in the upper-bound price distribution for each counterfactual information exchange considered in Section 8.

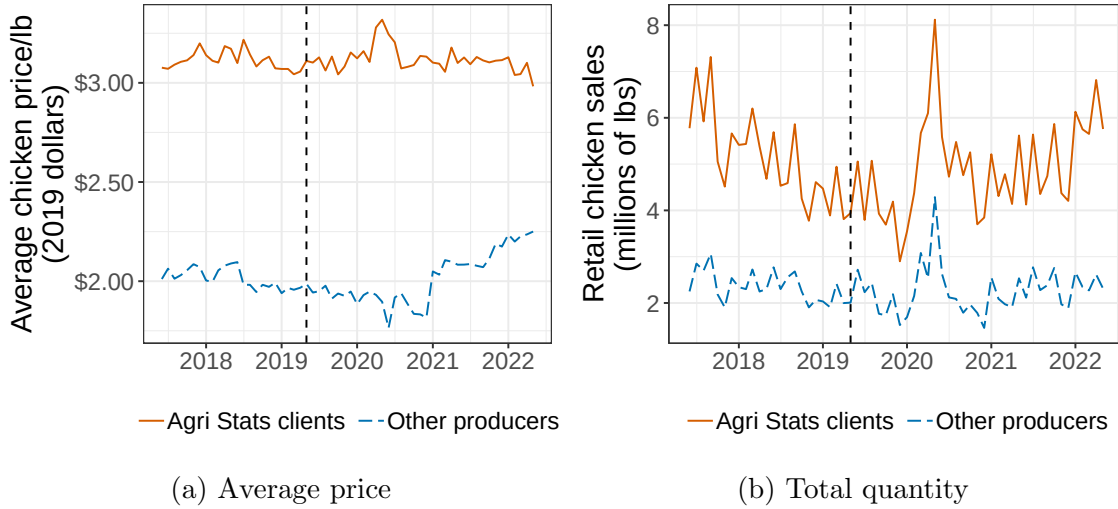


Figure A.1: Average chicken price and sales quantity over time

Note: Average prices are weighted by total within-UPC-store quantity throughout the time period.

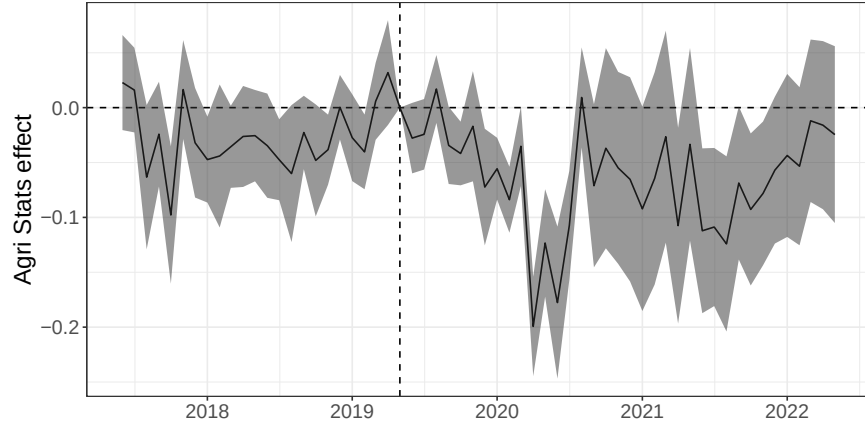


Figure A.2: Triple-differences estimate of the Agri Stats effect

Note: Estimates come from a regression of log prices on UPC-region fixed effects, month-meat and month-Agri Stats indicator fixed effects, and a month-turkey indicator-Agri Stats indicator interaction, which are plotted along with their 95% confidence interval, with standard errors clustered by UPC-region. Regression is weighted by total within-UPC-region quantity throughout the time period. The vertical dashed line represents May 2019.

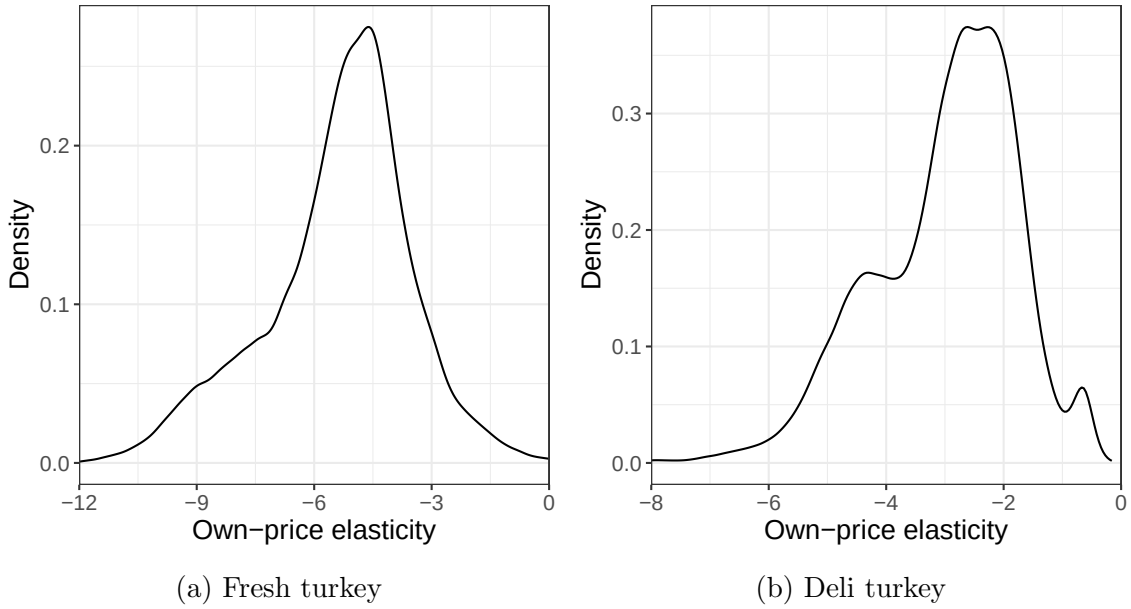


Figure A.3: Distribution of estimated own-price elasticities

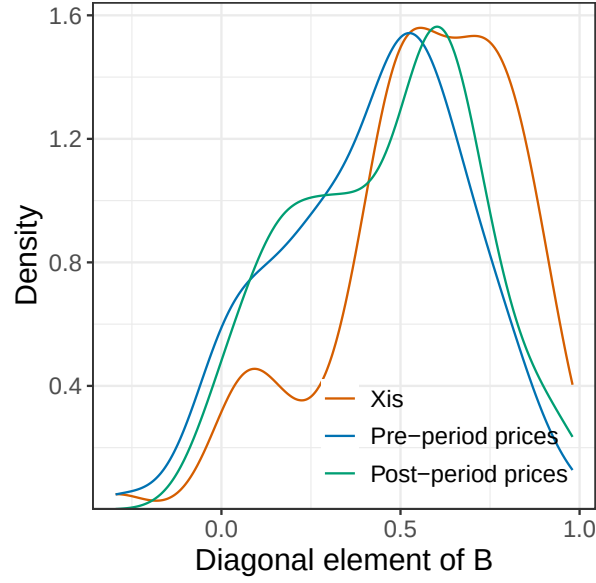


Figure A.4: Distribution of diagonal elements of B^ξ and B^p

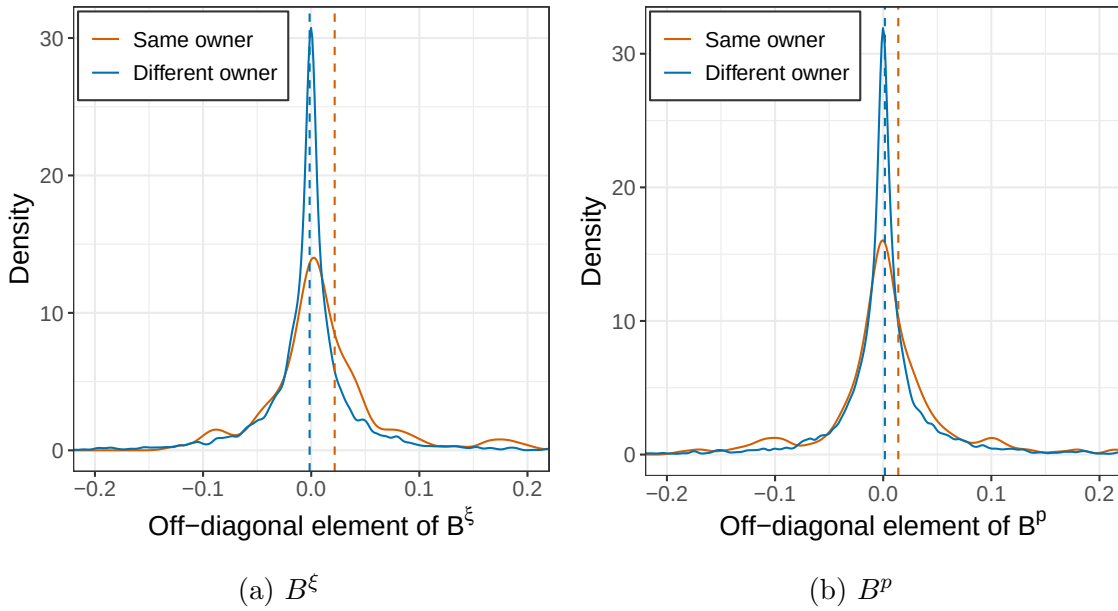


Figure A.5: Distribution of off-diagonal elements of B^ξ and B^p

Note: To be included, the off-diagonal must correspond to a pair of products with more than five markets in common. Off-diagonal elements of B^p only include coefficients of lagged prices on prices, not lagged demand shocks.

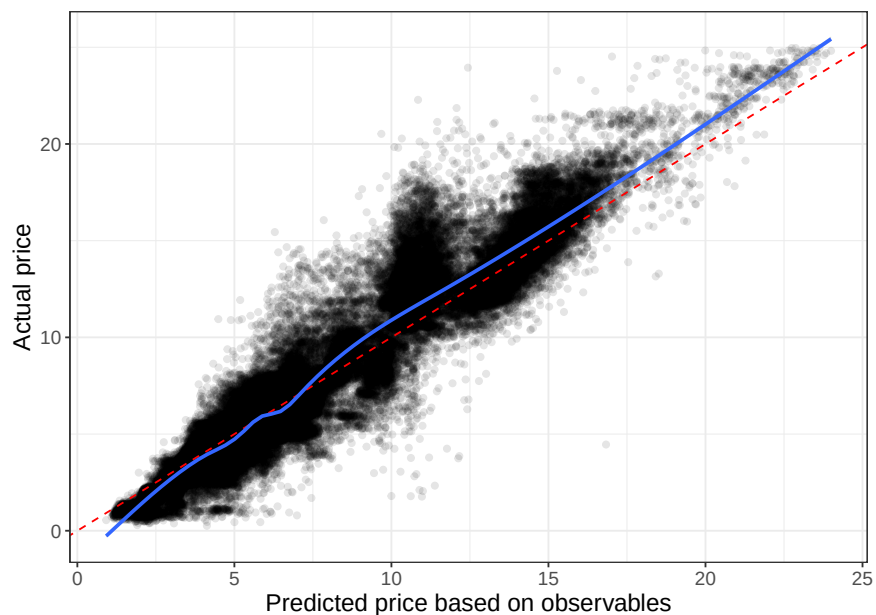


Figure A.6: Prices predicted by observables in timeseries model

Note: The red dashed line is the 45-degree line and the blue solid line represents the fitted predictions based on a generalized additive model with a cubic spline.

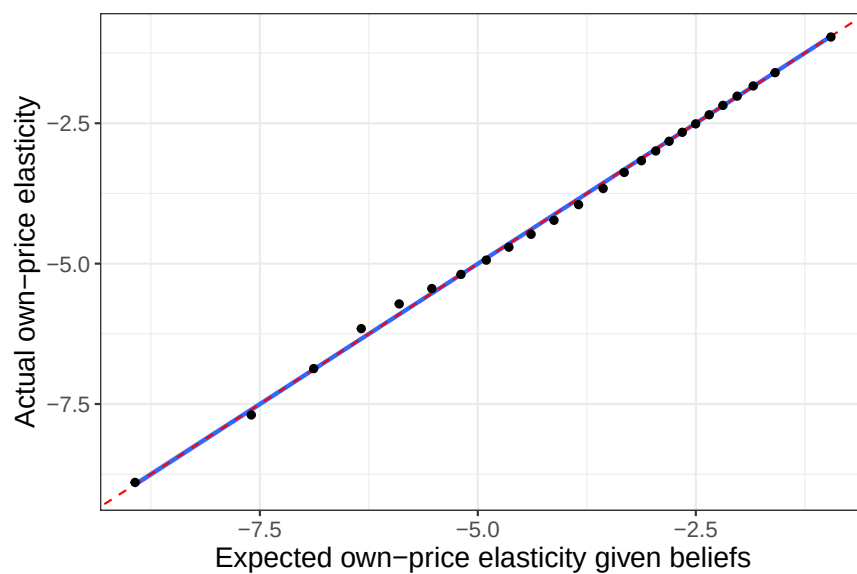
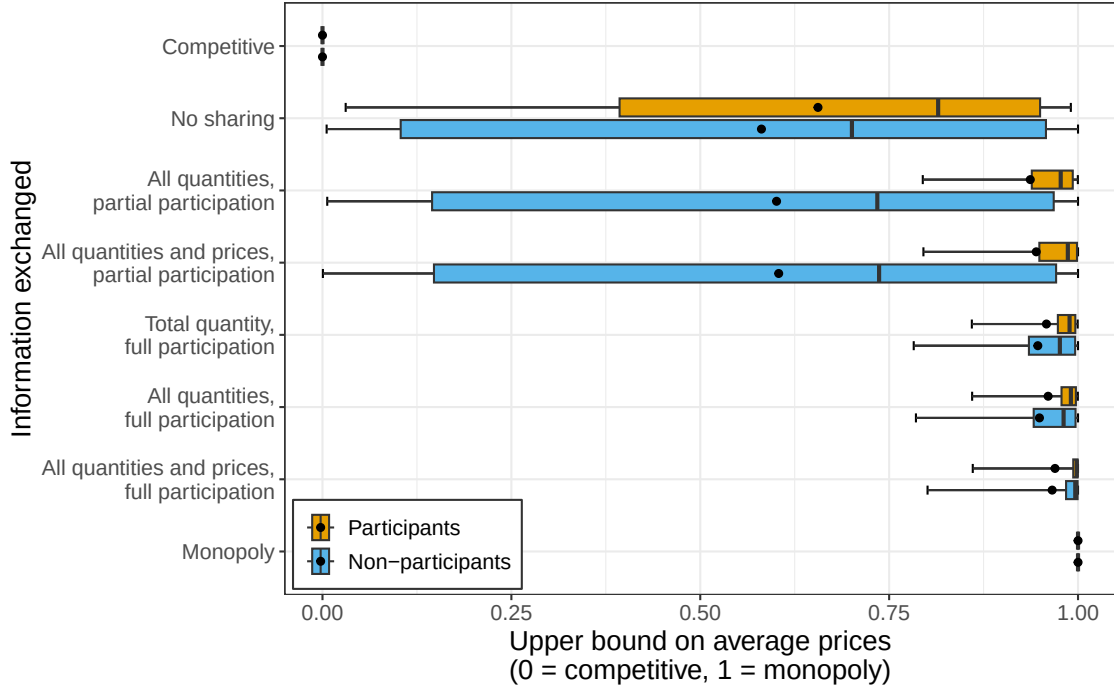
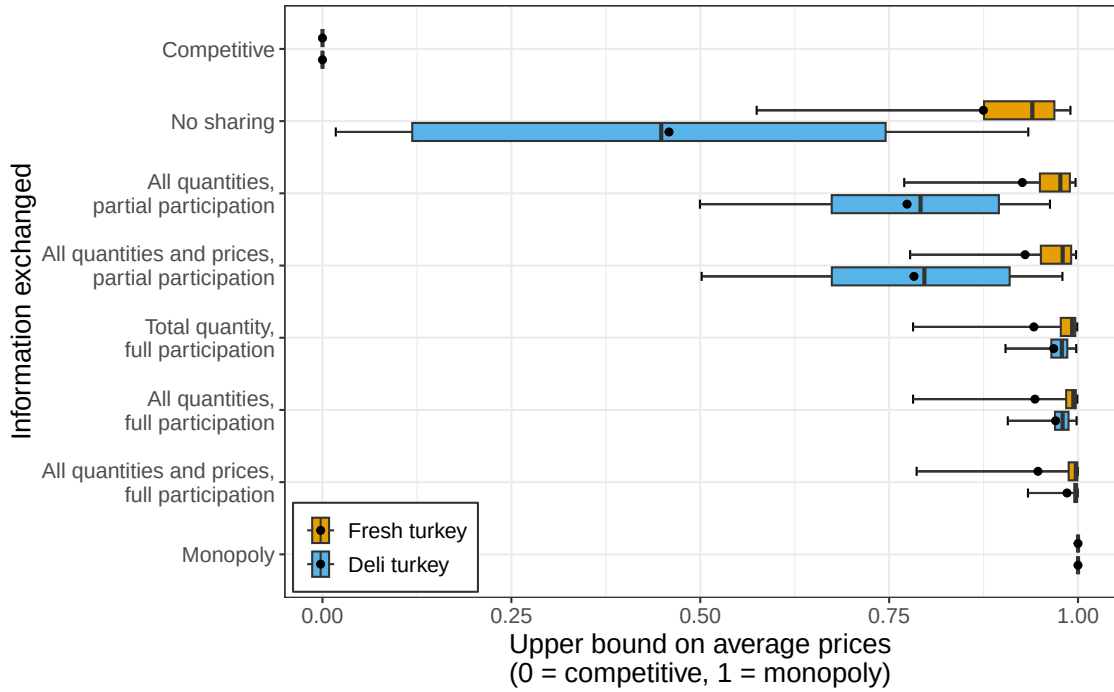


Figure A.7: Expected own-price elasticity against actual own-price elasticity

Note: Observations are divided into 25 equally sized bins based on the expected own-price elasticity implied by their producer's beliefs, with average expected and actual own-price elasticities then computed within bin. Expected own-price elasticities were computed using a quasi-Monte Carlo simulation with 1,000 draws for each firm-market. The red dashed line is the 45-degree line and the blue solid line is the line of best fit.



(a) Participants vs. non-participants



(b) Fresh vs. deli turkey

Figure A.8: Average price upper bounds by participation status and module

Note: I compute the upper bound on average prices within each market (region-module), then plot the distribution across markets. The whiskers are the 10th and 90th percentiles, the edges of the box are the 25th and 75th percentiles, the middle line is the median, and the point is the average. Participants are involved in all exchanges, while non-participants are only involved in those with full participation.

Scenario	Average prices			Profits (\$m/year)			CS (\$m/year)	TS (\$m/year)
	AS	Others	Total	AS	Others	Total		
<i>A. Fresh turkey</i>								
No sharing (actual)	3.54	3.61	3.57	129.0	53.5	182.5	265.6	448.1
Information effect	3.44	3.60	3.49	126.7	52.2	178.9	277.0	455.9
Info + conduct effect	3.68	3.63	3.66	135.3	54.7	190.0	254.3	444.3
<i>B. Deli turkey</i>								
No sharing (actual)	6.09	6.10	6.09	81.1	28.1	109.2	130.5	239.8
Information effect	6.10	6.10	6.10	81.3	28.1	109.4	130.3	239.7
Info + conduct effect	6.29	6.10	6.24	82.0	28.6	110.6	124.2	234.8

Table A.1: Agri Stats effects by module

Note: Average prices are weighted by no-sharing (actual) total quantity. In all scenarios except no sharing, Agri Stats clients share prices, quantities, costs, and inventories, and other producers just observe their own prices and quantities. In the “Information effect” scenario, firms maximize their own expected profits. In the “Info + conduct effect” scenario, Agri Stats clients maximize (15) using the estimated conduct parameter, while other producers continue to maximize their own expected profits.

Info exchanged	Share of markets w/ prob <...				
	0.0001	0.0005	0.0010	0.0050	0.0100
No sharing	0.820	0.890	0.904	0.934	0.947
All quantities, partial participation	0.833	0.895	0.908	0.930	0.934
All quantities and prices, partial participation	0.864	0.904	0.912	0.934	0.939
Total quantity, full participation	0.939	0.943	0.947	0.947	0.952
All quantities, full participation	0.904	0.930	0.930	0.943	0.943
All quantities and prices, full participation	0.947	0.961	0.965	0.965	0.969
Total	0.885	0.920	0.928	0.942	0.947

Table A.2: Distribution of smallest probability across upper-bound calculations

Note: Each cell shows the unweighted share of markets (region-module combinations) where the smallest probability on any node is less than the probability in the column, given the information environment in the row.

B Data and computation details

B.1 Data processing

I begin with scanner data from two NielsenIQ product modules: “fresh meat” and “lunchmeat-sliced-refrigerated.” I identify UPCs pertaining to chicken and turkey products using string searches on the native UPC descriptions; I exclude UPCs that contain multiple kinds of meats, such as ham and turkey deli combo packs. I also use the UPC descriptions to define NielsenIQ data fields in a consistent manner, since private-label UPCs typically do not have populated product description fields.

I aggregate prices and sales quantities by the pound. While most UPCs have their sales units denoted in pounds, some UPCs have units denominated in counts rather than in weight. I impute these products’ weight based on their price by first computing the average per-unit price of each weight of product with at least a 1% sales share in the same month-region-retailer combination, and then imputing each UPC’s weight as the one that minimizes the mean-squared difference between that product’s price and the average price of products of that size across month-region-retailers in which that UPC is found. I then aggregate the data to the UPC-month-store level by computing sales-weighted average prices. From there, I omit “value-added” processors by removing all UPCs made by firms that are not linked to a poultry processing plant—as it is then unclear whether their turkey originates from an Agri Stats subscriber—removing less than 1% of the sample by volume in this way. For private-label products, I identify the original processor from publicly available sources and drop those whose processor I cannot determine. I also remove all stores that are not featured in the data in each year from 2017 to 2022, and deflate all prices to May 2019 values using the CPI-U. This results in the dataset used for the reduced-form analyses.

I next geocode all poultry processing plants and find the driving distance between them and each store in which a product from that poultry processing plant’s owner was sold. The most granular geographic information about stores in the NielsenIQ data is their three-digit ZIP code, or ZIP3; I compute distances to the centroid of the store’s ZIP3, unless the centroid does not lie close enough to a road for the Google Maps API to compute driving distances to it, in which case I manually adjust the point to the closest road to the centroid. I then compute driving distances between each plant-centroid pair using the Google Maps API, and compute the driving distance for each firm to each store under the assumption that firms supply stores using the nearest available plant. In keeping with the relatively fast spoilage of meat products, I assume that firms can only use the set of licensed plants in a given month to supply sales in that month; that is, if a firm opens or closes a plant, that

affects the driving distances contemporaneously. Occasionally, firms have sales before they have a processing plant or after their processing plants drop out of the data; in these cases, I take the set of available processing plants as their first or last one in the data respectively.

To move to the dataset used for structural estimation, I start by collapsing to the region-product-month level, where a product is defined as a unique value of brand- $\mathbb{1}(\text{breast meat})$ - $\mathbb{1}(\text{whole smoked turkey})$ for fresh turkey and brand- $\mathbb{1}(\text{low fat})$ for deli turkey, and removing all regions without at least one inside product purchase in each month. I then remove all region-products that never have an inside share of at least 1% in any month, and remove region-product-months with an inside share of less than 0.01%, as these are likely products that are either discontinued mid-month or not widely offered. I define Gandhi-Houde instruments as in the main text, using a random forest to compute the exogenous price variable $\mathbb{E}[p_{jrt} \mid x_{jrt}, z_{jrt}]$. I then identify inside purchases in the panel data and construct sample weights for computing micro-moments using NielsenIQ’s native projection weights, accounting for multiple panelist purchases in a given month by splitting a panelist’s weight across their purchases.

B.2 Estimation

B.2.1 Timeseries

Here I present more detail on my implementation of the [Blundell and Bond \(1998\)](#) estimator for the timeseries estimation of (12). I follow the [Sigmund and Ferstl \(2021\)](#) extension to panel VAR settings. The demand shock relation to estimate is

$$\xi_{rt} = \bar{\xi}_r + B^\xi \xi_{r(t-1)} + e_{rt}^\xi. \quad (21)$$

The [Blundell and Bond \(1998\)](#) estimator combines moments from two equations: a transformed (difference) equation instrumented by lagged levels of the outcome, and a level equation instrumented by lagged differences of the outcome. For any variable x , let Δx_{rt} denote its forward orthogonal deviation ([Sigmund and Ferstl, 2021](#)), and let $\mathbb{E}[\cdot]$ denote the expectation across regions r . Because ξ_{rt} and e_{rt}^ξ are J -vectors (where J is the number of products), each condition below is a $J \times J$ matrix of moments, with one scalar moment for each pairing of an instrument product and an outcome equation. The valid moments are then, for any period t and any lag $\ell \geq 1$,

$$\begin{aligned} \mathbb{E} \left[\xi_{r(t-\ell)} (\Delta e_{rt}^\xi)' \right] &= 0, \quad (\text{difference equation}) \\ \mathbb{E} \left[\Delta \xi_{r(t-\ell)} (e_{rt}^\xi)' \right] &= 0. \quad (\text{level equation}) \end{aligned} \quad (22)$$

The instruments for the difference equation expand triangularly in the number of time periods, and many are weak (Roodman, 2009). I therefore restrict to the first two lags (i.e., $\ell \leq 2$) and collapse these moments following Roodman (2009), summing each lag's moment across the periods t at which it applies. The level equation instead uses only the most recent lagged difference as an instrument in each period, so its moments grow linearly rather than triangularly in the number of periods and require no such reduction. The final set of moments I use contains the sample analogues of

$$\begin{aligned} \sum_t \mathbb{E} \left[\xi_{r(t-\ell)} (\Delta e_{rt}^\xi)' \right] &= 0, \quad \ell \in \{1, 2\} \\ \mathbb{E} \left[\Delta \xi_{r(t-1)} (e_{rt}^\xi)' \right] &= 0 \quad \forall t. \end{aligned} \tag{23}$$

I estimate $\bar{\xi}_r$ and B^ξ using two-step GMM. While the direct implementation requires a balanced panel, the presence of regional products (and, to a lesser extent, changes in product assortment over time) results in observation of an unbalanced panel. As recommended by Roodman (2009), I balance the panel by imputing missing values of ξ_{rt} as zeros, which results in valid moments.

To estimate Σ^ξ , I first compute fitted residuals $\hat{e}_{rt}^\xi$ given the panel estimates, and compute pairwise covariances of $\hat{e}_{jrt}^\xi, \hat{e}_{krt}^\xi$ for all pairs of products (j, k) , treating the covariance as zero if the two products have fewer than two markets in common. As this may not result in a positive definite covariance matrix due to differences in product assortment across markets, I then project this matrix onto the space of positive definite matrices using the following procedure. Letting $\bar{\Sigma}^\xi$ be the matrix of pairwise covariances, I eigendecompose $\bar{\Sigma}^\xi = Q\Lambda Q'$. I then floor eigenvalues in Λ at a very small positive number, the result of which is Λ_+ , and back out $\hat{\Sigma}^\xi = Q\Lambda_+Q'$. This gives the closest positive definite and symmetric covariance matrix to $\bar{\Sigma}^\xi$ in the Frobenius norm, which I then take into the remaining estimation steps.

Computation of the price timeseries follows similar steps, with three differences. First, the contemporaneous price depends on other observables nonlinearly through the function $g(w_{rt}, z_{rt}^{GH})$. I use the semiparametric two-step estimator of Robinson (1988) by residualizing p_{rt} and $\xi_{r(t-1)}$ against nonparametric functions of (w_{rt}, z_{rt}^{GH}) using a random forest and computing the Blundell and Bond (1998) estimator on the residuals. I also add a time trend and an Agri Stats indicator, although the latter is eventually absorbed by the product-region fixed effect $\bar{p}_r$. Second, there is the additional linear component in the slope matrix for $\xi_{r(t-1)}$. I instrument this regressor in the level equation with its lagged differences, adding the moment $\mathbb{E} [\Delta \xi_{r(t-1)} (e_{rt}^p)'] = 0$ for all t . Third, I estimate the relation separately before and after May 2019, as I assume firms play a different Markov equilibrium in pricing rules

in each period.

B.2.2 Beliefs

To estimate beliefs, I first derive the stationary beliefs implied by the timeseries estimates. I treat (ξ, p) jointly as one random vector and write $B = \begin{pmatrix} B^\xi & 0 \\ B_\xi^p & B_p^p \end{pmatrix}$ as the matrix of VAR slope coefficients and $\Sigma = \text{BlockDiag}(\Sigma^\xi, \Sigma^p)$ as the covariance matrix of the innovations of that vector. The distribution at the very beginning of the sample is given by $(\xi_{r1}, p_{r1}) \sim \mathcal{N}(\mu_{r1}, \Sigma_{r1})$, where

$$\mu_{r1} = (I - B)^{-1} \begin{pmatrix} \bar{\xi}_r \\ \bar{p}_r + g(w_{r1}, z_{r1}^{GH}) \end{pmatrix} \quad (24)$$

$$\Sigma_{r1} = \{\Sigma_r : B\Sigma_r B' - \Sigma_r + \Sigma = 0\} \quad (25)$$

where Σ_{r1} solves the discrete-time Lyapunov equation. I seed initial beliefs for all firms in the first period as $\mu_{f r1} = \mu_{r1}$, $\Sigma_{f r1} = \Sigma_{r1}$. Then, fixing a region r and for all periods $t = 1, \dots, T$, where t^* corresponds to May 2019 and T corresponds to May 2022, I use the following procedure to iteratively compute beliefs.

1. For firm f producing in market rt , let $\mathcal{F}_{f rt}^{\text{obs}} = \{f' : f' \in AS \setminus \{f\}\}$ if $f \in AS$ and $t \leq t^*$; otherwise, let $\mathcal{F}_{f rt}^{\text{obs}} = \emptyset$.
2. For all $j = 1, \dots, J_{rt}$ such that $j \in \mathcal{J}_{f' rt}$ for some $f' \in \mathcal{F}_{f rt}^{\text{obs}} \cup \{f\}$: replace the $(J + j)$ th element of $\mu_{f rt}$ with $p_{j rt}$ and the $(J + j)$ th row and column of $\Sigma_{f rt}$ with zero. This corresponds to the firm knowing for sure that the price of good j will equal $p_{j rt}$. (J is added because the first J components correspond to values of ξ .)
3. If $f \in AS$ and $t \leq t^*$, define the signal functions $y_f(\xi_{rt}, p_{rt}) = (p_{AS, rt}, s_{AS, rt}(p_{rt}, \xi_{rt}))$; otherwise, define $y_f(\xi_{rt}, p_{rt}) = (p_{f rt}, s_{f rt}(p_{rt}, \xi_{rt}))$. Define the realizations of the signal analogously.
4. Given $(\mu_{f rt}, \Sigma_{f rt}, y_{f rt})$, compute $(\mu_{f rt}^{\text{post}}, \Sigma_{f rt}^{\text{post}})$ using Laplace's method, according to which

$$\mu_{f rt}^{\text{post}} = \arg \min_{(\xi, p)} ((\xi, p) - \mu_{f rt})' \Sigma_{f rt}^+ ((\xi, p) - \mu_{f rt}) \quad \text{s.t.} \quad y_f(\xi, p_{-f}) = y_{f rt}$$

$$\Sigma_{f rt}^{\text{post}} = U(U' \Sigma_{f rt}^+ U)^{-1} U'$$

where (with an obvious abuse of notation) $y_f(\xi, p_{-f}) = y_f(p_f, p_{-f}, s_{f rt}(p_f, p_{-f}, \xi))$ is firm f 's signal as a function of ξ and p_{-f} , and U is an orthonormal basis of the tangent space of the constraint manifold $\{(\xi, p_{-f}) : y_f(\xi, p_{-f}) = y_{f rt}\}$, evaluated at $\mu_{f rt}^{\text{post}}$. The

pseudoinverse Σ_{frt}^+ is used because Σ_{frt} is generally singular; all rows and columns corresponding to prices firm f knows in advance (which include at least its own prices) will be zero. Here, the pseudoinverse corresponds to taking the inverse of the submatrix associated with nondegenerate elements of firm f 's prior, while keeping the degenerate elements equal to zero.

5. Compute $(\mu_{fr(t+1)}, \Sigma_{fr(t+1)})$ according to the following:

$$\begin{aligned}\mu_{fr(t+1)} &= \begin{pmatrix} \bar{\xi}_r \\ \bar{p}_r + g(w_{r(t+1)}, z_{r(t+1)}^{GH}) \end{pmatrix} + B\mu_{frt}^{\text{post}} \\ \Sigma_{fr(t+1)} &= B\Sigma_{frt}^{\text{post}}B' + \Sigma\end{aligned}$$

If a new firm enters a particular region after period 1, I initialize that firm's beliefs at the stationary beliefs, unless that firm is an Agri Stats client and it is before May 2019, in which case I assign it the same beliefs as all other Agri Stats clients in that region in that period.

Finally, given that beliefs are persistent over time, one approximation error or particularly strange set of observed values can affect all future downstream beliefs. To lessen the chances of this happening, in each period I winsorize each firm's prior beliefs such that its mean belief over a particular variable x (where x is an element of either p or ξ) falls within $[\underline{x} - 3\sigma_x, \bar{x} + 3\sigma_x]$, where $\underline{x}$ and $\bar{x}$ are the minimum and maximum values of x in the data, and σ_x is its sample standard deviation. For prices, I additionally floor the mean belief at zero.

B.2.3 Marginal costs

Once firm f 's beliefs are computed, I can recover its costs under Bertrand-Markov from its first-order conditions, (11). For a given firm f in market rt , I first take M draws $(\xi_{rt}^m, p_{rt}^m)_{m=1}^M$ from $\mathcal{N}(\mu_{frt}, \Sigma_{frt})$ using quasi-Monte Carlo methods (shifted Halton draws with antithetic variates). I then find c_{frt} using

$$c_{frt} = p_{frt} + \left(\frac{1}{M} \sum_{m=1}^M D_{p_f} s_{frt}(p_{rt}^m, \xi_{rt}^m)' \right)^{-1} \left(\frac{1}{M} \sum_{m=1}^M s_{frt}(p_{rt}^m, \xi_{rt}^m) \right) \quad (26)$$

where $D_{p_f} s_{frt}(p_{rt}^m, \xi_{rt}^m) = \left[\frac{\partial s_{jrt}(p_{rt}^m, \xi_{rt}^m)}{\partial p_{krt}} \right]_{j,k \in \mathcal{J}_{frt}}$. Applying this formula for all f and rt such that $t > t^*$ gives all post-May 2019 costs. To compute pre-May 2019 costs under a particular value of τ for Agri Stats subscribers, I use the fact that Agri Stats clients have common

priors while sharing information by solving for their costs jointly, according to

$$c_{AS,rt} = p_{AS,rt} + \left(\mathcal{H}_{AS,rt}(\tau) \odot \frac{1}{M} \sum_{m=1}^M D_{p_{AS}} s_{AS,rt}(p_{rt}^m, \xi_{rt}^m)' \right)^{-1} \left(\frac{1}{M} \sum_{m=1}^M s_{AS,rt}(p_{rt}^m, \xi_{rt}^m) \right) \quad (27)$$

where $\mathcal{H}_{AS,rt}(\tau)$ is the within-Agri Stats ownership matrix—element jk equals 1 if products j and k are produced by the same firm, and τ otherwise—and $\odot$ is the element-by-element multiplication operator. Other firms' costs follow (26).

Finally, to estimate the cost function $h(w_{jrt})$ and recover the residuals ω_{jrt} , I first residualize all variables against product, region, and month (i.e., time) fixed effects, and then apply a random forest to the residuals. I do so jointly in the pre- and post-periods, separately for each value of τ .

B.2.4 Bootstrap

There are three sources of asymptotic variance in my estimation pipeline: the demand parameters, the timeseries parameters (which in part depend on the demand parameters, through ξ), and the cost function. To account for asymptotic variation in the demand parameters, I use a parametric bootstrap by pulling draws from the asymptotic distribution of the parameters implied by the two-step GMM robust covariance matrix, and then re-solving for the values of ξ that make predicted and actual shares match for each draw of the parameters. Then, for each draw of the parameters, I account for other sources of asymptotic variation by drawing regions with replacement, following the approach of [Kapetanios \(2008\)](#) approach for bootstrapping panel estimation routines with timeseries dependence. I construct 500 such joint draws of demand parameters and sets of regions after estimating demand, then run each draw through the preceding estimation steps and the conduct-testing routine. This process provides the standard errors I report in the main text.

B.3 Counterfactual Markov equilibria in pricing rules

For a given profile of signal functions $\{y_f\}$, computing a counterfactual Markov equilibrium in pricing rules requires finding a long-run price kernel Π^+ that is long-run consistent in the sense of (9): if firms conjecture that rivals' prices follow Π^+ , set their prices to maximize their expected profits, and update their beliefs given the signals those prices realize, then the induced long-run law of motion of prices must reproduce Π^+ . As described in Section 8, this is a fixed point of the map $\Pi^+ \mapsto \Phi(\Pi^+)$, where Φ returns the long-run law of motion

of the expected-profit-maximizing prices generated by the beliefs that the conjectured Π^+ induces. I recover this fixed point by iterating on Φ : given a conjectured kernel, I simulate the post-May 2019 prices and beliefs it implies, reestimate the kernel from those simulated prices, and repeat until the conjectured and reestimated kernels coincide.

In practice, the parameterization of the price timeseries I use is defined by a tuple $(g(\cdot), \bar{p}_r, B^p, \Sigma^p)$. I use a random forest to estimate $g(\cdot)$, which is not continuous in the inputs, and hence would make convergence difficult if reestimated within the fixed-point iteration. I therefore take the approach of only adjusting $g(\cdot)$ from the estimated function by a constant for each product-region combination. As any constant shift in $g(\cdot)$ is isomorphic to a shift in the product-region fixed effect $\bar{p}_r$, this is equivalent to keeping $g(\cdot)$ fixed in counterfactuals and only recomputing $(\bar{p}_r, B^p, \Sigma^p)$ for each counterfactual iteration. Holding $g(\cdot)$ fixed, I stack the remaining free parameters into a single vector

$$\Pi^+ = (\{\bar{p}_r\}_{r=1}^R, B_\xi^p, B_p^p, \Sigma^p), \quad (28)$$

where $B^p = (B_\xi^p \ B_p^p)$ is partitioned conformably with $(\xi_{r(t-1)}, p_{r(t-1)})$ as in Appendix B.2.2. One evaluation of Φ unpacks Π^+ , simulates the implied equilibrium prices and beliefs market by market, and reestimates (12) on the simulated prices using system GMM to return $\Phi(\Pi^+)$. A counterfactual equilibrium is a zero of the residual $G(\Pi^+) = \Phi(\Pi^+) - \Pi^+$. Because B^p and Σ^p are pooled across regions, evaluating Φ couples all markets, so the fixed point cannot be solved region by region.

To compute equilibrium prices in a particular market rt given the vector of costs c_{rt} and a system of prior beliefs $\{\mu_{f_{rt}}, \Sigma_{f_{rt}}\}_{f \in \mathcal{F}_{rt}}$, I adapt the [Morrow and Skerlos \(2011\)](#) fixed-point iteration for this setting. Decompose $D_p s_{rt}(p_{rt}; \xi_{rt}) = \Lambda_{rt}(p_{rt}, \xi_{rt}) - \Gamma_{rt}(p_{rt}, \xi_{rt})$, where Λ_{rt} is a diagonal $J \times J$ matrix with the j th element of the diagonal equal to

$$\Lambda_{jrt}(p_{rt}, \xi_{rt}) = \int \alpha_i(\theta) s_{ijrt}(p_{rt}; \xi_{rt}, \theta) dF_{rt}(\theta) \quad (29)$$

and Γ_{rt} is a dense $J \times J$ matrix with jk th element equal to

$$\Gamma_{jkrt}(p_{rt}, \xi_{rt}) = \int \alpha_i(\theta) s_{ijrt}(p_{rt}; \xi_{rt}, \theta) s_{ikrt}(p_{rt}; \xi_{rt}, \theta) dF_{rt}(\theta) \quad (30)$$

where $F_{rt}(\theta)$ is the distribution of structural demand parameters (α_i, β_i) in market rt . One may then rewrite the first-order condition for firm f as $p_{f_{rt}} = c_{f_{rt}} + \zeta_{f_{rt}}(p_{f_{rt}})$, where

$$\zeta_{f_{rt}}(p_{f_{rt}}) = \mathbb{E}_{f_{rt}} [\Lambda_{f_{rt}}(p_{rt}, \xi_{rt})]^{-1} (\mathbb{E}_{f_{rt}} [\Gamma_{f_{rt}}(p_{rt}, \xi_{rt})'] (p_{f_{rt}} - c_{f_{rt}}) - \mathbb{E}_{f_{rt}} [s_{f_{rt}}(p_{rt}; \xi_{rt})]). \quad (31)$$

Here, $\Lambda_{f_{rt}}(\cdot)$ and $\Gamma_{f_{rt}}(\cdot)$ are the submatrices of $\Lambda_{rt}(\cdot)$ and $\Gamma_{rt}(\cdot)$ associated with firm f 's products. The advantage of writing it in this way is that $\zeta_{f_{rt}}$ is a local contraction, and hence iterating on $p_{f_{rt}} \leftarrow c_{f_{rt}} + \zeta_{f_{rt}}(p_{f_{rt}})$ reliably converges to prices satisfying firm f 's first-order conditions. To apply the method, I compute expectations with respect to firm f_{rt} 's beliefs using quasi-Monte Carlo, replacing the draws corresponding to firm f 's prices as appropriate. If a set of firms S share cost information, then I replace $\mathbb{E}_{f_{rt}}[\Lambda_{f_{rt}}(\cdot)]$ with $\mathbb{E}_{S_{rt}}[\Lambda_{S_{rt}}(\cdot)]$ and $\mathbb{E}_{f_{rt}}[\Gamma_{f_{rt}}(\cdot)']$ with $\mathbb{E}_{S_{rt}}[\mathcal{H}_{S_{rt}} \odot \Gamma_{S_{rt}}(\cdot)']$, where $\mathcal{H}_{S_{rt}}$ is the within- S ownership matrix and $\mathbb{E}_{S_{rt}}[\cdot]$ is the operator that, in row j , computes the expectation of the matrix with respect to the beliefs of the firm that owns product j . Finally, given a set of equilibrium prices in market rt , I compute posterior beliefs using Laplace's method, and then compute priors in the following period using the estimated law of motion of ξ and the candidate Π^+ , as in Appendix B.2.2. Looping over all markets gives the full set of equilibrium prices, and then $\Phi(\Pi^+)$ is computed using system GMM (Appendix B.2.1).

My baseline solver is a damped fixed-point iteration $\Pi_{k+1}^+ = (1 - q)\Pi_k^+ + q\Phi(\Pi_k^+)$, iterated until the largest change across the $(B_\xi^p, B_p^p, \Sigma^p)$ blocks of Π^+ , or the change in equilibrium prices across iterates, falls below a tolerance. I have found that damping ($q < 1$) helps stabilize the iteration at the cost of convergence speed, although Φ is not guaranteed to be a contraction. Note that while generally Markov equilibria in pricing rules are separable by region (but not time periods), the way in which I parameterize the equilibrium law of motion of prices makes the computation of counterfactual Markov equilibria in pricing rules completely nonseparable across regions or markets. This is due to the common slopes B^p and innovation covariance matrix Σ^p . As such, there may not exist a Π^+ that makes $\Phi(\Pi^+) = \Pi^+$ in practice. Nevertheless, for all the counterfactuals I consider, damped fixed-point iteration either finds a Π^+ such that the sup-norm $\|\Phi(\Pi^+) - \Pi^+\|_\infty$ is less than 0.001, or settles in a basin in which changes to Π^+ vary equilibrium prices by at most 1 cent.

B.4 Upper bound

This appendix details the computation of the collusive upper bound described in Section 8.1. Throughout I condition on a single market and suppress the rt subscript. Each firm f enters with the belief $\xi \sim \mathcal{N}(\mu_f^\xi, \Sigma_f^\xi)$, derived from the stationary belief implied by the timeseries estimates. Thus,

$$\mathbb{E}_f[\pi_f(p)] = \mathbb{E}_{\xi \sim \mathcal{N}(\mu_f^\xi, \Sigma_f^\xi)} [(p_f - c_f)' s_f(p, \xi)] \quad (32)$$

is firm f 's expected profit given price vector p . As in Appendix B.2.2, I approximate all expectations against firm f 's beliefs using a quasi-Monte Carlo average over M draws $\{\xi_f^m\}_{m=1}^M$ from $\mathcal{N}(\mu_f^\xi, \Sigma_f^\xi)$, generated with shifted Halton sequences and antithetic variates. Finally,

let $\kappa = \delta/(1 - \delta)$, and let p^C and p^M be the competitive and monopoly prices, which I solve for using the iteration scheme described in Appendix B.3.

B.4.1 The discretized problem

I approximate the occupation measure α by a finite mixture over N price nodes, $\alpha = \{(p_n, a_n)\}_{n=1}^N$, with $p_n \in \mathbb{R}^J$ and probability weights $a_n \geq 0$ summing to one. Given price weights proportional to actual shares $\psi_j \geq 0$ with $\sum_j \psi_j = 1$, the program is

$$\alpha^*(y) = \arg \max_{(p_n, a_n)_{n=1}^N} \sum_{n=1}^N a_n \sum_j \psi_j p_{n,j} \quad (33)$$

$$\begin{aligned} \text{s.t.} \quad & \sum_{n=1}^N a_n = 1, \quad a_n \geq 0, \quad p_n \in [p^C, p^M] \quad \forall n, \\ & C_{n,f}^{\text{Nash}}(\alpha) \leq 0 \quad \forall f, \forall n, \end{aligned} \quad (34)$$

$$C_f^{\chi^2}(\alpha) \leq \kappa \text{Var}_f(\alpha) \quad \forall f, \quad (35)$$

$$C_f^{TV}(\alpha) \leq 0 \quad \forall f, \quad (36)$$

where $\text{Var}_f(\alpha) = \text{Var}_{p \sim \alpha}(\mathbb{E}_f[\pi_f(p)])$. The objective (33) is bilinear, with gradient $\partial/\partial a_n = \sum_j \psi_j p_{n,j}$ and $\partial/\partial p_{n,j} = a_n \psi_j$. The constraints are defined below.

B.4.2 Nash IC (34)

Let the punishment value be $\pi_f^C = \mathbb{E}_f[\pi_f(p^C)]$, which is independent of the node. If firms are recommended price p_n , firm f 's best one-shot deviation profit is

$$\pi_f^D(p_n) = \max_{\tilde{p}_f} \mathbb{E}_f[\pi_f(\tilde{p}_f, p_{n,-f})], \quad (37)$$

the static best response to $p_{n,-f}$ under f 's belief. The value of not deviating at node n is $(1 - \delta)\mathbb{E}_f[\pi_f(p_n)] + \delta\bar{\pi}_f(\alpha)$, where $\bar{\pi}_f(\alpha) = \sum_n a_n \mathbb{E}_f[\pi_f(p_n)]$ is the average equilibrium profit. The constraint is that for every firm f and node n ,

$$C_{n,f}^{\text{Nash}}(\alpha) = (1 - \delta) (\pi_f^D(p_n) - \mathbb{E}_f[\pi_f(p_n)]) + \delta (\pi_f^C - \bar{\pi}_f(\alpha)) \leq 0. \quad (38)$$

I solve (37) using a modified [Morrow and Skerlos \(2011\)](#) iteration scheme, where only firm f 's prices are updated in each iteration.

The derivative of expected profits with respect to prices is standard:

$$\frac{\partial \mathbb{E}_f[\pi_f(p)]}{\partial p_j} = \frac{1}{M} \sum_{m=1}^M \left[\mathbb{1}\{j \in \mathcal{J}_f\} s_j(p, \xi^m) + \sum_{k \in \mathcal{J}_f} (p_k - c_k) \frac{\partial s_k(p, \xi^m)}{\partial p_j} \right]. \quad (39)$$

The derivatives of π_f^D are

$$\frac{\partial \pi_f^D(p_n)}{\partial p_{n,j}} = \mathbb{1}\{j \notin \mathcal{J}_f\} \frac{\partial \mathbb{E}_f[\pi_f(p_f^D(p_n), p_{n,-f})]}{\partial p_j} \quad (40)$$

where $p_f^D(p_n)$ is the argmax of (37). The derivative when $j \in \mathcal{J}_f$ is zero by optimality of $p_f^D(p_n)$. Differentiating (38), the gradients of the Nash IC constraints are

$$\begin{aligned} \frac{\partial C_{n,f}^{\text{Nash}}}{\partial a_{n'}} &= -\delta \mathbb{E}_f[\pi_f(p_{n'})], \\ \frac{\partial C_{n,f}^{\text{Nash}}}{\partial p_{n',j}} &= \begin{cases} (1 - \delta) \left(\frac{\partial \pi_f^D(p_n)}{\partial p_j} - \frac{\partial \mathbb{E}_f[\pi_f(p_n)]}{\partial p_j} \right) - \delta a_n \frac{\partial \mathbb{E}_f[\pi_f(p_n)]}{\partial p_j}, & n' = n, \\ -\delta a_{n'} \frac{\partial \mathbb{E}_f[\pi_f(p_{n'})]}{\partial p_j}, & n' \neq n. \end{cases} \end{aligned} \quad (41)$$

B.4.3 χ^2 -detectability constraint (35)

The function in the constraint (35) is defined as

$$C_f^{\chi^2}(\alpha) = \max_{m_f, q_f} \frac{(g_f(m_f, \alpha)_+)^2}{\chi_f^2(m_f, q_f, \alpha)} \quad (42)$$

where $k_+ = \max\{k, 0\}$, and the manipulation (m_f, q_f) , the gain $g_f(\cdot)$, and the χ^2 -detectability $\chi_f^2(\cdot)$ are to be defined. This corresponds to squaring both sides of the original constraint (18), while noting that the RHS is nonnegative and hence the constraint is satisfied vacuously whenever $g_f/\sqrt{\chi_f^2}$ is negative; the squared version appears to perform better numerically.

Given $\alpha = (p_n, a_n)_{n=1}^N$, let $K \leq N$ be the number of unique prices for firm f , and let $k(n)$ map the set of indices $\{1, \dots, N\}$ corresponding to a unique price for firm f to a single index; that is, $p_{n,f} = p_{n',f}$ if and only if $k(n) = k(n')$. An action manipulation $m_f = (\tilde{p}_{k,f})_{k=1}^K$ is a set of K $|\mathcal{J}_f|$ -vectors of prices, where the interpretation is that firm f sets price $\tilde{p}_{k(n),f}$

whenever it is supposed to set price $p_{n,f}$.⁵⁸ The expected gain from m_f is

$$g_f(m_f, \alpha) = \sum_{n=1}^N a_n (\mathbb{E}_f[\pi_f(\tilde{p}_{k(n),f}, p_{n,-f})] - \mathbb{E}_f[\pi_f(p_n)]). \quad (43)$$

To define χ_f^2 , I must first state the way in which I approximate the signals. The signal entering detectability is the vector of shares, and under different information exchanges a firm may be able to manipulate either the entire vector of its own shares, or the vector of its own shares up to the total share. Thus, the goal is to construct a bijective transformation of the vector of shares that is (a) as close to linear in the demand indices as possible so that the transformed version is approximately conditionally normal, and (b) such that both firm f 's vector of shares and its total share are manipulable without affecting the remaining shares, which are reported truthfully. I therefore approximate the signals as follows. Fix one reference product $j_f^* \in \mathcal{J}_f$ and $j_{-f}^* \in \mathcal{J}_{-f}$. Let $s_f = \sum_{j \in \mathcal{J}_f} s_j$ be the total share for firm f , and likewise s_{-f} is the total share of other firms. I define the signal $y = (y_j)_{j=1}^J$ as

$$y_j(s) = \begin{cases} \log(s_j/s_{j_f^*}), & j \in \mathcal{J}_f, j \neq j_f^*, \\ \log(s_f/(1-s_f)), & j = j_f^*, \\ \log(s_j/s_{j_{-f}^*}), & j \notin \mathcal{J}_f, j \neq j_{-f}^*, \\ \log(s_{-f}/(1-s_{-f})), & j = j_{-f}^*, \end{cases}$$

and a firm's signal y_f is simply the vector containing the signals of all of its products. If s_j comes from a plain logit model with a normally distributed ξ , then $y_j \mid p$ is precisely normally distributed for $j \notin \{j_f^*, j_{-f}^*\}$.⁵⁹ The reference products are nonlinear functions of ξ and therefore are not precisely normally distributed. However, the bijection that guarantees that $y_j \mid p$ is normally distributed for all j — $y_j = \log(s_j/s_0)$, where $s_0 = 1 - \sum_j s_j$ —means that rivals' shares enter into firm f 's signals, and therefore the meaning of a manipulated signal for firm f is unclear. Thus, I adopt the above-displayed transformation, and treat $y \mid p \sim \mathcal{N}(\mu(p), \Sigma(p))$, with $\mu(p)$ and $\Sigma(p)$ estimated via simulation: $\mu(p) = M^{-1} \sum_m y(s(p, \xi^m))$ and $\Sigma(p) = M^{-1} \sum_m (y(s(p, \xi^m)) - \mu(p))(y(s(p, \xi^m)) - \mu(p))'$.⁶⁰

⁵⁸The original result allows for manipulations to be distributions of prices. For computational reasons, I only consider deterministic action manipulations. This weakly loosens the upper bound.

⁵⁹The specific choice of reference product does not matter for detectability, as two different $y_j(s)$ with two reference products are bijective affine maps of each other, and hence the underlying divergences are identical for any choice of reference.

⁶⁰For one counterfactual—the full share vector being exchanged publicly—the transformation $y_j = \log(s_j/s_0)$ is valid. I use the same approximation for this counterfactual as other counterfactuals for consistency, but find that the choice of approximations does not matter much: the median and 90th percentile absolute

I next define a message manipulation q_f . In theory, a message manipulation is a function that, given a “recommended” price p_f , a deviation price $\tilde{p}_f$, and a received signal y_f , returns a distribution over manipulated signals $\tilde{y}_f$. In practice, allowing this to be an arbitrary function results in an infinite-dimensional problem. I therefore restrict attention to affine manipulations chosen separately for each own-price block: $q_f = (A_k, b_k, \Omega_k)_{k=1}^K$, where if firm f is recommended price $p_{k(n),f}$, sets price $\tilde{p}_{k(n),f}$, and receives signal y_f , it manipulates its signal according to $\tilde{y}_f = A_{k(n)}y_f + b_{k(n)} + \varepsilon$, with $\varepsilon \sim \mathcal{N}(0, \Omega_{k(n)})$ independent of y_f . Restricting the space of manipulations in this way results in a valid, if slightly weaker, upper bound, while preserving the distribution of manipulated signals as a Gaussian, and hence facilitates stating the χ^2 -divergence in closed form.⁶¹

Suppose that under a particular information structure, firm f can manipulate signals with indices $\mathcal{M}$. If firm f is not sharing information at all, then $\mathcal{M} = \mathcal{J}_f$; if total shares are shared, then $\mathcal{M} = \mathcal{J}_f \setminus \{j_f^*\}$; and if it shares its full share vector, then $\mathcal{M} = \emptyset$. Given a deviation price $\tilde{p} = (\tilde{p}_f, p_{-f})$ and a manipulation (A, b, Ω) for the corresponding recommended price, the “true” signal is distributed $y \mid \tilde{p} \sim \mathcal{N}(\mu(\tilde{p}), \Sigma(\tilde{p}))$ while the manipulated signal distribution is $\tilde{y} \mid \tilde{p} \sim \mathcal{N}(\tilde{\mu}(\tilde{p}), \tilde{\Sigma}(\tilde{p}))$, where

$$\begin{aligned}\tilde{\mu}_{\mathcal{M}}(\tilde{p}) &= A\mu_{\mathcal{M}}(\tilde{p}) + b, \\ \tilde{\Sigma}_{\mathcal{M},\mathcal{M}}(\tilde{p}) &= A\Sigma_{\mathcal{M},\mathcal{M}}(\tilde{p})A' + \Omega, \\ \tilde{\Sigma}_{\mathcal{M},-\mathcal{M}}(\tilde{p}) &= A\Sigma_{\mathcal{M},-\mathcal{M}}(\tilde{p}),\end{aligned}$$

where $\Sigma_{\mathcal{M},\mathcal{M}}$ and $\tilde{\Sigma}_{\mathcal{M},\mathcal{M}}$ are the submatrices that firm f can manipulate (with $\Sigma_{\mathcal{M},-\mathcal{M}}$ and $\tilde{\Sigma}_{\mathcal{M},-\mathcal{M}}$ analogously defined), and $\tilde{\mu}_{-\mathcal{M}}(\tilde{p}) = \mu_{-\mathcal{M}}(\tilde{p})$ and $\tilde{\Sigma}_{-\mathcal{M},-\mathcal{M}}(\tilde{p}) = \Sigma_{-\mathcal{M},-\mathcal{M}}(\tilde{p})$ are unaffected by (A, b, Ω) .

Fix an index n , and let $(\tilde{p}_{k(n),f}, A_{k(n)}, b_{k(n)}, \Omega_{k(n)})$ be all the relevant components of the manipulation (m_f, q_f) . Let $\mu_n = \mu(p_n)$ and $\Sigma_n = \Sigma(p_n)$ be the parameters of the signal distribution on path, $\check{\mu}_n = \mu(\tilde{p}_{k(n),f}, p_{n,-f})$ and $\check{\Sigma}_n = \Sigma(\tilde{p}_{k(n),f}, p_{n,-f})$ be the parameters of the

percentage difference in upper bounds produced by these two approximations are 0.01% and 0.11% respectively for this counterfactual.

⁶¹There is theoretical and computational evidence that this family of manipulations is nearly optimal. First, [Kreps \(2026\)](#) proves that a message manipulation that reports $\tilde{y}_f = y_f + \Delta(\tilde{p}_f - p_f)$ is optimal in a single-product log-linear demand system. This manipulation is affine with $A_{k(n)} = I$, $b_{k(n)} = \Delta(\tilde{p}_{k(n),f} - p_{k(n),f})$, and $\Omega_{k(n)} = 0$. Second, [Kreps \(2026\)](#) derives theoretical bounds on the detectability of a given manipulation: it is bounded below by (but does not necessarily attain) the detectability of m_f in rivals’ signals only and above by the detectability of m_f in all signals if firm f tells the truth (i.e., sets $A_{k(n)} = I$, $b_{k(n)} = 0$, and $\Omega_{k(n)} = 0$). I find that the optimal affine message manipulation nearly uniformly attains the theoretical lower bound: if rivals-only detectability and detectability under truth-telling are rescaled to 0 and 1 respectively, then the median and 90th percentile minimum detectability from an affine manipulation are 0.001 and 0.008 respectively with no information sharing. Thus, the potential loss from this restriction is theoretically limited.

unmanipulated signal distribution under the deviation, and $\tilde{\mu}_n = \tilde{\mu}(\tilde{p}_{k(n),f}, p_{n,-f})$ and $\tilde{\Sigma}_n = \tilde{\Sigma}(\tilde{p}_{k(n),f}, p_{n,-f})$ be the parameters of the manipulated signal distribution. Finally, define $\Lambda = 2\tilde{\Sigma}_n^{-1} - \Sigma_n^{-1}$, $\beta = 2\tilde{\Sigma}_n^{-1}\tilde{\mu}_n - \Sigma_n^{-1}\mu_n$, $\mu^* = \Lambda^{-1}\beta$, and $\gamma = 2\tilde{\mu}_n'\tilde{\Sigma}_n^{-1}\tilde{\mu}_n - \mu_n'\Sigma_n^{-1}\mu_n - \beta'\mu^*$. The χ^2 -divergence between the on-path and manipulated signal distributions is

$$\chi_{n,f}^2 = |\Sigma_n|^{1/2}|\tilde{\Sigma}_n|^{-1}|\Lambda|^{-1/2} \exp(-0.5\gamma) - 1. \quad (44)$$

If Λ is not positive definite, then $\chi_{n,f}^2$ diverges to $+\infty$. Then the overall χ^2 -detectability is

$$\chi_f^2(m_f, q_f, \alpha) = \sum_{n=1}^N a_n \chi_{n,f}^2, \quad (45)$$

where the dependence of $\chi_{n,f}^2$ on p_n , $\tilde{p}_{k(n),f}$, $A_{k(n)}$, $b_{k(n)}$, and $\Omega_{k(n)}$ is suppressed. Equations (43) and (45) define the functions entering into the maximization problem (42).

Solving (42). I solve it as a nested optimization problem. First, I solve for the optimal message manipulation given an action manipulation $q_f^*(m_f, \alpha) = \arg \min_{q_f} \chi_f^2(m_f, q_f, \alpha)$. This problem is separable in blocks k . Then, I solve for $m_f^*(\alpha) = \arg \max_{m_f} \frac{g_f(m_f, \alpha)}{\sqrt{\chi_f^2(m_f, q_f^*(m_f, \alpha), \alpha)}}$. Both problems are solved with a gradient-based limited-memory BFGS optimizer.

In the inner problem, I specify $\Omega_k = L_k L_k'$, for a lower-triangular matrix L_k , which guarantees positive semidefiniteness and symmetry of Ω_k . Thus, the choice set in each problem consists of (A_k, b_k, L_k) , which includes $|\mathcal{M}|^2 + |\mathcal{M}| + |\mathcal{M}|(|\mathcal{M}| + 1)/2$ parameters. (Recall that $|\mathcal{M}| \leq |\mathcal{J}_f|$ for a given firm f , so this is a relatively small number of parameters—it equals 3 for a single-product firm.) The gradient of the objective with respect to a given parameter $\theta_k \in \{A_k, b_k, L_k\}$ is

$$\frac{\partial \chi_f^2(m_f, q_f, \alpha)}{\partial \theta_k} = \sum_{n:k(n)=k} a_n \frac{\partial \chi_{n,f}^2}{\partial \theta_k},$$

with, letting $\bar{A}_k$ be the block-diagonal matrix that applies A_k to coordinates $\mathcal{M}$ and the

identity to the rest and $\tilde{u}_n = \tilde{\mu}_n - \mu^*$,

$$\begin{aligned}
\frac{\partial \chi_{n,f}^2}{\partial A_{k(n)}} &= \left(\frac{\partial \chi_{n,f}^2}{\partial \tilde{\mu}_n} \right)_{\mathcal{M}} \check{\mu}'_{n,\mathcal{M}} + 2 \left(\frac{\partial \chi_{n,f}^2}{\partial \tilde{\Sigma}_n} \bar{A}_{k(n)} \check{\Sigma}_n \right)_{\mathcal{M},\mathcal{M}}, \\
\frac{\partial \chi_{n,f}^2}{\partial b_{k(n)}} &= \left(\frac{\partial \chi_{n,f}^2}{\partial \tilde{\mu}_n} \right)_{\mathcal{M}}, \\
\frac{\partial \chi_{n,f}^2}{\partial L_{k(n)}} &= 2 \left(\frac{\partial \chi_{n,f}^2}{\partial \tilde{\Sigma}_n} \right)_{\mathcal{M},\mathcal{M}} L_{k(n)}, \\
\frac{\partial \chi_{n,f}^2}{\partial \tilde{\mu}_n} &= -2(1 + \chi_{n,f}^2) \tilde{\Sigma}_n^{-1} \tilde{u}_n, \\
\frac{\partial \chi_{n,f}^2}{\partial \tilde{\Sigma}_n} &= (1 + \chi_{n,f}^2) \tilde{\Sigma}_n^{-1} (\Lambda^{-1} - \tilde{\Sigma}_n + \tilde{u}_n \tilde{u}_n') \tilde{\Sigma}_n^{-1}.
\end{aligned} \tag{46}$$

Only the lower triangle of $\partial \chi_{n,f}^2 / \partial L_{k(n)}$ enters the gradient, as the upper triangle of $L_{k(n)}$ is fixed at zero. This problem is relatively cheap to compute, as the true simulated signal values implied by the deviation are held fixed throughout. The main starting value matches the conditional distribution of the manipulated coordinates given the truthfully reported coordinates to its on-path counterpart; that is, with all moments evaluated at a representative node n of block k , $B = \Sigma_{n,\mathcal{M},-\mathcal{M}} \Sigma_{n,-\mathcal{M},-\mathcal{M}}^{-1}$, and $\check{B} = \check{\Sigma}_{n,\mathcal{M},-\mathcal{M}} \check{\Sigma}_{n,-\mathcal{M},-\mathcal{M}}^{-1}$,

$$\begin{aligned}
A_k &= B \check{B}^+, \\
b_k &= \mu_{n,\mathcal{M}} - A_k \check{\mu}_{n,\mathcal{M}} + B(\check{\mu}_{n,-\mathcal{M}} - \mu_{n,-\mathcal{M}}), \\
\Omega_k &= (\Sigma_{n,\mathcal{M},\mathcal{M}} - B \Sigma_{n,-\mathcal{M},\mathcal{M}}) - A_k \left(\check{\Sigma}_{n,\mathcal{M},\mathcal{M}} - \check{B} \check{\Sigma}_{n,-\mathcal{M},\mathcal{M}} \right) A_k',
\end{aligned}$$

with $\check{B}^+$ denoting the Moore-Penrose pseudoinverse, Ω_k projected to PSD, and L_k extracted from Ω_k using a Cholesky decomposition. This choice makes detectability arise only from the marginal distribution of the truthfully reported coordinates, and attains the theoretical floor of rivals-only detectability if m_f simply mean-shifts the signals. To ensure robustness to local minima, I also start this problem at truthful reporting ($A_k = I, b_k = 0, L_k = 0$), at the solution at the previous iterate if available, at the value that makes rivals' expected posterior match the on-path expected posterior (which is more robust to situations in which Ω_k is not PSD before it is projected), and at convex combinations of these starting values.

I then import the solution $q_f^*(m_f, \alpha)$ to (42), which optimizes over $m_f = (\tilde{p}_{k,f})_{k=1}^K$. The envelope theorem implies that $\frac{\partial \chi_f^2(m_f, q_f^*(m_f, \alpha), \alpha)}{\partial m_f} = \frac{\partial \chi_f^2(m_f, q_f, \alpha)}{\partial m_f} \Big|_{q_f = q_f^*(m_f, \alpha)}$. Let $g_{n,f} =$

$\mathbb{E}_f[\pi_f(\tilde{p}_{k(n),f}, p_{n,-f})] - \mathbb{E}_f[\pi_f(p_n)]$, so that $g_f = \sum_n a_n g_{n,f}$. The gradients are

$$\begin{aligned}
\frac{\partial \frac{g_f}{\sqrt{\chi_f^2}}}{\partial \tilde{p}_{k,j}} &= \frac{\sum_{n:k(n)=k} a_n \frac{\partial g_{n,f}}{\partial \tilde{p}_{n,j}}}{\sqrt{\chi_f^2}} - \frac{g_f \sum_{n:k(n)=k} a_n \frac{\partial \chi_{n,f}^2}{\partial \tilde{p}_{n,j}}}{2\chi_f^2 \sqrt{\chi_f^2}}, \\
\frac{\partial g_{n,f}}{\partial \tilde{p}_{n,j}} &= \frac{\partial \mathbb{E}_f[\pi_f(\tilde{p}_{n,f}, p_{n,-f})]}{\partial p_j} \quad (\text{see (39)}), \\
\frac{\partial \chi_{n,f}^2}{\partial \tilde{p}_{n,j}} &= \frac{\partial \chi_{n,f}^2}{\partial \tilde{\mu}_n} \cdot \frac{\partial \tilde{\mu}_n}{\partial \tilde{p}_{n,j}} + \left\langle \frac{\partial \chi_{n,f}^2}{\partial \tilde{\Sigma}_n}, \frac{\partial \tilde{\Sigma}_n}{\partial \tilde{p}_{n,j}} \right\rangle, \\
\frac{\partial \tilde{\mu}_{n,i}}{\partial \tilde{p}_{n,j}} &= \sum_{\ell=1}^J \bar{A}_{k(n),i\ell} \frac{\partial \mu_\ell(\tilde{p}_{n,f}, p_{n,-f})}{\partial p_j}, \\
\frac{\partial \tilde{\Sigma}_{n,i\ell}}{\partial \tilde{p}_j} &= \sum_{r=1}^J \sum_{s=1}^J \bar{A}_{k(n),ir} \bar{A}_{k(n),\ell s} \frac{\partial \Sigma_{rs}(\tilde{p}_{n,f}, p_{n,-f})}{\partial p_j}, \\
\frac{\partial \mu_i(p)}{\partial p_j} &= \frac{1}{M} \sum_{m=1}^M \frac{\partial y_{i,m}(p)}{\partial p_j}, \\
\frac{\partial \Sigma_{i\ell}(p)}{\partial p_j} &= \frac{1}{M} \sum_{m=1}^M \left(\frac{\partial y_{i,m}(p)}{\partial p_j} y_{\ell,m}(p) + \frac{\partial y_{\ell,m}(p)}{\partial p_j} y_{i,m}(p) \right) - \frac{\partial \mu_i(p)}{\partial p_j} \mu_\ell(p) - \frac{\partial \mu_\ell(p)}{\partial p_j} \mu_i(p), \\
y_{j,m}(p) &= y_j(s(p, \xi^m)), \\
\frac{\partial y_{i,m}(p)}{\partial p_j} &= \begin{cases} \frac{\partial s_i(p, \xi^m)/\partial p_j}{s_i(p, \xi^m)} - \frac{\partial s_{j_f^*}(p, \xi^m)/\partial p_j}{s_{j_f^*}(p, \xi^m)}, & i \in \mathcal{J}_f, i \neq j_f^*, \\ \frac{\sum_{\ell \in \mathcal{J}_f} \partial s_\ell(p, \xi^m)/\partial p_j}{(1 - \sum_{\ell \in \mathcal{J}_f} s_\ell(p, \xi^m)) (\sum_{\ell \in \mathcal{J}_f} s_\ell(p, \xi^m))}, & i = j_f^*, \\ \frac{\partial s_i(p, \xi^m)/\partial p_j}{s_i(p, \xi^m)} - \frac{\partial s_{j_{-f}^*}(p, \xi^m)/\partial p_j}{s_{j_{-f}^*}(p, \xi^m)}, & i \notin \mathcal{J}_f, i \neq j_{-f}^*, \\ \frac{\sum_{\ell \in \mathcal{J}_{-f}} \partial s_\ell(p, \xi^m)/\partial p_j}{(1 - \sum_{\ell \in \mathcal{J}_{-f}} s_\ell(p, \xi^m)) (\sum_{\ell \in \mathcal{J}_{-f}} s_\ell(p, \xi^m))}, & i = j_{-f}^*, \end{cases} \quad (47)
\end{aligned}$$

where $\langle A, B \rangle = \sum_{ij} A_{ij} B_{ij}$ is the Frobenius inner product. This problem is also initialized at multiple points: the solution at the previous iterate, the competitive price, the monopoly price, and four points between the competitive and monopoly prices. Solving this problem results in the optimal deviation $m_f^*(\alpha)$ and the implied optimal message manipulation $q_f^*(\alpha) = q_f^*(m_f^*(\alpha), \alpha)$.

Finally, the solution to this problem enters into the overall constraint, (35), along with the profit variance

$$\text{Var}_f(\alpha) = \sum_{n=1}^N a_n (\mathbb{E}_f[\pi_f(p_n)])^2 - \left(\sum_{n=1}^N a_n \mathbb{E}_f[\pi_f(p_n)] \right)^2. \quad (48)$$

As before, the envelope theorem implies that the partial derivatives of $C_f^{\chi^2}(\alpha)$ with respect to elements of α are equal to the partial derivatives of the maximand $(g_f(m_f, \alpha)_+)^2 / \chi_f^2(m_f, q_f, \alpha)$, evaluated at $m_f^*(\alpha), q_f^*(\alpha)$. Evaluating all functions at this point, omitting arguments for readability, and defining $u_n = \mu_n - \mu^*$, I obtain

$$\begin{aligned}
\frac{\partial C_f^{\chi^2}}{\partial a_n} &= 2 \frac{g_f}{\chi_f^2} g_{n,f} - \left(\frac{g_f}{\chi_f^2} \right)^2 \chi_{n,f}^2, \\
\frac{\partial C_f^{\chi^2}}{\partial p_{n,j}} &= a_n \left(2 \frac{g_f}{\chi_f^2} \cdot \frac{\partial g_{n,f}}{\partial p_{n,j}} - \left(\frac{g_f}{\chi_f^2} \right)^2 \frac{\partial \chi_{n,f}^2}{\partial p_{n,j}} \right), \\
\frac{\partial g_{n,f}}{\partial p_{n,j}} &= \mathbb{1}\{j \notin \mathcal{J}_f\} \frac{\partial \mathbb{E}_f[\pi_f(\tilde{p}_{k(n),f}, p_{n,-f})]}{\partial p_{n,j}} - \frac{\partial \mathbb{E}_f[\pi_f(p_n)]}{\partial p_{n,j}} \quad (\text{see (39)}), \\
\frac{\partial \chi_{n,f}^2}{\partial p_{n,j}} &= \mathbb{1}\{j \notin \mathcal{J}_f\} \left(\frac{\partial \chi_{n,f}^2}{\partial \tilde{\mu}_n} \cdot \frac{\partial \tilde{\mu}_n}{\partial p_{n,j}} + \left\langle \frac{\partial \chi_{n,f}^2}{\partial \tilde{\Sigma}_n}, \frac{\partial \tilde{\Sigma}_n}{\partial p_{n,j}} \right\rangle \right) + \frac{\partial \chi_{n,f}^2}{\partial \mu_n} \cdot \frac{\partial \mu(p_n)}{\partial p_j} + \left\langle \frac{\partial \chi_{n,f}^2}{\partial \Sigma_n}, \frac{\partial \Sigma(p_n)}{\partial p_j} \right\rangle, \\
\frac{\partial \chi_{n,f}^2}{\partial \mu_n} &= (1 + \chi_{n,f}^2) \Sigma_n^{-1} u_n, \\
\frac{\partial \chi_{n,f}^2}{\partial \Sigma_n} &= \frac{1}{2} (1 + \chi_{n,f}^2) (\Sigma_n^{-1} - \Sigma_n^{-1} (\Lambda^{-1} + u_n u_n') \Sigma_n^{-1}), \\
\frac{\partial \text{Var}_f}{\partial a_n} &= \mathbb{E}_f[\pi_f(p_n)]^2 - 2 \mathbb{E}_f[\pi_f(p_n)] \sum_{n'=1}^N a_{n'} \mathbb{E}_f[\pi_f(p_{n'})], \\
\frac{\partial \text{Var}_f}{\partial p_{n,j}} &= 2 a_n \left(\mathbb{E}_f[\pi_f(p_n)] - \sum_{n'=1}^N a_{n'} \mathbb{E}_f[\pi_f(p_{n'})] \right) \frac{\partial \mathbb{E}_f[\pi_f(p_n)]}{\partial p_j},
\end{aligned}$$

with $\frac{\partial \mu(p)}{\partial p_j}$ and $\frac{\partial \Sigma(p)}{\partial p_j}$ defined as in (47).

B.4.4 TV-detectability constraint (36)

C_f^{TV} is defined as

$$C_f^{TV}(\alpha) = \max_{\tilde{p}_f} \sum_n a_n \left(\mathbb{E}_f[\pi_f(\tilde{p}_f, p_{n,-f})] - \mathbb{E}_f[\pi_f(p_n)] - 2\sqrt{2}\kappa \|\pi_f\|_\infty H_{-f}(p_n, \tilde{p}_f) \right) \quad (49)$$

where the profit difference is the expected gain from a constant deviation to $\tilde{p}_f$, and $H_{-f}(p_n, \tilde{p}_f)$ is the Hellinger distance between the distribution of the signals observed by firm f 's rivals under p_n and under $(\tilde{p}_f, p_{n,-f})$. This is implied by the original condition in [Kreps \(2026\)](#), which involves total variation distance $TV(\cdot)$, as $TV(\cdot) \leq \sqrt{2}H(\cdot)$; imposing the Hellinger version therefore weakly loosens the upper bound, and the same strategy is used in [Awaya and Krishna \(2020\)](#) to solve for bounds implied by [Awaya and Krishna \(2019\)](#). I set $\|\pi_f\|_\infty$,

the sup-norm of firm f 's profits, to its monopoly profits if it were the only firm that produced anything; equivalently, $\|\pi_f\|_\infty = \lim_{p_{-f} \rightarrow \infty} \max_{p_f} \mathbb{E}_f[\pi_f(p_f, p_{-f})]$.

Define the rival-observed signal $y_{-f}(s)$ as a function of the share vector s as follows. First, rivals always observe their own shares, s_{-f} . Second, if total quantities are publicly shared, then rivals observe the total share of firm f , $s_f = \sum_{j \in \mathcal{J}_f} s_j$.⁶² Third, if firm f shares all its shares with any of its rivals, then the vector of firm f 's shares, $(s_j)_{j \in \mathcal{J}_f}$, is included in the signal, so that the signal contains the full share vector s . Let $v_{-f}(s)$ be the function collecting the shares, or aggregates of shares, that firm f 's rivals can observe. Then, the transformed signal entering detectability is defined as

$$y_{-f}(s) = \log(v_{-f}(s)) - \log(1 - \mathbf{1}'v_{-f}(s))$$

where $\mathbf{1}$ is a conformable vector of ones. This is the additive log ratio transformation, which means that when $v_{-f}(s) = s$ and there are no random coefficients, $y_{-f}(s)$ is an affine transformation of ξ and hence is conditionally normally distributed. This transformation is usable here but not for the χ^2 -detectability condition because the TV condition involves no message manipulation, and hence does not imply a measurability restriction on which signals firm f can affect. To compute this signal, I treat $y_{-f} \mid p \sim \mathcal{N}(\mu(p), \Sigma(p))$ with $\mu(p) = M^{-1} \sum_m y_{-f}(s(p, \xi^m))$ and $\Sigma(p) = M^{-1} \sum_m (y_{-f}(s(p, \xi^m)) - \mu(p))(y_{-f}(s(p, \xi^m)) - \mu(p))'$. Then, detectability of the deviation $\tilde{p}_f$ from p_n is the Hellinger distance between $\mathcal{N}(\mu(p_n), \Sigma(p_n))$ and $\mathcal{N}(\mu(\tilde{p}_f, p_{n,-f}), \Sigma(\tilde{p}_f, p_{n,-f}))$:

$$H_{-f}(p_n, \tilde{p}_f) = \sqrt{1 - \exp(-D_B(\mu(p_n), \Sigma(p_n), \mu(\tilde{p}_f, p_{n,-f}), \Sigma(\tilde{p}_f, p_{n,-f})))},$$

$$D_B(\mu_1, \Sigma_1, \mu_2, \Sigma_2) = \frac{1}{8}(\mu_1 - \mu_2)' \left(\frac{\Sigma_1 + \Sigma_2}{2} \right)^{-1} (\mu_1 - \mu_2) + \frac{1}{2} \log \left(\frac{|(\Sigma_1 + \Sigma_2)/2|}{\sqrt{|\Sigma_1||\Sigma_2|}} \right), \quad (50)$$

where $D_B(\cdot)$ is the Bhattacharyya distance between two normals.

The inner problem over $\tilde{p}_f$ is solved using a gradient-based limited-memory BFGS solver. Let $C_f^{TV}(\alpha, \tilde{p}_f)$ be the maximand, $\mu_n = \mu(p_n)$, $\Sigma_n = \Sigma(p_n)$, $\tilde{\mu}_n = \mu(\tilde{p}_f, p_{n,-f})$, $\tilde{\Sigma}_n = \Sigma(\tilde{p}_f, p_{n,-f})$, and $d = \dim \mu_n$. Finally, let $\mathcal{J}_k^v$ be the set of products whose share is included in signal k ; this contains a single product for all signals associated with rivals' shares, as well as for firm f 's share if it shares its quantities, while it includes all of firm f 's products for

⁶²This is isomorphic to firm f 's rivals actually observing the total inside share, as firm f 's share can be backed out as the total inside share less all other shares.

the signal corresponding to its total share. The gradients of the inner problem are

$$\begin{aligned}
\frac{\partial C_f^{TV}}{\partial \tilde{p}_j} &= \sum_{n=1}^N a_n \left(\frac{\partial \mathbb{E}_f[\pi_f(\tilde{p}_f, p_{n,-f})]}{\partial p_j} - 2\sqrt{2}\kappa \|\pi_f\|_\infty \frac{\partial H_{-f}(p_n, \tilde{p}_f)}{\partial \tilde{p}_j} \right), \\
\frac{\partial H_{-f}}{\partial \tilde{p}_j} &= \frac{\exp(-D_B(\mu_n, \Sigma_n, \tilde{\mu}_n, \tilde{\Sigma}_n))}{2H_{-f}(p_n, \tilde{p}_f)} \sum_{k=1}^d \left(\frac{\partial \mu_k(\tilde{p}_f, p_{n,-f})}{\partial p_j} \cdot \frac{\partial D_B(\mu_n, \Sigma_n, \tilde{\mu}_n, \tilde{\Sigma}_n)}{\partial \mu_{2,k}} + \right. \\
&\quad \left. \sum_{\ell=1}^d \frac{\partial \Sigma_{k\ell}(\tilde{p}_f, p_{n,-f})}{\partial p_j} \cdot \frac{\partial D_B(\mu_n, \Sigma_n, \tilde{\mu}_n, \tilde{\Sigma}_n)}{\partial \Sigma_{2,k\ell}} \right), \\
\frac{\partial D_B}{\partial \mu_2} &= \frac{1}{4} \left(\frac{\Sigma_1 + \Sigma_2}{2} \right)^{-1} (\mu_2 - \mu_1), \\
\frac{\partial D_B}{\partial \Sigma_2} &= \frac{1}{4} \left(\left(\frac{\Sigma_1 + \Sigma_2}{2} \right)^{-1} - \Sigma_2^{-1} \right) - \frac{1}{16} \left(\frac{\Sigma_1 + \Sigma_2}{2} \right)^{-1} (\mu_2 - \mu_1)(\mu_2 - \mu_1)' \left(\frac{\Sigma_1 + \Sigma_2}{2} \right)^{-1}, \\
\frac{\partial \mu_k(p)}{\partial p_j} &= \frac{1}{M} \sum_{m=1}^M \frac{\partial y_{k,m}}{\partial p_j}, \\
\frac{\partial \Sigma_{k\ell}(p)}{\partial p_j} &= \frac{1}{M} \sum_{m=1}^M \left(\frac{\partial y_{k,m}(p)}{\partial p_j} y_{\ell,m} + \frac{\partial y_{\ell,m}(p)}{\partial p_j} y_{k,m}(p) \right) - \frac{\partial \mu_k(p)}{\partial p_j} \mu_\ell(p) - \frac{\partial \mu_\ell(p)}{\partial p_j} \mu_k(p), \\
y_{k,m}(p) &= y_{-f,k}(s(p, \xi^m)), \\
\frac{\partial y_{k,m}(p)}{\partial p_j} &= \frac{\sum_{r \in \mathcal{J}_k^v} \partial s_r(p, \xi^m) / \partial p_j}{v_k(s(p, \xi^m))} + \frac{\sum_{r \in \mathcal{J}^v} \partial s_r(p, \xi^m) / \partial p_j}{1 - \mathbf{1}' v_{-f}(s(p, \xi^m))},
\end{aligned}$$

where $\mathcal{J}^v = \cup_k \mathcal{J}_k^v$ collects the products entering any component of the signal, v_k is the k -th component of v_{-f} , and $\partial \mathbb{E}_f[\pi_f(p)] / \partial p_j$ is defined as in (39). I take as starting values the solution to the prior iterate if available, the competitive price, the monopoly price, and the Bertrand best response at the highest-weight node.

Then, the gradients of the function entering into the constraint are

$$\begin{aligned}
\frac{\partial C_f^{TV}}{\partial a_n} &= \mathbb{E}_f[\pi_f(\tilde{p}_f, p_{n,-f})] - \mathbb{E}_f[\pi_f(p_n)] - 2\sqrt{2}\kappa \|\pi_f\|_\infty H_{-f}(p_n, \tilde{p}_f), \\
\frac{\partial C_f^{TV}}{\partial p_{n,j}} &= a_n \left(\mathbb{1}\{j \notin \mathcal{J}_f\} \frac{\partial \mathbb{E}_f[\pi_f(\tilde{p}_f, p_{n,-f})]}{\partial p_j} - \frac{\partial \mathbb{E}_f[\pi_f(p_n)]}{\partial p_j} - 2\sqrt{2}\kappa \|\pi_f\|_\infty \frac{\partial H_{-f}(p_n, \tilde{p}_f)}{\partial p_j} \right), \\
\frac{\partial H_{-f}}{\partial p_j} &= \frac{\exp(-D_B(\mu_n, \Sigma_n, \tilde{\mu}_n, \tilde{\Sigma}_n))}{2H_{-f}(p_n, \tilde{p}_f)} \sum_{k=1}^d \left(\frac{\partial \mu_{n,k}}{\partial p_j} \cdot \frac{\partial D_B}{\partial \mu_{1,k}} + \mathbb{1}\{j \notin \mathcal{J}_f\} \frac{\partial \tilde{\mu}_{n,k}}{\partial p_j} \cdot \frac{\partial D_B}{\partial \mu_{2,k}} + \right. \\
&\quad \left. \sum_{\ell=1}^d \left(\frac{\partial \Sigma_{n,k\ell}}{\partial p_j} \cdot \frac{\partial D_B}{\partial \Sigma_{1,k\ell}} + \mathbb{1}\{j \notin \mathcal{J}_f\} \frac{\partial \tilde{\Sigma}_{n,k\ell}}{\partial p_j} \cdot \frac{\partial D_B}{\partial \Sigma_{2,k\ell}} \right) \right),
\end{aligned}$$

with all functions evaluated at the optimal $\tilde{p}_f$ solving the inner problem, $\partial \mu_{n,k} / \partial p_j =$

$\partial\mu_k(p_n)/\partial p_j$, $\partial\tilde{\mu}_{n,k}/\partial p_j = \partial\mu_k(\tilde{p}_f, p_{n,-f})/\partial p_j$, $\partial\Sigma_{n,k\ell}/\partial p_j = \partial\Sigma_{k\ell}(p_n)/\partial p_j$, $\partial\tilde{\Sigma}_{n,k\ell}/\partial p_j = \partial\Sigma_{k\ell}(\tilde{p}_f, p_{n,-f})/\partial p_j$, and $\partial D_B/\partial\mu_1$ and $\partial D_B/\partial\Sigma_1$ the same as $\partial D_B/\partial\mu_2$ and $\partial D_B/\partial\Sigma_2$ with the indices switched.

Delayed signals. If part of the signal is observed after a delay of L periods, then the results of [Kreps \(2026\)](#) implies that I can replace $C_f^{TV}(\alpha)$ in the constraint with $C_{f,\text{eff}}^{TV}(\alpha)$, which depends on the *effective detectability*

$$H_{-f,\text{eff}}(p, \tilde{p}_f) = (1 - \delta^L)H_{-f}(p, \tilde{p}_f; y^0) + \delta^L H_{-f}(p, \tilde{p}_f),$$

where $H_{-f}(\cdot; y^0)$ computes detectability assuming firms only observe the contemporaneously observed signal. In practice, firms always observe their own shares contemporaneously, so detectability in period 0 is computed with the shares entering the signal being s_{-f} . The baseline analysis assumes the information exchange occurs contemporaneously, and hence this formula collapses to the original constraint; sensitivities incorporating delay assume that any exchanged information is observed after two months. Finally, notice that although the same substitution is available for the χ^2 -detectability constraint if part of the signal is delayed, it has no bite. If firms lie about a signal that is then reported truthfully in a future period, then the χ^2 analogue of effective detectability is infinite. Thus, firms optimally tell the truth about their signals contemporaneously, and χ^2 -detectability and effective χ^2 -detectability are equal for the delayed-signal sensitivities considered here.

C Robustness checks

This section discusses results under alternative assumptions. For the robustness checks described in [Appendices C.1–C.4](#), I reestimate demand (if applicable), the timeseries process (if applicable), beliefs, and marginal costs under the Bertrand-Markov assumption. I then use the same procedure as in the main text to estimate the discontinuity in $\mathbb{E}[\omega_{jrt}]$ for Agri Stats clients and other producers at May 2019, and use this to compute the difference-in-discontinuities. Finally, I reestimate the conduct parameter that makes the estimated difference-in-discontinuities equal to zero. [Table C.1](#) summarizes the results. While I discuss results specific to each sensitivity below, I ultimately conclude that the conduct test is robust to alternative assumptions about the information structure and equilibrium. In all sensitivities, the estimated discontinuity for Agri Stats clients is statistically distinguishable from zero at the 5% level, while the estimated discontinuity for non-subscribers is statistically insignificant throughout. The difference-in-discontinuities and the conduct parameter estimate are also statistically significant for all sensitivities except optimal demand instru-

Sensitivity	Discontinuity in ω		Difference	Conduct parameter
	AS clients	Others		
Baseline	0.213 [0.071, 0.341]	0.000 [-0.088, 0.059]	0.213 [0.060, 0.371]	0.475 [0.178, 0.724]
<i>A. Information without sharing</i>				
Average price and total quantity	0.285 [0.116, 0.405]	-0.029 [-0.158, 0.049]	0.315 [0.131, 0.478]	0.582 [0.299, 0.746]
All prices and quantities	0.256 [0.104, 0.387]	-0.025 [-0.140, 0.033]	0.281 [0.126, 0.469]	0.546 [0.311, 0.800]
<i>B. Inventory signal with sharing</i>				
$\lambda = 0.1$	0.192 [0.062, 0.324]	0.003 [-0.085, 0.060]	0.188 [0.057, 0.343]	0.459 [0.191, 0.722]
$\lambda = 0.25$	0.171 [0.048, 0.300]	0.008 [-0.088, 0.064]	0.163 [0.036, 0.317]	0.443 [0.144, 0.756]
$\lambda = 0.5$	0.151 [0.023, 0.286]	0.012 [-0.077, 0.070]	0.139 [0.012, 0.293]	0.418 [0.057, 0.786]
<i>C. Transition between equilibria</i>				
3 months	0.214 [0.092, 0.340]	-0.008 [-0.079, 0.041]	0.222 [0.086, 0.353]	0.490 [0.261, 0.707]
6 months	0.163 [0.092, 0.295]	-0.018 [-0.074, 0.041]	0.181 [0.085, 0.318]	0.418 [0.222, 0.659]
12 months	0.126 [0.082, 0.275]	-0.028 [-0.086, 0.037]	0.154 [0.069, 0.298]	0.366 [0.205, 0.644]
<i>D. Initial conditions</i>				
Mean $- 1$ SD	0.221 [0.086, 0.350]	0.017 [-0.077, 0.058]	0.204 [0.073, 0.373]	0.457 [0.205, 0.696]
Mean $+ 1$ SD	0.214 [0.080, 0.337]	0.000 [-0.092, 0.057]	0.214 [0.076, 0.375]	0.472 [0.219, 0.709]
Variance $\times 0.5$	0.212 [0.078, 0.351]	0.000 [-0.086, 0.058]	0.213 [0.075, 0.370]	0.474 [0.215, 0.709]
Variance $\times 2$	0.212 [0.081, 0.339]	-0.001 [-0.087, 0.065]	0.214 [0.074, 0.377]	0.472 [0.209, 0.706]
<i>E. Optimal demand instruments</i>	0.068 [0.008, 0.142]	0.018 [-0.045, 0.064]	0.050 [-0.027, 0.143]	0.268 [0.000, 0.782]

Table C.1: Robustness of conduct results to alternative assumptions

Note: Brackets contain the 95% confidence intervals, computed with 500 bootstrap draws for the baseline and 200 bootstrap draws for sensitivities. The conduct parameter $\tau^* \in [0, 1]$ is estimated to minimize the square of the left-hand side of (14), assuming Agri Stats subscribers maximized the objective in (15). Discontinuities are estimated using a local linear regression before and after the cutoff, with an Epanechnikov kernel and a bandwidth computed to minimize placebo discontinuities over June 2017–February 2019 and December 2019–May 2022 separately for each sensitivity.

ments. Finally, Appendix C.5 discusses reduced-form evidence from my data that variation in vertical structure is unlikely to play a large role in mediating the Agri Stats effect.

C.1 Information structure

C.1.1 Information without sharing

The baseline assumption is that while not sharing information, firms only observe their own prices and shares. This assumption is defensible: according to litigation, firms paid millions of dollars for access to the Agri Stats reports and testified that their detail and timeliness are unmatched, while retailers attempted to purchase the reports and were denied (hence there must be additional information in the reports not available to retailers). However, it is possible that without Agri Stats, firms would be able to ascertain aggregates of past prices and shares from government indices, or more detailed data by purchasing retail scanner data.

I continue to assume that Agri Stats shared the full vector of prices and shares among its subscribers, and that firms observe their own prices and shares. In the first sensitivity, I also assume that all firms have access to average prices and total inside shares; that is, they append to their own prices and shares the signal

$$y_{f_{rt}}^{\text{gov}}(p_{rt}, s_{rt}) = \left(\frac{1}{J_{rt}} \sum_{j \in \mathcal{J}_{rt}} p_{jrt}, \sum_{j \in \mathcal{J}_{rt}} s_{jrt} \right). \quad (51)$$

In the second sensitivity, I assume that all firms have access to the full vector of past prices and quantities; that is, $y_{f_{rt}}(p_{rt}, s_{rt}) = (p_{rt}, s_{rt})$ for all f_{rt} . In this case, the only informational benefit of Agri Stats is informing firms about rivals' costs and inventories.

Panel A of Table C.1 shows the results with these assumptions. These alternative assumptions strengthen the results, increasing the estimated difference-in-discontinuities and conduct parameter.

C.1.2 Noisy within-group price conjectures

The baseline analysis assumes that firms share full information about costs, enabling Agri Stats clients to perfectly conjecture their fellow subscribers' prices. In this subsection, I weaken this assumption by assuming that the shared cost information yields a noisy signal ι_{jrt} of the price rather than pinning it down exactly. In particular, the signal firms receive is ι , not ω , and firms conjecture that

$$\iota_{jrt} \mid p_{jrt} \sim \mathcal{N}(p_{jrt}, \lambda \Sigma_{jj}^p) \quad (52)$$

where Σ_{jj}^p is the jj entry of the estimated covariance matrix of price innovations. Firms update their priors accordingly. Then, $p_{jrt} = \iota_{jrt}$ is realized.⁶³ The variance of the signal is proportional to the diagonal entries of Σ^p , so different values of λ can be interpreted as the imprecision of the conjecture firms form from the shared cost information relative to the empirical variance of prices over time.

Suppose Agri Stats client f , while sharing information, enters period t with the prior $(\xi_{rt}, p_{(-f)rt}) \sim \mathcal{N}(\mu_{frt}, \Sigma_{frt})$. Let $\mu_{frt}^{AS \setminus f}$ and $\Sigma_{frt}^{AS \setminus f}$ be the subvector and submatrix associated with other Agri Stats subscribers' prices, let $\Sigma_{frt}^{\circ, AS \setminus f}$ be the covariance block between those prices and the rest of the state (that is, ξ_{rt} and non-subscribers' prices), and let $\Sigma_{AS \setminus f}^p$ be the diagonal of the submatrix of Σ^p corresponding to other Agri Stats clients' products. Define the gain $G_{frt} = \Sigma_{frt}^{AS \setminus f} \left(\Sigma_{frt}^{AS \setminus f} + \lambda \Sigma_{AS \setminus f}^p \right)^{-1}$. Then, before setting p_{frt} , firm f updates its prior over $p_{AS \setminus f, rt}$ to $\mathcal{N}(\mu_{frt}^\lambda, \Sigma_{frt}^\lambda)$, where

$$\begin{aligned}\mu_{frt}^\lambda &= \mu_{frt}^{AS \setminus f} + G_{frt} \left(p_{AS \setminus f, rt} - \mu_{frt}^{AS \setminus f} \right), \\ \Sigma_{frt}^\lambda &= \Sigma_{frt}^{AS \setminus f} - G_{frt} \Sigma_{frt}^{AS \setminus f},\end{aligned}$$

and adjusts the covariance between these prices and the rest of the state to $\Sigma_{frt}^{\circ, AS \setminus f} (I - G'_{frt})$, holding the marginal belief over ξ_{rt} and non-subscribers' prices fixed.

The baseline model corresponds to $\lambda = 0$, where $G_{frt} = I$: the firm learns $p_{AS \setminus f, rt}$ exactly, and both Σ_{frt}^λ and the covariances of these prices with the rest of the state collapse to zero. As λ increases, the conjecture becomes noisier. As in the baseline, the price conjecture does not directly inform the firm about non-sharing firms' prices or ξ ; the firm does not revise the marginal beliefs over these objects, and the cross-covariances shrink only as much as is required to keep the firm's beliefs consistent with its sharpened belief over subscriber prices.

Panel B in Table C.1 displays the results with $\lambda = 0.1, 0.25$, and 0.5 . While a less precise conjecture slightly lowers the conduct parameter estimate, Bertrand-Markov conduct and $\tau^* = 0$ continue to be rejected.

C.2 Transition between equilibria

The moment condition is a difference-in-discontinuities at May 2019, meaning local conditions around May 2019 are important. While firms' beliefs vary continuously around this

⁶³I note two caveats about this exercise. First, I emphasize the assumption that realizations of ι_{jrt} equal p_{jrt} ; it implies that firms' priors over their rivals' prices always move towards the actual prices, rather than ever moving away due to an unlucky signal draw (the chances of which vanish as $\lambda \rightarrow 0$). Second, while sharing information and when $\tau > 0$, firms still incorporate rivals' profits into their objective as though they perfectly knew $\{\omega_{f'rt}\}_{f' \in \mathcal{F}_f^{\text{obs}}}$. Thus, for the purposes of this exercise, firms that share information perfectly know their rivals' profit function, but only imperfectly conjecture their rivals' prices.

date, in the sense that June 2019 beliefs are derived from May 2019 beliefs despite them coming from separate equilibria, the binary assumption of two separate equilibria with different price Markov processes before and after May 2019 may be driving some of the estimated discontinuity. Although the distributions of the slope and innovation matrices appear similar before and after (Table 6), there may also be differences in the intercepts, or in particularly high-leverage slope coefficients.

To assess the sensitivity of the conduct tests to the transition between pricing processes, I smooth the transition between the estimated price timeseries. Let t^* correspond to May 2019, and $T > 0$ be an integer-valued transition length. For $t \in [t^* + 1, t^* + T]$, I assume that firms' conjectured Markov pricing rules are based on the linear mixture of the estimated pre- and post-May 2019 timeseries:

$$p_{rt} = \frac{T + t^* - t}{T}(\bar{p}_r^{\text{pre}} + g^{\text{pre}}(w_{rt}, z_{rt}^{GH}) + B^{p,\text{pre}}(\xi_{r(t-1)}, p_{r(t-1)})) + \frac{t - t^*}{T}(\bar{p}_r^{\text{post}} + g^{\text{post}}(w_{rt}, z_{rt}^{GH}) + B^{p,\text{post}}(\xi_{r(t-1)}, p_{r(t-1)})) + e_{rt}^p,$$

$$e_{rt}^p \sim \mathcal{N}\left(\mathbf{0}, \frac{T + t^* - t}{T}\Sigma^{p,\text{pre}} + \frac{t - t^*}{T}\Sigma^{p,\text{post}}\right).$$

For product-regions that do not have pre-period observations, and hence do not have estimated fixed effects $\bar{p}_r^{\text{pre}}$, I use $\bar{p}_r^{\text{post}}$ without mixing. The baseline analysis corresponds to $T = 1$; I reestimate beliefs, costs, and the conduct test with $T = 3, 6$, and 12 months.

Results are shown in Panel C of Table C.1. Bertrand-Markov conduct is rejected for all values of T . Notably, while the conduct parameter estimate does shrink for $T = 12$, the lower end of the 95% confidence interval is close to the baseline. This suggests that the mixing also has some precision benefits for the estimate. Ultimately, the assumption of an immediate transition to a new Markov equilibrium in pricing rules after May 2019 does not appear to be critical to the results.

C.3 Initial conditions

Identifying the system of beliefs requires an assumption about their initial conditions. My baseline assumption starts firms' beliefs at the stationary distribution, as defined in (24) and (25). I verify robustness by perturbing this assumption in two ways. First, I adjust mean beliefs by one standard deviation in either direction, to $\mu_{fr1} \pm \sqrt{\text{Diag}(\Sigma_{fr1})}$. Second, I multiply Σ_{fr1} by either 0.5, representing tighter initial beliefs, or 2, representing wider initial beliefs. I then iteratively find beliefs, recover marginal costs, and estimate conduct as usual.

Panel D of Table C.1 contains the results. Perturbing the initial conditions leads to

Parameter	Fresh turkey				Deli turkey		
	Constant	Price	Breast	WS	Constant	Price	Low fat
β		-2.233 (0.149)	2.306 (0.504)	-3.131 (1.600)		-0.560 (0.050)	-0.532 (0.033)
diag(Σ)	0.152 (0.783)	0.165 (0.033)	1.357 (0.254)	0.350 (4.510)	0.000 (0.431)	0.131 (0.020)	0.000 (0.136)
II: Medium income	-1.653 (0.278)	0.665 (0.084)	-2.323 (0.194)	1.765 (0.243)	1.075 (0.060)	-0.074 (0.008)	-0.354 (0.041)
II: High income	-3.419 (0.261)	1.056 (0.081)	-2.398 (0.199)	2.232 (0.251)	0.052 (0.073)	0.064 (0.010)	-0.415 (0.039)
II: Children	0.996 (0.129)	-0.043 (0.033)	-0.725 (0.077)	0.003 (0.134)	1.134 (0.050)	-0.109 (0.006)	-0.044 (0.030)
Number of observations		57,616				109,820	
Median own-price elasticity		-6.41				-3.17	
Median aggregate elasticity		-1.06				-0.62	
First stage R^2		0.71				0.83	
First stage F -statistic		2,742.7				406.1	

Table C.2: Demand estimation results with approximate optimal instruments

Note: Heteroskedasticity-robust standard errors in parentheses. “WS” refers to whole/smoked.

essentially identical conduct parameters. It makes sense that initial conditions would not affect the results much: firms are sharing information at the start of the sample, leading to a quick convergence of firms’ beliefs. Regardless, this sensitivity confirms that the initial conditions assumption does not affect the ultimate results.

C.4 Optimal demand instruments

Approximating the [Chamberlain \(1987\)](#) optimal instruments for use in BLP-style demand estimation was originally suggested by [Berry et al. \(1995\)](#), and recent papers have shown its ability to improve the performance of mixed logit demand models ([Reynaert and Verboven, 2014](#); [Conlon and Gortmaker, 2020](#)). To test whether using optimal instruments affects my results, I follow the “approximate” optimal IV construction of [Conlon and Gortmaker \(2020\)](#), as initially conceived in [Berry et al. \(1999\)](#). I compute a pilot estimate of the structural parameters $\hat{\theta}^{1s}$ and residuals $\hat{\xi}^{1s}$ using one-step GMM and the baseline instruments. I then approximate $\mathbb{E} \left[\frac{\partial s(\theta; \xi)}{\partial \theta} \middle| z \right]$ with $\mathbb{E} \left[\frac{\partial s(\hat{\theta}^{1s}; \mathbb{E}[\hat{\xi}^{1s}])}{\partial \theta} \middle| z \right]$ and reestimate demand using two-step GMM with these as instruments.

Table C.2 shows the estimated demand parameters using the feasible approximation to the optimal instruments. The primary difference between these results and the baseline results is that estimated demand is more price-sensitive, particularly in fresh turkey. The

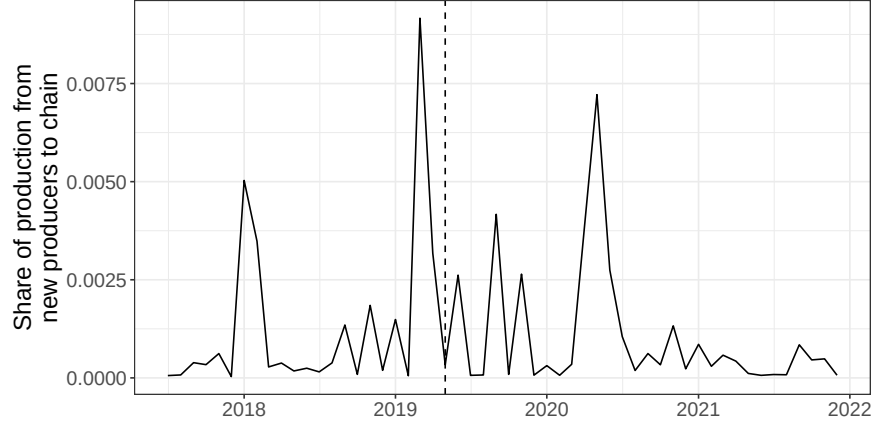


Figure C.1: Churn rates over time

Note: The vertical dashed line represents May 2019.

aggregate elasticity also moves further away from USDA’s estimates ([Maples et al., 2022](#)). Many of the other coefficients, particularly among the demographic interactions, remain similar.

I run these estimates through the remainder of the estimation pipeline and report the conduct results in Panel E of Table C.1. Notably, this is the only sensitivity where Bertrand-Markov conduct is not rejected at the 5% level. If I impose the alternative moment restriction of no discontinuity in Agri Stats clients’ costs, without differencing others’ costs, then Bertrand-Markov conduct is nonetheless rejected at the 5% level. If nothing else, these results demonstrate that the conduct test with the baseline supply-side model is falsifiable; the rejection of Bertrand-Markov conduct in the baseline analysis is not imposed by the supply-side assumptions. However, it does show that the results are at least somewhat sensitive to the demand estimates.

C.5 Vertical structure

This subsection discusses reduced-form evidence suggesting that processor-retailer bargaining was not a significant factor in the Agri Stats effect.

First, one could imagine that processors used the Agri Stats reports to learn which of their competitors were contracting with different retailers, and that therefore processors used the reports to segment the market. If this dynamic occurred, then the rate of churn of brand offerings at the retailer level might have been impacted by the end of the Agri Stats exchange. Figure C.1 plots churn—as measured by the share of production from producers that are new to a retail chain—by month over time. The first thing to notice is that churn is

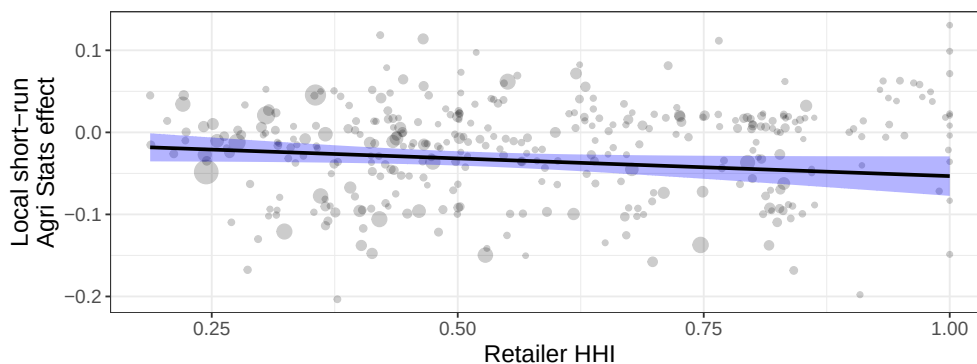


Figure C.2: Short-run Agri Stats effect by local retailer concentration

Note: Each bubble is a market (region-module), and its size is proportional to total quantity in that market. The black line and shaded region are the predictions and 95% confidence intervals from a weighted regression of local short-run Agri Stats effect on the retail-level HHI, with standard errors computed using a Murphy-Topel correction.

extremely small: less than 1% of production is from new producers to the chain. Second, this churn does not appreciably rise after May 2019, suggesting that the end of the information exchange had no effect on churn. This suggests that I can limit the effects of the exchange to the intensive margin of prices, as opposed to the extensive margin of product assortment.

Second, I examine whether local retailer concentration explains heterogeneity in the short-run Agri Stats effect on prices. Since downstream bargaining power can countervail upstream collusion (Chassang and Ortner, 2023), one would expect that retailers with more market power would be less affected by any pre-May 2019 collusion, and therefore should have Agri Stats effects with a smaller magnitude. Figure C.2 reveals that this is not the case (using local concentration as a proxy for market power): if anything, areas with more concentrated retail sectors had slightly larger-magnitude short-run Agri Stats effects, with a 100-point retail HHI increase being associated with a 0.04pp larger Agri Stats effect (although this is not statistically significant at the 5% level). This suggests that heterogeneity in bargaining power is not a significant factor in explaining price effects of information sharing in this context.

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